
Intersection Theory

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

August 4, 2026

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Introduction

A reader who has worked through *Algebraic Geometry* [4] can already intersect. On a smooth projective surface two curves meet in a well-defined number of points, counted with multiplicity; that number depends only on the divisor classes, it is symmetric, and it computes things the naive count cannot, such as the self-intersection $E^2 = -1$ of an exceptional curve, which no set-theoretic count of points could produce. The construction was built by hand, and it was built for divisors on a surface. It says nothing about two surfaces inside a fourfold, nothing about a curve and a surface inside a threefold, and nothing about what happens when the two subvarieties share a component instead of meeting transversally.

Those are not exotic questions. They are the questions the subject was born from. How many lines meet four general lines in $\mathbb{P}^3$? How many conics are tangent to five general conics? Nineteenth century enumerative geometry answered such questions in quantity, by a principle of conservation of number: deform the configuration until the answer becomes visible, and trust that the count did not change along the way. The answers were, for the most part, right. The method was not a proof. Schubert's calculus of 1879 systematized it into a symbolic algebra of conditions so effective, and so unjustified, that Hilbert made its rigorous foundation the fifteenth of his 1900 problems.

The gap is precisely the one the surface theory papers over. Conservation of number needs an equivalence relation on cycles under which the count is invariant, and it needs the count to be defined even when the two cycles fail to meet properly. Rational equivalence supplies the first: two cycles are equivalent when one deforms to the other along a family parametrized by $\mathbb{P}^1$, and the quotient of the cycles by that relation is the Chow group. Deformation to the normal cone, introduced by Fulton and MacPherson in the 1970s, supplies the second: rather than moving the cycles until they meet properly, degenerate the ambient embedding to its normal cone, where the intersection can be read off the zero section of a bundle. That construction replaced the moving lemma, which had been the classical route and which is genuinely hard to prove in the generality one wants.

What comes out is more than a rigorous count. The Chow group is a functor, covariant for proper maps and contravariant for flat ones, and on a smooth variety it carries a ring structure, so questions about subvarieties become questions in a graded ring. Vector bundles acquire Chern classes valued in that ring, computed from Segre classes and the splitting principle. And Grothendieck's Riemann-Roch theorem relates the Chern character of a pushforward to the pushforward of a Chern character,

subsuming Riemann-Roch for curves, which this reader has already seen [4, Riemann-Roch for curves], into a statement about arbitrary proper morphisms.

Goal

The book opens with cycles and rational equivalence, and establishes the Chow group $A_*(X)$ together with the two functorialities that make it usable: proper pushforward and flat pullback, with the localization sequence and homotopy invariance that compute it in the first examples ($\mathbb{A}^n$, $\mathbb{P}^n$, curves). Divisors come next, in the form the rest of the book needs them: not as the divisor class group of a normal variety, but as an operator $c_1(L) \cap -$ on cycle classes, commutative and compatible with both functorialities. That operator is the first Chern class, and it is the germ of everything that follows.

Segre classes of cones generalize it. They give the Chern classes of a vector bundle, the Whitney sum formula, the splitting principle, and with them the projective bundle formula, which computes the Chow groups of $\mathbb{P}(E)$ and closes the one debt the opening chapter leaves open. Deformation to the normal cone then delivers the Gysin map for a regular embedding, and the intersection product follows by reduction to the diagonal: on a smooth variety X , the product of two classes is the Gysin pullback along $X \hookrightarrow X \times X$. This is the point at which $A^*(X)$ becomes a ring, Bézout’s theorem becomes a corollary, and the surface pairing built by hand upstream is recovered as a special case rather than an analogy.

The last movement turns outward. Comparing rational equivalence with algebraic, homological and numerical equivalence puts the Chow ring next to cohomology through the cycle class map, which is what makes algebraic cycles a Hodge-theoretic object and what the standard conjectures are about; correspondences, cycles on $X \times Y$ acting as operators, are the entry point to motives. Riemann-Roch is proved in Grothendieck’s form and specialized back down to Hirzebruch-Riemann-Roch and Noether’s formula. The book closes where it began, on enumerative geometry: the Chow ring of the Grassmannian, Schubert calculus done properly, degeneracy loci, and the classical counting problems that motivated the whole apparatus, now with proofs.

0.0.1 Prerequisites and Place in the Series This volume builds on *Algebraic Geometry* [4]. No later volume currently lists it as a prerequisite. The reader has access to every completed volume of the series but is not assumed to have read all of them; prerequisites are cited where invoked.

Notation and Conventions

Throughout, a scheme is a scheme of finite type over a fixed algebraically closed field $\bar{k}$; such a scheme is called an *algebraic scheme*. A point is a closed point unless stated otherwise, and a subvariety of a scheme is a closed integral subscheme. The treatment follows Fulton’s *Intersection Theory* [2] as its spine, with Vakil’s intersection-theory notes [6] for exposition, adapted to a reader who already owns scheme theory and sheaf cohomology in full [4].

Part I

The Chow Group

Cycles are easy to write down and useless as they stand: the group of formal integer combinations of subvarieties is enormous, is not an invariant of anything, and admits no product. This part cuts it down to something that is. Rational equivalence identifies two cycles when one sweeps out to the other in a family over $\mathbb{P}^1$, and the quotient, the Chow group, turns out to carry exactly the structure the classical geometers were assuming when they deformed a configuration and trusted the count. Proper pushforward and flat pullback make it functorial, the localization sequence and homotopy invariance make it computable, and intersecting with a divisor gives the first operation that lowers dimension. That last operation, the first Chern class, is the seed of everything the rest of the book builds.

1

Rational Equivalence

Recall from [1] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x, y, z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x, y)], [g(x, y)] \in k[x, y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves f, g let F, G denote the resulting variety. The intersection of two (affine or projective) plane curves F, G at P was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}_{\mathbb{A}_k^2, P}}{(f, g)} = \dim_K \mathcal{O}_{Z, P}$$

where $Z = \mathcal{Z}(f, g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$
2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) + i(P, F_2 \cdot G)$
3. (invariance of representation): if F' is given by $f + gh$ for some $h \in k[x, y]$:

$$i(P, F' \cdot G) = i(P, F \cdot G)$$

4. If F, G have a common component through P (i.e. the dimension of the irreducible variety in $F \cap G$ containing p is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.
5. If $P \notin F \cap G$:

$$i(P, F \cdot G) = 0$$

6. if $f(x, y) = x - a$ and $g(x, y) = y - b$ (more generally $\frac{\partial(f, g)}{\partial(x, y)} \neq 0$ at p) then:

$$i(P, F \cdot G) = 1$$

7. If F has no common component with G and H and P is a simple point of F :

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H)$$

1.1 Cycles and Rational Equivalence

The classical intersection number recalled above attaches an integer to a pair of plane curves meeting at a point. Turning that local count into a theory valid on any variety and in every dimension calls for a home for “formal combinations of subvarieties” together with an equivalence relation flexible enough to make intersection numbers well defined. That home is the Chow group, the higher-dimensional descendant of the divisor class group of [4, Divisor Class Group]: where the divisor class group records codimension-one subvarieties modulo the divisors of rational functions, the Chow group records subvarieties of *every* dimension modulo the same mechanism.

This section builds the three objects in turn, the group of cycles $Z_k(X)$, the subgroup $\text{Rat}_k(X)$ of cycles rationally equivalent to zero, and the quotient $A_k(X)$. Throughout the chapter X is an algebraic scheme (a scheme of finite type over the algebraically closed field $\bar{k}$), a *subvariety* means a closed integral subscheme, and a point means a closed point unless stated otherwise.

The construction rests on the order of vanishing of a rational function along a codimension-one subvariety, introduced for divisors in [4, Divisors: Geometric Representation of Line Bundles]. On a variety W a prime divisor is a codimension-one subvariety V , and the local ring $\mathcal{O}_{W,V}$ is a one-dimensional local domain with fraction field the function field $k(W)$. When W is regular in codimension one this ring is a discrete valuation ring and its valuation ν_V records orders of vanishing ([4, Principal Divisor]); the length-theoretic definition below removes the regularity hypothesis, which matters as soon as W is an arbitrary subvariety of X .

Definition 1.1.1: Order Along a Subvariety

Let W be a variety and $V \subseteq W$ a codimension-one subvariety, and set $A = \mathcal{O}_{W,V}$, a one-dimensional local domain with fraction field $k(W)$. For nonzero $a \in A$ the quotient $A/(a)$ has finite length, and the *order of a along V* is

$$\text{ord}_V(a) = \text{length}_A(A/(a))$$

Because $\text{ord}_V(ab) = \text{ord}_V(a) + \text{ord}_V(b)$, this extends to a homomorphism $\text{ord}_V: k(W)^* \rightarrow \mathbb{Z}$ by $\text{ord}_V(a/b) = \text{ord}_V(a) - \text{ord}_V(b)$.

When $\mathcal{O}_{W,V}$ is a discrete valuation ring, in particular whenever W is normal, this length equals the valuation $\nu_V(a)$, so definition 1.1.1 agrees with the order function of [4, Divisors: Geometric Representation of Line Bundles]. For a fixed $r \in k(W)^*$ the integer $\text{ord}_V(r)$ vanishes for all but finitely many V ([4, Almost All Valuations Vanish]), so each sum $\sum_V \text{ord}_V(r)[V]$ below is finite.

Definition 1.1.2: Cycle

Let X be an algebraic scheme. A k -cycle on X is a finite formal $\mathbb{Z}$ -linear combination $\sum_i n_i [V_i]$ of k -dimensional subvarieties $V_i \subseteq X$ with $n_i \in \mathbb{Z}$. The k -cycles form the free abelian group

$$Z_k(X) = \bigoplus_{\dim V=k} \mathbb{Z}[V]$$

on the set of k -dimensional subvarieties of X , and the group of all cycles is $Z_*(X) = \bigoplus_{k \geq 0} Z_k(X)$.

A cycle remembers only which subvarieties occur and with what integer multiplicity; it detects neither embedded nor nonreduced structure, a point made precise by the equality $Z_k(X) = Z_k(X_{\text{red}})$. Two cases sit at the extremes. A 0-cycle is a finite formal sum of closed points. At the top, if X has dimension n with n -dimensional irreducible components $X_1, \dots, X_r$, then $Z_n(X)$ is free on $[X_1], \dots, [X_r]$; the *fundamental cycle*

$$[X] = \sum_i \text{length}_{\mathcal{O}_{X, X_i}}(\mathcal{O}_{X, X_i}) [X_i]$$

weights each component by the multiplicity with which X carries it, and reduces to $\sum_i [X_i]$ when X is reduced.

The one construction that produces relations among cycles is the divisor of a rational function. Given a $(k+1)$ -dimensional subvariety $W \subseteq X$ and a nonzero $r \in k(W)^*$, its *divisor* is the k -cycle

$$[\text{div}(r)] = \sum_V \text{ord}_V(r) [V] \in Z_k(X)$$

the sum running over the codimension-one subvarieties V of W , each of which is a k -dimensional subvariety of X . The sum is finite by [4, Almost All Valuations Vanish], and $[\text{div}(rs)] = [\text{div}(r)] + [\text{div}(s)]$ because ord_V is a homomorphism.

Definition 1.1.3: Rational Equivalence

A k -cycle $\alpha \in Z_k(X)$ is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are finitely many $(k+1)$ -dimensional subvarieties $W_i \subseteq X$ and rational functions $r_i \in k(W_i)^*$ with

$$\alpha = \sum_i [\text{div}(r_i)]$$

Two k -cycles are *rationally equivalent*, written $\alpha \sim \beta$, when $\alpha - \beta \sim 0$. The cycles rationally equivalent to zero form a subgroup $\text{Rat}_k(X) \leq Z_k(X)$: it is closed under addition by definition and under negation because $[\text{div}(r^{-1})] = -[\text{div}(r)]$.

Geometrically, a subvariety W together with a rational function r presents $[\text{div}(r)]$ as the difference between the zeros and the poles of r , that is, between the fibers over 0 and ∞ of the rational map $r: W \dashrightarrow \mathbb{P}^1$. Rational equivalence is thus the relation generated by sweeping one cycle to another through a family parametrized by the line. In the one codimension seen before, $k = n - 1$ on an n -dimensional variety, a cycle is a Weil divisor and rational equivalence is exactly the linear equivalence of [4, Divisor Class Group]; the Chow group below carries $\text{Cl } X$ to all dimensions at once.

Definition 1.1.4: Chow Group

The k -th Chow group of an algebraic scheme X is the quotient

$$A_k(X) = Z_k(X)/\text{Rat}_k(X)$$

Its elements are k -cycle classes, and the class of a cycle α is written $[\alpha]$, or simply α when no confusion results. The total Chow group is

$$A_*(X) = \bigoplus_{k \geq 0} A_k(X) = Z_*(X)/\text{Rat}_*(X), \quad \text{Rat}_*(X) = \bigoplus_{k \geq 0} \text{Rat}_k(X)$$

Following the cohomological convention, much of the literature grades by codimension, writing $A^p(X) = A_{n-p}(X)$ or $CH^p(X)$ when X is equidimensional of dimension n ; the two indexings carry the same information, and definition 1.1.4 is the promised general form of the construction previewed for surfaces in [4, Chow Rings and the General Intersection Product]. A cycle $\sum_i n_i [V_i]$ is *positive* (or *effective*) if it is nonzero with every $n_i \geq 0$, and a cycle class is positive if it has a positive representative. Positivity is the algebraic trace of an actual geometric subvariety, and deciding which classes are positive is among the central difficulties of the subject.

Passing to the quotient collapses exactly the ambiguity that a complete variety imposes on its points.

Example 1.1.5: Zero-Cycles on the Line

On $X = \mathbb{P}^1$ any two closed points are rationally equivalent. Write t for the affine coordinate and ∞ for the point at infinity. Distinct points $p, q \in \mathbb{A}^1 \subseteq \mathbb{P}^1$ are cut out by $t - p$ and $t - q$, and the rational function $r = (t - p)/(t - q)$ has

$$\text{ord}_p(r) = 1, \quad \text{ord}_q(r) = -1, \quad \text{ord}_x(r) = 0 \text{ for } x \neq p, q$$

since r takes the finite nonzero value 1 at ∞ . Hence $[\text{div}(r)] = [p] - [q]$ and $[p] \sim [q]$; taking $r = t - p$ instead gives $[\text{div}(t - p)] = [p] - [\infty]$, so $[\infty]$ lies in the same class. Every point of $\mathbb{P}^1$ therefore represents a single class h . A rational function on $\mathbb{P}^1$ has as many zeros as poles, so $\sum_V \text{ord}_V(r) = 0$ for all r , and the total-coefficient map $\sum_i n_i [p_i] \mapsto \sum_i n_i$ descends to an isomorphism

$$A_0(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{Z}, \quad h \mapsto 1$$

The affine line behaves oppositely. On $X = \mathbb{A}^1$ the function $t - p$ has the single zero p and no pole, so $[\text{div}(t - p)] = [p]$ and every point satisfies $[p] \sim 0$; thus $A_0(\mathbb{A}^1) = 0$. On an affine line a point can be swept away to nothing, whereas on the complete line it cannot, and the surviving invariant of a class is its degree.

The contrast in example 1.1.5 is why completeness recurs throughout the theory: the degree of a zero-cycle is an invariant of its rational equivalence class only on a complete scheme, a point taken up in section 1.2. The remaining first properties of Chow groups are collected next; each is either immediate or follows from the localization sequence established later in the chapter.

Example 1.1.6: First Properties of Chow Groups

Let X be an algebraic scheme.

1. *Reduction.* Since cycles see only subvarieties, $A_k(X) = A_k(X_{\text{red}})$ for every k .

2. *Disjoint unions.* If $X = X' \sqcup X''$ is a disjoint union of open-and-closed subschemes, then $A_k(X) = A_k(X') \oplus A_k(X'')$, as every subvariety lies in one piece.
3. *Top-dimensional cycles.* If $\dim X = n$, there is no subvariety of dimension $n + 1$, so $\text{Rat}_n(X) = 0$ and $A_n(X) = Z_n(X)$ is free on the n -dimensional components of X . In particular the coefficient of such a component in a class $\alpha \in A_n(X)$ is well defined.
4. *Right-exactness.* For closed subschemes $X_1, X_2 \subseteq X$ there is an exact sequence

$$A_k(X_1 \cap X_2) \rightarrow A_k(X_1) \oplus A_k(X_2) \rightarrow A_k(X_1 \cup X_2) \rightarrow 0$$

with no left-exactness in general. This is a consequence of the localization sequence developed later in the chapter.

Exercise 1.1.1

1. Let X be the reduced scheme consisting of two distinct points. Compute $Z_k(X)$ and $A_k(X)$ for all k .
2. Show that $A_0(\mathbb{A}^n) = 0$ for every $n \geq 1$.
3. Prove from the definitions that $Z_k(X) = Z_k(X_{\text{red}})$ and $\text{Rat}_k(X) = \text{Rat}_k(X_{\text{red}})$, and deduce $A_k(X) = A_k(X_{\text{red}})$.
4. Let X have dimension n with n -dimensional irreducible components $X_1, \dots, X_r$. Show that $A_n(X) = \bigoplus_{i=1}^r \mathbb{Z}[X_i]$.
5. Let C be a smooth projective curve over $\bar{k}$. Show that $A_0(C) \cong \text{Cl } C = \text{Pic } C$, and that the degree gives a short exact sequence $0 \rightarrow \text{Pic}^0 C \rightarrow A_0(C) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$.

1.2 Proper Pushforward and the Degree

Cycles and their classes change under morphisms, and the first of these functorialities is covariant: a proper morphism pushes cycles forward, respects rational equivalence, and on a complete scheme produces the degree of a zero-cycle.

A proper morphism $f: X \rightarrow Y$ carries closed sets to closed sets, so the image of a subvariety $V \subseteq X$ is again a subvariety $W = f(V) \subseteq Y$: the image of an irreducible set is irreducible, and properness (universal closedness) makes it closed. Without properness the image can fail to be closed, as for the projection of [4, Non-Proper Morphisms], and pushforward is not defined. The inclusion of function fields $k(W) \hookrightarrow k(V)$ is a finite extension exactly when $\dim V = \dim W$, and its degree counts the points of V over a general point of W .

Definition 1.2.1: Proper Pushforward

Let $f: X \rightarrow Y$ be a proper morphism and $V \subseteq X$ a subvariety with image $W = f(V)$. Set

$$\deg(V/W) = \begin{cases} [k(V) : k(W)] & \dim V = \dim W \\ 0 & \dim V > \dim W \end{cases}$$

(the extension $k(W) \hookrightarrow k(V)$ is finite in the first case), and put $f_*[V] = \deg(V/W)[W]$. Extending $\mathbb{Z}$ -linearly gives the *pushforward* $f_*: Z_k(X) \rightarrow Z_k(Y)$.

Pushforward is covariant: for composable proper morphisms $(g \circ f)_* = g_* \circ f_*$, since degrees of function-field extensions multiply in towers. When $\dim V = \dim W$ the number $\deg(V/W)$ counts the sheets of the generically finite cover $V \rightarrow W$, matching the topological degree over $\mathbb{C}$. That f_* descends to Chow groups is the content of this section, and rests on how it acts on the divisor of a rational function.

Proposition 1.2.2: Pushforward of Divisor

Let $f: X \rightarrow Y$ be a proper surjective morphisms between varieties, and let $r \in k(X)^*$. Then:

1. if $\dim(Y) < \dim(X)$, then

$$f_*[\operatorname{div}(r)] = 0$$

2. if $\dim(Y) = \dim(X)$, then

$$f_*[\operatorname{div}(r)] = [\operatorname{div}(N(r))]$$

where $N(r)$ is the *norm* of r (computed as the determinant of the $R(Y)$ -linear endomorphism of $R(X)$ given by multiplying by r).

Proof:

The order function and the norm are multiplicative, so both sides of each identity are homomorphisms in r . Both parts reduce to two model computations; write $n = \dim X$.

Step 1: the line over a point. Let $Y = \operatorname{Spec} k$ and $X = \mathbb{P}^1$, with f the structure map. Every closed point of $\mathbb{P}^1$ has residue field $k = \bar{k}$, so $f_*[p] = [Y]$ for each p and $f_*[\operatorname{div}(r)] = (\sum_V \operatorname{ord}_V(r))[Y]$. Writing $r = c \prod_j (t - a_j)^{m_j}$ in the affine coordinate t , the order of r at a_j is m_j and its order at ∞ is $-\sum_j m_j$ (in $s = 1/t$ one has $r = s^{-\sum_j m_j} u$ with u a unit at $s = 0$), so $\sum_V \operatorname{ord}_V(r) = 0$ and $f_*[\operatorname{div}(r)] = 0$. The same count shows that on any proper curve C over a field K a principal divisor has degree zero: normalize C and choose a finite map $\tilde{C} \rightarrow \mathbb{P}_K^1$, then apply Step 2 to that map and this computation to $\mathbb{P}_K^1 \rightarrow \operatorname{Spec} K$.

Step 2: finite morphisms of equal dimension. Suppose f is finite with $\dim Y = \dim X$, so that $K = k(Y) \subseteq L = k(X)$ is a finite extension. Fix a codimension-one subvariety $W \subseteq Y$ and let $A = \mathcal{O}_{Y,W}$, a one-dimensional local domain with fraction field K . Its integral closure B in L is a finite A -module whose maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_s$ correspond to the codimension-one subvarieties $V_i \subseteq X$ over W , with $B_{\mathfrak{m}_i} = \mathcal{O}_{X,V_i}$ and residue degree $[k(V_i) : k(W)] = [B/\mathfrak{m}_i : A/\mathfrak{m}]$. Only these V_i contribute to the coefficient of $[W]$ in $f_*[\operatorname{div}(r)]$, which is therefore $\sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$, while the coefficient of $[W]$ in $[\operatorname{div}(N(r))]$ is $\operatorname{ord}_W(N(r))$. For $r \in B$ a nonzerodivisor these

agree:

$$\text{ord}_W(N(r)) = \text{length}_A(A/N(r)A) = \text{length}_A(B/rB) = \sum_i [k(V_i) : k(W)] \text{ord}_{V_i}(r)$$

the middle equality is the determinant-length formula, since $N(r)$ is the determinant of multiplication by r on the finite A -module B , and the last is additivity of length across the semilocal decomposition of B , weighted by residue degrees (see [2, Appendix A] for this length bookkeeping). As $L = \text{Frac}(B)$ and both sides are homomorphisms, the identity holds for all $r \in L^*$, which is part (2) for finite f .

Step 3: the general case. A proper surjective f with $\dim Y = \dim X$ is generically finite, and the coefficient of a codimension-one $W \subseteq Y$ in $f_*[\text{div}(r)]$ is computed over the generic point of W , where f becomes finite; Step 2 gives (2). For (1), let $\dim Y < n$. Since $f_*[\text{div}(r)] \in Z_{n-1}(Y)$ vanishes when $\dim Y < n-1$, only $\dim Y = n-1$ remains, and there $f_*[\text{div}(r)] = c[Y]$ with c its coefficient over the generic point of Y . Base-changing to $K = k(Y)$ replaces f by a proper curve $C = X_K$ over the field K and r by a function in $k(C)^*$, and $c = \sum_{P \in C} \text{ord}_P(r) [k(P) : K]$ is the degree of its principal divisor, which vanishes by Step 1. Hence $f_*[\text{div}(r)] = 0$, completing the proof.

Theorem 1.2.3: Pushforward Preserves Rational Equivalence

Let $f: X \rightarrow Y$ be proper. If $\alpha \in Z_k(X)$ is rationally equivalent to zero, then so is $f_*\alpha$. Consequently f_* descends to a homomorphism

$$f_*: A_k(X) \rightarrow A_k(Y)$$

and $A_*(-)$ equipped with proper pushforward is a covariant functor.

Proof:

It suffices to push forward a generator $\alpha = [\text{div}(r)]$, where $r \in k(W)^*$ for some $(k+1)$ -dimensional subvariety $W \subseteq X$. Replacing X by W and Y by $f(W)$ makes f proper and surjective, and proposition 1.2.2 identifies $f_*[\text{div}(r)]$ with 0 (when $\dim f(W) < \dim W$) or with $[\text{div}(N(r))]$ (when $\dim f(W) = \dim W$). Either value is a principal divisor, hence rationally equivalent to zero, so $f_*\alpha \sim 0$. Functoriality on classes then follows from $(g \circ f)_* = g_* \circ f_*$ on cycles, as we sought to show.

Definition 1.2.4: Degree of a Zero-Cycle

Let X be complete (proper over $\text{Spec } k$), with structure morphism $\pi: X \rightarrow \text{Spec } k$, and let $\alpha = \sum_P n_P [P]$ be a zero-cycle on X . The *degree* of α is

$$\deg(\alpha) = \int_X \alpha = \sum_P n_P [k(P) : k]$$

Since $\text{Spec } k$ is a single point with no lower-dimensional subvarieties, $A_0(\text{Spec } k) = Z_0(\text{Spec } k) = \mathbb{Z}[\text{Spec } k] \cong \mathbb{Z}$, and under this identification $\deg(\alpha) = \pi_*\alpha$. When the scheme is clear one writes $\int \alpha$ for $\int_X \alpha$.

By theorem 1.2.3 rationally equivalent zero-cycles have the same degree, so $\deg: A_0(X) \rightarrow \mathbb{Z}$ is well defined on a complete scheme. Extending by $\int_X \alpha = 0$ for $\alpha \in A_k(X)$ with $k > 0$ and using $\pi_X = \pi_Y \circ f$, any morphism $f: X \rightarrow Y$ of complete schemes satisfies

$$\int_X \alpha = \int_Y f_* \alpha$$

for every $\alpha \in A_*(X)$, a special case of the functoriality of pushforward.

1.2.5 Assuming Pushforward with subschemes Let $Y_1, \dots, Y_r$ be closed subschemes of a scheme X . Let Y be a closed subscheme of X which contains all the Y_i . Given $\alpha_i \in A_* Y_i$, $i = 1, \dots, r$, and $\beta \in A_* Y$, the notation “ $\beta = \sum_{i=1}^r \alpha_i$ in $A_* Y$ ” stands in for the precise equation $\beta = \sum_{i=1}^r \varphi_{i*}(\alpha_i)$ where φ_i is the inclusion of Y_i in Y .

Example 1.2.6: Degrees, Bézout, and the Role of Properness

1. *Bézout’s theorem.* Two curves $C, D \subseteq \mathbb{P}^2$ of degrees c, d with no common component meet in a zero-cycle $C \cdot D$ of degree cd . Pushing forward to a point, $\int_{\mathbb{P}^2} [C \cdot D] = cd$: the classical intersection number from the chapter opening is the degree of a zero-cycle, and its independence of the choice of C and D in their linear systems is rational equivalence together with theorem 1.2.3.
2. *Properness is necessary.* For the non-proper projection of [4, Non-Proper Morphisms] the image of a subvariety can fail to be closed, so f_* is not even defined; and on $\mathbb{A}^1$ every point is rationally equivalent to zero (example 1.1.5), so no degree survives. The degree is an invariant of a rational equivalence class only on a complete scheme.
3. *A finite cover of curves.* A finite morphism $\varphi: C \rightarrow C'$ of smooth projective curves satisfies $\varphi_*[P] = [k(P) : k(\varphi(P))] [\varphi(P)]$. For a point $Q \in C'$ away from the branch locus, $\varphi^{-1}(Q)$ consists of $\deg \varphi = [k(C) : k(C')]$ reduced points, so $\varphi_*[\varphi^{-1}(Q)] = (\deg \varphi) [Q]$; since $\deg(\varphi_* \alpha) = \deg(\alpha)$ on the complete curves, the sheet count is exactly the degree recorded by φ .

Exercise 1.2.1

1. For the finite morphism $\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^1$, $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (in characteristic $\neq 2$), compute $\varphi_*[p]$ for a closed point p and $\varphi_*[\varphi^{-1}(q)]$ for a closed point q . Verify that $\deg(\varphi_* \alpha) = \deg(\alpha)$ for every zero-cycle α , and explain where the degree $\deg \varphi = 2$ appears.
2. Let $\alpha = \sum_i n_i [P_i]$ be a zero-cycle on $\mathbb{P}^n$ over $\bar{k}$. Compute $\deg(\alpha)$ and prove it is invariant under rational equivalence.
3. Deduce from proposition 1.2.2 that a nonzero rational function on a smooth projective curve C has principal divisor of degree zero, and show by example that this fails on $\mathbb{A}^1$.
4. For the structure map $\pi: \mathbb{P}^1 \rightarrow \text{Spec } \bar{k}$ and a rational function $r \in k(\mathbb{P}^1)^*$, verify $\int_{\mathbb{P}^1} [\text{div}(r)] = \int_{\text{Spec } \bar{k}} \pi_*[\text{div}(r)]$, and say which hypothesis of definition 1.2.4 would fail were $\mathbb{P}^1$ replaced by $\mathbb{A}^1$.

1.3 Flat Pullback

Proper pushforward of section 1.2 is covariant: it moves cycles along a morphism in the direction of the arrow, lowering nothing and raising nothing in dimension. The second functoriality of the Chow group runs the other way. A morphism $f: X \rightarrow Y$ pulls a subvariety $V \subseteq Y$ back to its preimage $f^{-1}(V) \subseteq X$, and one would like to read that preimage as a cycle on X . For an arbitrary f this fails: the preimage can jump in dimension from fiber to fiber, can acquire embedded components, and carries no canonical multiplicities. The hypothesis that repairs every one of these defects at once is flatness. A flat morphism of relative dimension n has equidimensional fibers, so the preimage of a k -dimensional subvariety is pure of dimension $k + n$, and the scheme-theoretic preimage supplies the multiplicities with which its components must be counted. The resulting map $f^*: Z_k(Y) \rightarrow Z_{k+n}(X)$ raises dimension by n , is contravariant, and descends to Chow groups.

This section constructs f^* , records its three structural properties (it is a graded homomorphism, it composes contravariantly, and it commutes with proper pushforward across a fiber square), and proves that it respects rational equivalence. The two guiding examples are the ones every later computation rests on: restriction to an open subscheme, and the projection $X \times \mathbb{A}^n \rightarrow X$ of a trivial affine bundle.

Recall that a morphism $f: X \rightarrow Y$ of algebraic schemes is flat if each local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. Flatness is the algebraic incarnation of the geometric demand that the fibers of f vary without collapsing or jumping, and its precise dimensional consequence is what makes pullback of cycles possible.

Definition 1.3.1: Flat Morphism of Relative Dimension n

A morphism $f: X \rightarrow Y$ of algebraic schemes is *flat of relative dimension n* if f is flat and, for every subvariety $W \subseteq Y$ and every irreducible component Z of $f^{-1}(W)$,

$$\dim Z = \dim W + n$$

Equivalently, f is flat and every nonempty fiber of f has pure dimension n .

The equivalence stated in the definition is the mechanism the whole construction turns on, so it is worth isolating. Flatness forces the fiber dimension to be locally constant, and it forces the preimage of a subvariety to inherit that constant as an additive shift; neither holds without it.

Proposition 1.3.2: Flat Preimages Are Equidimensional

Let $f: X \rightarrow Y$ be a flat morphism of algebraic schemes with Y a variety, and suppose every nonempty fiber of f has pure dimension n . Then for every subvariety $W \subseteq Y$ each irreducible component of $f^{-1}(W)$ has dimension $\dim W + n$. In particular $f^{-1}(W)$ is either empty or pure of dimension $\dim W + n$.

Proof:

Write $g: f^{-1}(W) \rightarrow W$ for the restriction of f ; it is the base change of f along the closed immersion $W \hookrightarrow Y$, hence flat, and its fibers are fibers of f , so each nonempty fiber of g has pure dimension n . Thus it suffices to prove the statement for $W = Y$: that a flat morphism $g: X' \rightarrow W$ onto a variety W , with all nonempty fibers of pure dimension n , has $\dim Z = \dim W + n$ for

every irreducible component Z of X' .

Let $Z \subseteq X'$ be an irreducible component, with generic point η . Flatness satisfies going-down (the going-down property of flat local homomorphisms, [3, p. III.9]), so the minimal prime η of X' contracts to the minimal prime of W ; that is, $g(\eta)$ is the generic point of W , and g restricts to a dominant morphism $g|_Z: Z \rightarrow W$ of varieties. Both are integral of finite type over $\bar{k}$, so their dimensions are transcendence degrees of function fields, and the dimension of a fiber of a dominant morphism of varieties can be read at a closed point.

The fiber-dimension theorem for the dominant morphism $g|_Z: Z \rightarrow W$ of varieties ([3, Exercise II.3.22]) provides a dense open $U \subseteq Z$ such that for every closed point $z \in U$, the fiber $(g|_Z)^{-1}(g(z))$ has dimension exactly $\dim Z - \dim W$ at z . The complement in Z of the other irreducible components of X' is also a dense open, so their intersection is nonempty and contains a closed point z ; write $w = g(z) \in W$. Since z lies off the other components, X' coincides with Z near z , so the fiber $X'_w = g^{-1}(w)$ coincides with $(g|_Z)^{-1}(w)$ near z , and its dimension at z is $\dim Z - \dim W$. By hypothesis X'_w is nonempty of pure dimension n , so this local fiber dimension equals n . Therefore

$$n = \dim Z - \dim W$$

hence $\dim Z = \dim W + n$, as we sought to show.

The proposition is the geometric license behind the definition of flat pullback. If $V \subseteq Y$ is a k -dimensional subvariety and f is flat of relative dimension n , then $f^{-1}(V)$ is a closed subscheme of X , pure of dimension $k + n$, and one takes its associated cycle, weighting each component by the length that measures the multiplicity with which the scheme carries it. This is exactly the fundamental cycle of definition 1.1.2 applied to the scheme-theoretic preimage.

Definition 1.3.3: Flat Pullback

Let $f: X \rightarrow Y$ be a flat morphism of relative dimension n and $V \subseteq Y$ a k -dimensional subvariety. The scheme-theoretic preimage $f^{-1}(V) = X \times_Y V$ is a closed subscheme of X , pure of dimension $k + n$, with irreducible components $Z_1, \dots, Z_r$. The *flat pullback* of $[V]$ is the $(k + n)$ -cycle

$$f^*[V] = [f^{-1}(V)] = \sum_{i=1}^r \text{length}_{\mathcal{O}_{f^{-1}(V), Z_i}} (\mathcal{O}_{f^{-1}(V), Z_i}) [Z_i]$$

the length being taken in the local ring of the preimage scheme at the generic point of Z_i . Extending $\mathbb{Z}$ -linearly gives the flat pullback

$$f^*: Z_k(Y) \rightarrow Z_{k+n}(X)$$

The multiplicities are the point of the construction. The preimage as a topological set records only the components Z_i ; the scheme structure records how many times f “covers” each one, and it is these lengths that make f^* compatible with divisors and hence with rational equivalence. When f is an open immersion the preimage is reduced and every length is 1, so $f^*[V] = [V \cap X]$; this is the source of the restriction map computed in the next example. The relative dimension n is the amount by which f^* raises the dimension of a cycle, so f^* is graded of degree $+n$.

Example 1.3.4: Open Restriction and the Projection $X \times \mathbb{A}^n \rightarrow X$

Two flat morphisms recur throughout the theory, and their pullbacks can be read off the definition.

1. *Open immersion.* Let $j: U \hookrightarrow Y$ be the inclusion of an open subscheme. Such j is flat of relative dimension 0: it is a localization at the level of local rings, and its fibers are single reduced points (over $u \in U$) or empty (over $y \notin U$). For a k -dimensional subvariety $V \subseteq Y$ the preimage $j^{-1}(V) = V \cap U$ is the open subscheme of V , which is reduced. If $V \cap U \neq \emptyset$ it is a k -dimensional subvariety of U and $j^*[V] = [V \cap U]$; if $V \subseteq Y \setminus U$ the preimage is empty and $j^*[V] = 0$. Thus flat pullback along an open immersion is the *restriction of cycles* $Z_k(Y) \rightarrow Z_k(U)$ that forgets the part of a cycle supported off U , the map at the heart of the localization sequence of section 1.4.
2. *Trivial affine bundle.* Let $p: X \times \mathbb{A}^n \rightarrow X$ be the projection. It is flat, being a base change of the flat morphism $\mathbb{A}_k^n \rightarrow \text{Spec } \bar{k}$ along $X \rightarrow \text{Spec } \bar{k}$, and every fiber is a copy of $\mathbb{A}^n$, pure of dimension n ; so p is flat of relative dimension n . For a k -dimensional subvariety $V \subseteq X$ the preimage is $V \times \mathbb{A}^n$, which is integral of dimension $k + n$, so

$$p^*[V] = [V \times \mathbb{A}^n]$$

the cylinder over V . No multiplicity appears because $V \times \mathbb{A}^n$ is a variety. This is the computation that will make p^* an isomorphism on Chow groups (theorem 1.4.6).

Both examples exhibit the pattern the general theory generalizes: pullback restricts a cycle to an open part in the first case and thickens it uniformly along the fiber in the second. The structural properties below show that these behaviors are functorial.

The first property collects the ways in which f^* is well behaved as a map of cycle groups. It is a graded homomorphism raising dimension by the relative dimension, and it composes contravariantly, so that a tower of flat morphisms pulls back through the composite exactly as one expects.

Proposition 1.3.5: Basic Properties of Flat Pullback

Let $f: X \rightarrow Y$ be flat of relative dimension n .

1. *Homomorphism.* $f^*: Z_k(Y) \rightarrow Z_{k+n}(X)$ is a homomorphism of abelian groups for each k , and assembles to a graded homomorphism $f^*: Z_*(Y) \rightarrow Z_*(X)$ of degree n .
2. *Fundamental cycle.* If Y is a variety, then $f^*[Y] = [X]$: flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space.
3. *Composition.* If $g: Y \rightarrow Y'$ is flat of relative dimension m , then $g \circ f$ is flat of relative dimension $m + n$ and

$$(g \circ f)^* = f^* \circ g^*$$

as maps $Z_k(Y') \rightarrow Z_{k+m+n}(X)$.

4. *Identity.* $(\text{id}_X)^* = \text{id}$ on $Z_*(X)$.

Proof:

1. *Homomorphism.* By construction f^* is defined on generators $[V]$ and extended $\mathbb{Z}$ -linearly, so it is a homomorphism $Z_k(Y) \rightarrow Z_{k+n}(X)$ by fiat; proposition 1.3.2 is what guarantees the target group is $Z_{k+n}(X)$, since $f^{-1}(V)$ is pure of dimension $k+n$. Summing over k gives a degree- n graded homomorphism.
2. *Fundamental cycle.* Let Y be a variety of dimension d . Since Y is integral, its fundamental cycle is the single generator $[Y]$ with multiplicity 1, not a sum over several components. The preimage $f^{-1}(Y) = X$ is all of X , and $f^*[Y] = [X]$ is by definition the fundamental cycle of X , weighting each top-dimensional component of X by its length multiplicity. This is definition 1.1.2's fundamental cycle $[X]$, so the identity holds.
3. *Composition.* Composites of flat morphisms are flat, and relative dimensions add because fiber dimensions add along the tower: a fiber of $g \circ f$ over $y' \in Y'$ maps to the fiber $g^{-1}(y')$ of pure dimension m , with fibers of f of pure dimension n , so it has pure dimension $m+n$. Hence $g \circ f$ is flat of relative dimension $m+n$. For the cycle identity it suffices to check on a generator $[V]$ with $V \subseteq Y'$ a k -dimensional subvariety. Both sides are cycles supported on $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$, so the claim is that the multiplicities match component by component. The scheme-theoretic preimage is computed by iterated fiber product,

$$(g \circ f)^{-1}(V) = X \times_{Y'} V = X \times_Y (Y \times_{Y'} V) = f^{-1}(g^{-1}(V))$$

so the underlying scheme of $(g \circ f)^{-1}(V)$ equals the underlying scheme of $f^{-1}(g^{-1}(V))$. It remains to see that the length multiplicities agree, that is, that applying f^* to the cycle $g^*[V] = \sum_i \ell_i [W_i]$ (where W_i are the components of $g^{-1}(V)$ with lengths ℓ_i) reproduces the multiplicities of $(g \circ f)^{-1}(V)$. Fix a component Z of the common preimage, with generic point z lying over the generic point w of some W_i . Flatness of f makes $\mathcal{O}_{X,z}$ flat over $\mathcal{O}_{Y,w}$, and length is additive and multiplies through a flat local homomorphism along the fiber: the length of $\mathcal{O}_{(g \circ f)^{-1}(V),z}$ equals ℓ_i times the length of $\mathcal{O}_{f^{-1}(W_i),z}$. Summing z over the components of $f^{-1}(W_i)$ and i over the components of $g^{-1}(V)$ gives $(g \circ f)^*[V] = f^*(g^*[V])$, as required.

4. *Identity.* The identity map is flat of relative dimension 0 with $(\text{id})^{-1}(V) = V$ reduced, so $(\text{id}_X)^*[V] = [V]$ on generators, completing the proof.

Contravariant functoriality (part 3) is what will let a Chow-group computation be assembled from simpler flat maps, exactly as covariant functoriality of pushforward let degrees be computed in towers. The fundamental-cycle identity (part 2) is the seed of every computation in section 1.5: it identifies the generator of the top Chow group of the total space of a flat family in terms of the base.

The remaining two properties are the interaction of flat pullback with the machinery already built. The first is its compatibility with proper pushforward. Pullback and pushforward move cycles in opposite directions, and they do not commute for a single morphism; but across a fiber square, where one has a flat map and a proper map meeting at right angles, the two operations do commute. This is the cycle-level form of the projection formula, and it is the technical engine behind essentially every later compatibility.

Proposition 1.3.6: Compatibility of Flat Pullback and Proper Pushforward

Consider a fiber square of algebraic schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

that is Cartesian, $X' = X \times_Y Y'$, with f proper and g flat of relative dimension n . Then f' is proper, g' is flat of relative dimension n , and

$$g^* \circ f_* = f'_* \circ g'^*: Z_k(X) \rightarrow Z_{k+n}(Y')$$

Consequently the identity holds on Chow groups.

Proof:

Step 1: the maps in the base-changed square. Properness and flatness are stable under base change, so f' is proper and g' is flat; the fibers of g' over a point of X are fibers of g over its image in Y , so g' has the same pure relative dimension n as g . Both composites $g^* \circ f_*$ and $f'_* \circ g'^*$ therefore land in $Z_{k+n}(Y')$, and both are homomorphisms, so it suffices to check the identity on a generator $[V]$ with $V \subseteq X$ a k -dimensional subvariety.

Step 2: reduction to $X = V$. Replacing X by V , Y by $W = f(V)$, and forming the induced fiber squares, the four schemes become V , W , $V' = V \times_W W'$, $W' = W \times_Y Y'$, with $f: V \rightarrow W$ proper surjective and $g: W' \rightarrow W$ flat of relative dimension n . It suffices to prove

$$g^*(f_*[V]) = f'_*(g'^*[V])$$

where now $f_*[V] = \deg(V/W)[W]$ by definition 1.2.1 and $g'^*[V] = [g'^{-1}(V)] = [V']$ (with its multiplicities).

Step 3: the two dimension regimes. There are two cases according to whether f drops dimension.

If $\dim V > \dim W$, then $\deg(V/W) = 0$, so the left side is $g^*(0) = 0$. On the right, $V' = V \times_W W'$ maps to W' under f' with fibers the fibers of $V \rightarrow W$, which are positive dimensional over the generic point; hence $\dim V' > \dim W'$ and every component of V' has image of strictly smaller dimension in W' , so $f'_*[V'] = 0$ by the dimension clause of definition 1.2.1. Both sides vanish.

If $\dim V = \dim W$, write $d = \deg(V/W) = [k(V) : k(W)]$. The left side is $g^*(d[W]) = d[W']$, using $g^*[W] = [W'] = [g^{-1}(W)]$ (the base W is a variety and g is flat, so g^* of its fundamental cycle is the fundamental cycle of W' by proposition 1.3.5(2), which here is the total space $W' = g^{-1}(W)$). For the right side, $[V'] = g'^*[V]$ carries the pullback multiplicities flagged above, and $f'_*[V']$ must account for both those multiplicities and the residue-field degrees; flatness is what makes only the field degree survive. Fix a component W'_j of W' , with generic point η_j . Localizing at η_j , the fiber of $f': V' \rightarrow W'$ over η_j is $\text{Spec } k(V) \otimes_{k(W)} k(W'_j)$: the generic fiber of $f: V \rightarrow W$ is $\text{Spec } k(V)$, a $k(W)$ -algebra of dimension d since V, W are varieties, and flat base change along $k(W) \hookrightarrow k(W'_j)$ preserves that dimension, so $\dim_{k(W'_j)}(k(V) \otimes_{k(W)} k(W'_j)) = d$. This dimension is exactly the total multiplicity that $f'_*[V']$ records over W'_j : summing the pullback multiplicity of each component of V' dominating W'_j against its residue degree over $k(W'_j)$ recovers the length d of the fiber algebra, because flatness of g (hence of g') forbids any

length being created or lost in the base change. Thus $f'_*[V'] = \sum_j d[W'_j] = d[W']$, matching the left side. The two sides agree in both regimes, which establishes $g^*f_* = f'_*g'^*$ on cycles. Since each of g^* , f_* , f'_* , g'^* descends to Chow groups by theorem 1.2.3 and proposition 1.3.7 below, the identity descends as well, completing the proof.

The compatibility should be read as the statement that pushing a cycle forward and then restricting to a flat base change gives the same class as first restricting and then pushing forward within the base change. In the special case where g is an open immersion, it says that pushforward commutes with restriction to an open set, which is the reason the localization sequence of the next section is a sequence of A_* -functorial maps. In the case where f is the projection to a point, it recovers the invariance of the degree of a zero-cycle under a flat base change of the ground scheme.

Everything so far concerns cycles. To obtain a map of Chow groups, flat pullback must carry Rat_k into Rat_{k+n} . The mechanism is that flat pullback commutes with taking the divisor of a rational function, up to the multiplicities that flatness controls, so a cycle that is a sum of divisors pulls back to a sum of divisors.

Proposition 1.3.7: Flat Pullback Preserves Rational Equivalence

Let $f: X \rightarrow Y$ be flat of relative dimension n . If $\alpha \in Z_k(Y)$ is rationally equivalent to zero, then $f^*\alpha \in Z_{k+n}(X)$ is rationally equivalent to zero. Consequently f^* descends to a homomorphism

$$f^*: A_k(Y) \rightarrow A_{k+n}(X)$$

and $A_*(-)$ equipped with flat pullback is a contravariant functor on flat morphisms of algebraic schemes, raising dimension by the relative dimension.

Proof:

It suffices to treat a generator of $\text{Rat}_k(Y)$, namely $\alpha = [\text{div}(r)]$ for a $(k+1)$ -dimensional subvariety $W \subseteq Y$ and a nonzero $r \in k(W)^*$. The claim is that $f^*[\text{div}(r)]$ is a sum of divisors of rational functions on subvarieties of X .

Step 1: reduction to the total space over W . The cycle $[\text{div}(r)]$ is supported on W , and $f^*[\text{div}(r)]$ is supported on $f^{-1}(W)$. Let $W' = f^{-1}(W) = X \times_Y W$ with its scheme-theoretic (possibly non-reduced) structure, and let $f_W: W' \rightarrow W$ be the restriction of f , which is flat of relative dimension n by base change. Because the components of $f^{-1}(W)$ are the components of W' and the pullback multiplicities are computed inside W' , it is enough to prove that $f_W^*[\text{div}(r)] \sim 0$ in $Z_{k+n}(W')$; the class then remains rationally equivalent to zero in X under the proper (closed immersion) pushforward along $W' \hookrightarrow X$, which preserves rational equivalence by theorem 1.2.3. Thus one may assume $Y = W$ is a variety and $r \in k(Y)^*$.

Step 2: the key identity on components. With Y a variety and $r \in k(Y)^*$, the divisor $[\text{div}(r)] = \sum_V \text{ord}_V(r)[V]$ runs over the codimension-one subvarieties $V \subseteq Y$. The assertion to prove is the flat-pullback divisor formula

$$f^*[\text{div}(r)] = [\text{div}(f^\#r)] \quad \text{on each component of } X \tag{1.1}$$

interpreted componentwise as follows. Let $X_1, \dots, X_s$ be the irreducible components of X , each of dimension $n + \dim Y$ by proposition 1.3.2, with generic points dominating Y (flatness sends generic points to the generic point of the base). On each X_j the pullback $f^\#r$ of r is a nonzero

rational function, since $r \neq 0$ and X_j dominates Y . The claim is

$$f^*[\operatorname{div}(r)] = \sum_{j=1}^s m_j [\operatorname{div}(f^\# r|_{X_j})]$$

where $m_j = \operatorname{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$ is the multiplicity of the component X_j in the fundamental cycle $[X]$. Each term on the right is a divisor of a rational function on the subvariety $(X_j)_{\operatorname{red}}$, hence rationally equivalent to zero, so (1.1) gives $f^*[\operatorname{div}(r)] \sim 0$.

Step 3: verifying the identity. Fix a codimension-one subvariety $D \subseteq X_j$, a $(k+n)$ -dimensional subvariety of X , and let $V = \overline{f(D)} \subseteq Y$. Two cases arise by the relative-dimension count of proposition 1.3.2 applied to $f|_{X_j}$.

If $\dim V = \dim Y$, then $V = Y$ and D dominates Y ; the coefficient of $[D]$ in $[\operatorname{div}(f^\# r)]$ is $\operatorname{ord}_D(f^\# r)$, computed in the discrete-valuation-style local ring $\mathcal{O}_{X_j,D}$, while D does not appear in $f^*[\operatorname{div}(r)]$ because $[\operatorname{div}(r)]$ has no component equal to Y . But $\operatorname{ord}_D(f^\# r) = 0$. The load-bearing point is that D dominates Y : its generic point ξ maps to the generic point of Y , at which r is a nonzero rational function, that is, a unit of $k(Y)$. Since $f^\# r$ is the pullback of r along the local homomorphism $\mathcal{O}_{Y,Y} = k(Y) \rightarrow \mathcal{O}_{X_j,D}$ induced on stalks at ξ , the image $f^\# r$ is a unit of $\mathcal{O}_{X_j,D}$, so $\operatorname{ord}_D(f^\# r) = 0$ and no such D contributes to either side. (A nonzero rational function need not be a unit along a codimension-one subvariety in general; here it is one precisely because D dominates the base and so sees r only at the generic point of Y .)

If $\dim V = \dim Y - 1$, then V is one of the codimension-one subvarieties appearing in $[\operatorname{div}(r)]$, and D is a component of $f^{-1}(V)$, lying on one or more of the components X_j . Two kinds of local ring enter the count, and keeping them distinct is the crux. Write

$$A = \mathcal{O}_{Y,V}, \quad \tilde{B} = \mathcal{O}_{X,D}$$

Here A is a one-dimensional local domain and $\operatorname{ord}_V(r) = \operatorname{length}_A(A/rA)$; the ring $\tilde{B}$ is the local ring of the whole (possibly nonreduced) scheme X at the generic point of D , and it is flat over A because f is flat. The minimal primes of the one-dimensional ring $\tilde{B}$ correspond to the components X_j passing through D ; for such a component the quotient $B_j = \tilde{B}/\mathfrak{p}_j = \mathcal{O}_{X_j,D}$ is the discrete-valuation-style local ring of the reduced component X_j at D , in which $\operatorname{ord}_D(f^\# r|_{X_j}) = \operatorname{length}_{B_j}(B_j/(f^\# r)B_j)$ is computed. When X is irreducible along D there is a single such j , but in general several components may meet along D .

The coefficient of $[D]$ in $f^*[\operatorname{div}(r)]$ is $\operatorname{ord}_V(r) \cdot e_D$, where

$$e_D = \operatorname{length}_{\mathcal{O}_{f^{-1}(V),D}}(\mathcal{O}_{f^{-1}(V),D}) = \operatorname{length}_{\tilde{B}}(\tilde{B}/\mathfrak{m}_A \tilde{B})$$

is the multiplicity of D in the pullback cycle $f^*[V]$, the second equality holding because $\mathcal{O}_{f^{-1}(V),D} = \tilde{B} \otimes_A A/\mathfrak{m}_A = \tilde{B}/\mathfrak{m}_A \tilde{B}$. The coefficient of $[D]$ in $\sum_j m_j [\operatorname{div}(f^\# r|_{X_j})]$ is $\sum_{j: D \subseteq X_j} m_j \cdot \operatorname{ord}_D(f^\# r|_{X_j})$, the sum running over the components X_j that pass through D . That the two coefficients agree is two multiplicative facts chained together. First, flatness of $\tilde{B}$ over A makes length multiply through the base change: filtering A/rA by its $\operatorname{ord}_V(r)$ composition factors $A/\mathfrak{m}_A$ and applying the exact functor $- \otimes_A \tilde{B}$ gives

$$\operatorname{length}_{\tilde{B}}(\tilde{B}/r\tilde{B}) = \operatorname{length}_A(A/rA) \cdot \operatorname{length}_{\tilde{B}}(\tilde{B}/\mathfrak{m}_A \tilde{B}) = \operatorname{ord}_V(r) \cdot e_D$$

Second, order in the one-dimensional local ring $\tilde{B}$ distributes over its minimal primes with the generic multiplicities as weights: each minimal prime $\mathfrak{p}_j$ has residue ring $B_j = \mathcal{O}_{X_j, D}$ and localization $\tilde{B}_{\mathfrak{p}_j} = \mathcal{O}_{X, X_j}$ of length $m_j = \text{length}_{\mathcal{O}_{X, X_j}}(\mathcal{O}_{X, X_j})$, so the additivity of order over the minimal primes of a one-dimensional local ring ([2, App. A.3]) gives, for the nonzerodivisor $f^\#r$,

$$\text{length}_{\tilde{B}}(\tilde{B}/(f^\#r)\tilde{B}) = \sum_{j: D \subseteq X_j} m_j \cdot \text{length}_{B_j}(B_j/(f^\#r)B_j) = \sum_{j: D \subseteq X_j} m_j \cdot \text{ord}_D(f^\#r|_{X_j})$$

Since $f^\#r$ is the image of r in $\tilde{B}$, the left sides of the two displays coincide, whence $\text{ord}_V(r) \cdot e_D = \sum_{j: D \subseteq X_j} m_j \cdot \text{ord}_D(f^\#r|_{X_j})$: the two coefficients of $[D]$ match. When X is irreducible along D the sum has a single term and reduces to $e_D = m_j \cdot e_{D/V}$, where $e_{D/V} = \text{length}_{B_j}(B_j/\mathfrak{m}_A B_j)$ is the ramification-type multiplicity of D over V computed in the reduced component ring B_j ; the pullback multiplicity e_D over the full preimage scheme then exceeds it by exactly the factor m_j . Summing over the divisors D and, on the right, over the components X_j reproduces both sides of (1.1) term by term. Hence $f^*[\text{div}(r)] = \sum_j m_j [\text{div}(f^\#r|_{X_j})]$ is a sum of divisors of rational functions, so $f^*[\text{div}(r)] \sim 0$.

Descent to Chow groups follows: $f^*(\text{Rat}_k(Y)) \subseteq \text{Rat}_{k+n}(X)$, so f^* induces $A_k(Y) \rightarrow A_{k+n}(X)$. Functoriality on classes is inherited from proposition 1.3.5(3),(4) on cycles, giving a contravariant functor as claimed, completing the proof.

The proposition completes the package: flat pullback is a well-defined operation on Chow groups, contravariant and dimension-raising, dual to the covariant dimension-preserving pushforward of section 1.2. The pair (f_*, f^*) makes A_* into a functor covariant for proper maps and contravariant for flat maps, with the compatibility of proposition 1.3.6 tying the two together across fiber squares. This is the structure the localization sequence and homotopy invariance of section 1.4 exploit, and it is why the affine-bundle projection $p: X \times \mathbb{A}^n \rightarrow X$, whose pullback was computed in example 1.3.4, will turn out to induce an isomorphism on Chow groups.

Example 1.3.8: Pullback to a Product and to an Open Set

The two flat morphisms of example 1.3.4 descend to Chow groups explicitly.

1. *Cylinder map.* For $p: X \times \mathbb{A}^1 \rightarrow X$ the induced map $p^*: A_k(X) \rightarrow A_{k+1}(X \times \mathbb{A}^1)$ sends the class of a subvariety V to the class of the cylinder $V \times \mathbb{A}^1$. On $X = \text{Spec } k$ this reads $A_0(\text{Spec } k) = \mathbb{Z} \rightarrow A_1(\mathbb{A}^1)$, sending the class of the point to $[\mathbb{A}^1]$, the fundamental class of the affine line, consistent with $p^*[Y] = [X]$ from proposition 1.3.5(2).
2. *Restriction to an open set.* For an open immersion $j: U \hookrightarrow Y$ the map $j^*: A_k(Y) \rightarrow A_k(U)$ restricts a class to U , forgetting components supported on the complement $Y \setminus U$. On $Y = \mathbb{P}^1$ with $U = \mathbb{A}^1$ this is the surjection $A_0(\mathbb{P}^1) = \mathbb{Z} \rightarrow A_0(\mathbb{A}^1) = 0$ from example 1.1.5: the class of a point on $\mathbb{P}^1$ restricts to a point of $\mathbb{A}^1$, which is rationally trivial. The kernel is generated by the class of the point at infinity, the part of $\mathbb{P}^1$ removed. This surjectivity with the removed piece generating the kernel is the prototype of the localization sequence.

Exercise 1.3.1

1. Let $f: X \rightarrow Y$ be flat of relative dimension 0 (for instance a finite flat morphism, or an open

- immersion). Show directly from definition 1.3.3 that f^* maps $Z_k(Y) \rightarrow Z_k(X)$ for every k , preserving dimension, and that $f^*[V] = \sum_i e_i [Z_i]$ where Z_i are the components of $f^{-1}(V)$ and e_i their multiplicities in the fiber over the generic point of V .
2. Let $\pi: \mathbb{A}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}^n$ be the tautological $\mathbb{G}_m$ -bundle. It is flat of relative dimension 1. Compute $\pi^*[\mathbb{P}^k]$ for a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$, identifying its class as a cycle on $\mathbb{A}^{n+1} \setminus \{0\}$.
 3. Give an example of a flat morphism f for which $f^*: A_*(Y) \rightarrow A_*(X)$ is not injective, and one for which it is not surjective. (Hint: the projection $\mathbb{P}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ and the open immersion $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$.)
 4. Let $q: X \times \mathbb{A}^m \times \mathbb{A}^n \rightarrow X \times \mathbb{A}^m$ and $p: X \times \mathbb{A}^m \rightarrow X$ be the projections. Verify the composition law proposition 1.3.5(3) for $p \circ (\text{id} \times -)$ directly on the class of a subvariety $V \subseteq X$, confirming both routes give $[V \times \mathbb{A}^m \times \mathbb{A}^n]$.
 5. Let $f: X \rightarrow Y$ be proper and $j: U \hookrightarrow Y$ an open immersion, with fiber square $X_U = f^{-1}(U) \rightarrow X$ over j . Using proposition 1.3.6, prove that $(f|_{X_U})_*(\alpha|_{X_U}) = (f_*\alpha)|_U$ in $A_*(U)$ for every $\alpha \in A_*(X)$, that is, proper pushforward commutes with restriction to an open subset.
 6. Let $E \rightarrow X$ be a vector bundle of rank r over a variety X of dimension d , with projection π flat of relative dimension r . Compute $\pi^*[X]$ and identify the top Chow group $A_{d+r}(E)$, first showing that E is irreducible.

1.4 The Fundamental Sequences

The two functorialities already in hand, proper pushforward (section 1.2) and flat pullback (section 1.3), combine into the two structural results that make Chow groups computable. The first is a long-exact-sequence surrogate: cutting a scheme into a closed piece and its open complement relates the three Chow groups by a right-exact sequence, the algebraic analogue of the excision property of a homology theory. The second is homotopy invariance in its algebraic form: an affine bundle over X , and in particular the trivial bundle $X \times \mathbb{A}^n$, has the same Chow groups as X , shifted in degree by the fibre dimension. Neither result computes a single Chow group in isolation, but together they reduce the computation of $A_*(X)$ for the spaces of the next section to bookkeeping: stratify X by locally closed pieces on which the answer is known, and assemble the pieces through the sequence.

Both proofs rest on a single principle. A cycle on X is a formal combination of subvarieties; a subvariety either meets the open set U or is contained in the closed complement Z , and rational equivalence is generated by divisors of rational functions, which live on subvarieties and so obey the same dichotomy. The localization sequence is the exact translation of this observation into a statement about the three groups, and affine-bundle invariance is its first substantial payoff, extracted by Noetherian induction.

Throughout, $Z \subseteq X$ is a closed subscheme with open complement $U = X \setminus Z$, and

$$i: Z \hookrightarrow X, \quad j: U \hookrightarrow X$$

denote the inclusions. The closed immersion i is proper, so it pushes cycles forward by definition 1.2.1: a subvariety $V \subseteq Z$ is also a subvariety of X , and $i_*[V] = [V]$ since $\deg(V/V) = 1$. The open immersion j is flat of relative dimension 0, so it pulls cycles back by definition 1.3.3: for a subvariety

$V \subseteq X$ the restriction is $j^*[V] = [V \cap U]$ when V meets U and $j^*[V] = 0$ when $V \subseteq Z$. These two operations are what the sequence is built from.

Theorem 1.4.1: Localization Sequence

Let X be an algebraic scheme, $Z \subseteq X$ a closed subscheme, and $U = X \setminus Z$ its open complement, with $i: Z \hookrightarrow X$ and $j: U \hookrightarrow X$ the inclusions. For every k the sequence

$$A_k(Z) \xrightarrow{i_*} A_k(X) \xrightarrow{j^*} A_k(U) \longrightarrow 0$$

is exact: j^* is surjective, and $\ker(j^*) = \text{im}(i_*)$.

Proof:

The map i_* is proper pushforward along the closed immersion (theorem 1.2.3) and j^* is flat pullback along the open immersion (proposition 1.3.7); both are defined on Chow groups, so the sequence is a sequence of homomorphisms. Exactness is checked at the two nonzero spots.

Step 1: surjectivity of j^ .* A subvariety $V' \subseteq U$ is a k -dimensional closed integral subscheme of the open set U . Its closure $V = \overline{V'}$ in X is a k -dimensional subvariety of X with $V \cap U = V'$, since closure adds only points lying in the closed set Z and does not raise the dimension. Thus $j^*[V] = [V']$, and every generator of $Z_k(U)$ is hit by a cycle on X . Hence $j^*: Z_k(X) \rightarrow Z_k(U)$ is already surjective at the level of cycles, and a fortiori $j^*: A_k(X) \rightarrow A_k(U)$ is surjective.

Step 2: the inclusion $\text{im}(i_) \subseteq \ker(j^*)$.* For a subvariety $V \subseteq Z$ the intersection $V \cap U$ is empty, so $j^*i_*[V] = j^*[V] = 0$. By linearity $j^* \circ i_* = 0$ on $Z_k(Z)$, hence on $A_k(Z)$, which gives $\text{im}(i_*) \subseteq \ker(j^*)$.

Step 3: the inclusion $\ker(j^) \subseteq \text{im}(i_*)$.* This is the substance of the theorem. Let $\alpha \in Z_k(X)$ be a cycle whose class in $A_k(X)$ lies in $\ker(j^*)$, so $j^*\alpha \sim 0$ on U ; the goal is to modify α by a cycle rationally equivalent to zero on X until it is supported on Z , that is, until it lies in the image of $i_*: Z_k(Z) \rightarrow Z_k(X)$.

Split the cycle by whether its components meet U . Group the subvarieties occurring in α as those contained in Z and those meeting U , and write

$$\alpha = \alpha_Z + \beta, \quad \alpha_Z \in Z_k(Z), \quad \beta = \sum_i n_i [V_i]$$

where each V_i meets U . Since $\alpha_Z = i_*\alpha_Z$ already lies in the image of i_* , it may be discarded, and it suffices to treat β . Applying j^* gives $j^*\alpha = j^*\beta = \sum_i n_i [V'_i]$ with $V'_i = V_i \cap U$ the distinct k -dimensional subvarieties of U , and by hypothesis this cycle is rationally equivalent to zero on U :

$$\sum_i n_i [V'_i] = \sum_j [\text{div}(r'_j)] \quad \text{in } Z_k(U)$$

for finitely many $(k+1)$ -dimensional subvarieties $W'_j \subseteq U$ and rational functions $r'_j \in k(W'_j)^*$.

The idea is now to lift each W'_j and r'_j from U to X and compare divisors. Let $W_j = \overline{W'_j} \subseteq X$ be the closure, a $(k+1)$ -dimensional subvariety of X with $W_j \cap U = W'_j$. Restriction of functions to the dense open $W'_j \subseteq W_j$ is an isomorphism of function fields $k(W_j) \xrightarrow{\sim} k(W'_j)$, since a variety and a dense open subset have the same function field; let $r_j \in k(W_j)^*$ be the function corresponding to r'_j . Form the divisor $[\text{div}(r_j)] \in Z_k(X)$. Its restriction to U is governed by

the flat-pullback compatibility of the divisor construction: for a codimension-one subvariety $V \subseteq W_j$ meeting U , the local ring $\mathcal{O}_{W_j, V} = \mathcal{O}_{W'_j, V \cap U}$ is unchanged by passing to the open set, so $\text{ord}_V(r_j) = \text{ord}_{V \cap U}(r'_j)$, while a codimension-one subvariety of W_j contained in Z contributes a component of $[\text{div}(r_j)]$ that lies in $Z_k(Z)$. Splitting the sum over these two kinds of V ,

$$[\text{div}(r_j)] = [\text{div}(r'_j)] + \gamma_j, \quad \gamma_j \in Z_k(Z) \quad (1.2)$$

where the first term on the right is read inside $Z_k(X)$ via the identification $V \leftrightarrow \overline{V}$ of subvarieties of U with their closures, and γ_j collects the codimension-one subvarieties of W_j that lie in Z .

Step 3, conclusion. Sum (1.2) over j and use $\sum_j [\text{div}(r'_j)] = \beta$ as cycles on X under the closure identification (both sides are the cycle $\sum_i n_i [V_i]$, since $V_i = \overline{V'_i}$). This yields

$$\sum_j [\text{div}(r_j)] = \beta + \sum_j \gamma_j$$

so that

$$\beta = \sum_j [\text{div}(r_j)] - \sum_j \gamma_j$$

The first sum is a $\mathbb{Z}$ -combination of divisors of rational functions on X , hence lies in $\text{Rat}_k(X)$; the second sum $\gamma := \sum_j \gamma_j$ lies in $Z_k(Z)$. Therefore, in $A_k(X)$,

$$[\beta] = -[\gamma] = i_*(-[\gamma]) \in \text{im}(i_*)$$

Adding back the discarded $\alpha_Z = i_*[\alpha_Z]$ gives $[\alpha] = i_*([\alpha_Z] - [\gamma])$, exhibiting the class of α in the image of i_* . This proves $\ker(j^*) \subseteq \text{im}(i_*)$, and with Steps 1 and 2 the sequence is exact, completing the proof.

The mechanism of the proof is worth isolating, because it recurs whenever Chow groups are computed by stratification. Surjectivity on the right is elementary: a subvariety of the open set extends to its closure, so no class on U is unreachable from X . The content is in the middle, and it is entirely a statement about lifting rational equivalences. A relation $j^*\alpha \sim 0$ on U is witnessed by finitely many rational functions on subvarieties of U ; each such function extends across Z to a rational function on a subvariety of X (function fields do not see the boundary), and the extended divisor differs from the original only in components supported on Z . The failure of the sequence to be exact on the left, which is left as it stands, measures exactly how much rational equivalence is created by the extra room X provides over Z : a cycle on Z can become rationally equivalent to zero in X without being so in Z , because X contains $(k+1)$ -dimensional subvarieties running out of Z into U . The affine line makes this concrete below.

1.4.2 Remark: No Left-Exactness The map $i_*: A_k(Z) \rightarrow A_k(X)$ need not be injective. Take $X = \mathbb{A}^1$, $Z = \{0\}$, and $k = 0$. Then $A_0(Z) = \mathbb{Z}$, generated by $[0]$, but $[0] \sim 0$ in $A_0(\mathbb{A}^1)$ by example 1.1.5, so $i_*[0] = 0$ and i_* is the zero map on a nonzero group. The kernel of i_* records cycles on Z that bound in X but not in Z , and there is no functorial description of it. This is why the localization sequence is only a three-term right-exact sequence, not a long exact sequence: Bloch's higher Chow groups supply the missing terms, but that theory is outside the scope of this book.

The localization sequence is stated for one closed subscheme, but its most frequent use is for a pair. Two closed subschemes X_1, X_2 of an ambient X can be compared through their union and intersection, and the resulting sequence is the Mayer-Vietoris right-exactness that was promised without proof in example 1.1.6. It is a companion to the localization sequence rather than a corollary of it: the derivation below runs directly at the level of cycles, using only that a subvariety of the union lies in one of the two pieces, so no appeal to theorem 1.4.1 is needed.

Corollary 1.4.3: Right-Exactness for Two Closed Subschemes

Let X_1 and X_2 be closed subschemes of an algebraic scheme, and set $X_1 \cup X_2$ and $X_1 \cap X_2$ (with reduced or any fixed scheme structure; Chow groups see only the reduced structure by example 1.1.6). For every k the sequence

$$A_k(X_1 \cap X_2) \xrightarrow{(\iota_{1*}, -\iota_{2*})} A_k(X_1) \oplus A_k(X_2) \xrightarrow{j_{1*} + j_{2*}} A_k(X_1 \cup X_2) \longrightarrow 0$$

is exact, where $\iota_m: X_1 \cap X_2 \hookrightarrow X_m$ and $j_m: X_m \hookrightarrow X_1 \cup X_2$ are the inclusions. There is no exactness at $A_k(X_1 \cap X_2)$ in general.

Proof:

Write $Y = X_1 \cup X_2$. The argument is cleanest at the level of cycles, using the one structural fact that an irreducible closed subset of $Y = X_1 \cup X_2$ is contained in X_1 or in X_2 . Consequently a subvariety $V \subseteq Y$ lies in X_1 or in X_2 (or in both, in which case $V \subseteq X_1 \cap X_2$), and because $Z_k(Y)$ is free on subvarieties, a cycle expressible both by subvarieties of X_1 and by subvarieties of X_2 has all its components in $X_1 \cap X_2$.

Step 1: surjectivity of $j_{1} + j_{2*}$.* Every subvariety of Y lies in X_1 or X_2 , so at the level of cycles $Z_k(Y) = j_{1*}Z_k(X_1) + j_{2*}Z_k(X_2)$; the map $Z_k(X_1) \oplus Z_k(X_2) \rightarrow Z_k(Y)$, $(a_1, a_2) \mapsto a_1 + a_2$, is surjective. Passing to classes, $j_{1*} + j_{2*}: A_k(X_1) \oplus A_k(X_2) \rightarrow A_k(Y)$ is surjective.

Step 2: the composite is zero. For a subvariety $V \subseteq X_1 \cap X_2$ the two pushforwards $j_{1*}\iota_{1*}[V]$ and $j_{2*}\iota_{2*}[V]$ both equal $[V]$ in $Z_k(Y)$, since each inclusion sends $[V]$ to $[V]$. Hence $(j_{1*} + j_{2*}) \circ (\iota_{1*}, -\iota_{2*}) = j_{1*}\iota_{1*} - j_{2*}\iota_{2*} = 0$, giving $\text{im}(\iota_{1*}, -\iota_{2*}) \subseteq \ker(j_{1*} + j_{2*})$.

Step 3: exactness in the middle. Let $(\eta_1, \eta_2) \in A_k(X_1) \oplus A_k(X_2)$ satisfy $j_{1*}\eta_1 + j_{2*}\eta_2 = 0$ in $A_k(Y)$. Represent η_1 and η_2 by cycles $a_1 \in Z_k(X_1)$ and $a_2 \in Z_k(X_2)$. The hypothesis says $a_1 + a_2 \sim 0$ in $Z_k(Y)$, so there are finitely many $(k+1)$ -dimensional subvarieties $W_j \subseteq Y$ and $r_j \in k(W_j)^*$ with

$$a_1 + a_2 = \sum_j [\text{div}(r_j)] \quad \text{in } Z_k(Y)$$

Each W_j lies in X_1 or in X_2 , and the divisor $[\text{div}(r_j)]$ of a rational function on W_j is a cycle supported on W_j , hence on X_1 or on X_2 accordingly. Collect the terms:

$$b_1 = \sum_{W_j \subseteq X_1} [\text{div}(r_j)] \in \text{Rat}_k(X_1), \quad b_2 = \sum_{W_j \subseteq X_2, W_j \not\subseteq X_1} [\text{div}(r_j)] \in \text{Rat}_k(X_2)$$

where a W_j lying in both is assigned to the first group; then $\sum_j [\text{div}(r_j)] = b_1 + b_2$ and so $a_1 + a_2 = b_1 + b_2$ in $Z_k(Y)$. Rearranging,

$$c := a_1 - b_1 = b_2 - a_2$$

The left expression $a_1 - b_1$ is a cycle supported on X_1 , and the right expression $b_2 - a_2$ is a cycle supported on X_2 ; being the same cycle in the free group $Z_k(Y)$, it is supported on $X_1 \cap X_2$, that is $c \in Z_k(X_1 \cap X_2)$. Now compute classes. In $A_k(X_1)$, $b_1 \in \text{Rat}_k(X_1)$ is zero, so $\eta_1 = [a_1] = [a_1 - b_1] = \iota_{1*}[c]$; in $A_k(X_2)$, $b_2 \in \text{Rat}_k(X_2)$ is zero, so $\eta_2 = [a_2] = [a_2 - b_2] = -\iota_{2*}[c]$. Therefore

$$(\eta_1, \eta_2) = (\iota_{1*}[c], -\iota_{2*}[c]) = (\iota_{1*}, -\iota_{2*})([c])$$

lies in the image of $(\iota_{1*}, -\iota_{2*})$. This gives $\ker(j_{1*} + j_{2*}) \subseteq \text{im}(\iota_{1*}, -\iota_{2*})$, and with Step 2 exactness in the middle.

No left-exactness. The map $(\iota_{1*}, -\iota_{2*})$ need not be injective. Take X_1 and X_2 two distinct lines through the origin in $\mathbb{A}^2$, so $X_1 \cap X_2 = \{0\}$, in degree $k = 0$: then $A_0(X_1 \cap X_2) = \mathbb{Z}$ is generated by $[0]$, while $\iota_{1*}[0] = [0] \sim 0$ in $A_0(X_1) = A_0(\mathbb{A}^1)$ and likewise $\iota_{2*}[0] = 0$ in $A_0(X_2)$, so $(\iota_{1*}, -\iota_{2*})[0] = 0$ on a nonzero group. Thus there is no exactness at $A_k(X_1 \cap X_2)$, as we sought to show.

The corollary is the exact sequence recorded, without proof, as item (4) of example 1.1.6; the present derivation discharges that promissory note. Its shape is the Chow-group form of the Mayer-Vietoris sequence familiar from topology, truncated on the left for the same reason the localization sequence is: the intersection $X_1 \cap X_2$ can carry cycles that become rationally equivalent to zero once they are free to move in the larger $X_1 \cup X_2$. Where it is used most is in dévissage arguments, where a scheme is written as a union of simpler closed pieces and its Chow groups are assembled from those of the pieces and their overlaps.

A single application of the localization sequence recovers, from scratch, the computation of $A_0(\mathbb{A}^1)$ recorded in example 1.1.5, this time without touching a rational function by hand. The point is that $\mathbb{A}^1$ is $\mathbb{P}^1$ with a point removed, and the localization sequence turns the known $A_0(\mathbb{P}^1) = \mathbb{Z}$ into the answer for the open complement.

Example 1.4.4: Recomputing $A_0(\mathbb{A}^1)$ by Localization

Take $X = \mathbb{P}^1$, $Z = \{\infty\}$ the point at infinity, and $U = \mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. The localization sequence in degree 0 reads

$$A_0(\{\infty\}) \xrightarrow{i_*} A_0(\mathbb{P}^1) \xrightarrow{j^*} A_0(\mathbb{A}^1) \longrightarrow 0$$

The left group is $A_0(\{\infty\}) = \mathbb{Z}$, generated by the class of the single point ∞ , and $i_*[\infty] = [\infty] \in A_0(\mathbb{P}^1)$. By example 1.1.5 the group $A_0(\mathbb{P}^1) \cong \mathbb{Z}$ is generated by the class h of any point, and $[\infty] = h$ is a generator, so i_* is surjective. By exactness $\text{im}(i_*) = \ker(j^*)$, and since i_* is onto, $\ker(j^*) = A_0(\mathbb{P}^1)$; the quotient by the whole group is trivial:

$$A_0(\mathbb{A}^1) = A_0(\mathbb{P}^1) / \ker(j^*) \cong \text{coker}(i_*) = \mathbb{Z} / \mathbb{Z} = 0$$

Thus $A_0(\mathbb{A}^1) = 0$, in agreement with the direct computation. The single generator of $A_0(\mathbb{P}^1)$ is killed precisely because the point removed to form $\mathbb{A}^1$ was rationally equivalent to every remaining point; removing it collapses the whole group.

With the localization sequence available, the second structural result of the section follows by Noetherian induction. Homotopy invariance in algebraic geometry takes the form of affine-bundle invariance: pulling a cycle back along the projection of an affine bundle is an isomorphism on Chow groups. The motivating case is the trivial bundle $X \times \mathbb{A}^n \rightarrow X$, which asserts that appending an affine factor does not change the Chow group beyond the expected degree shift, the algebraic

counterpart of the fact that $\mathbb{A}^n$ is contractible.

The result is stated for a general affine bundle, of which the trivial bundle is the leading case.

Definition 1.4.5: Affine Bundle of Relative Dimension n

An *affine bundle of relative dimension n* over X is a scheme $p: E \rightarrow X$ that is locally on X a product with affine space: there is an open cover $\{U_\lambda\}$ of X and isomorphisms $p^{-1}(U_\lambda) \cong U_\lambda \times \mathbb{A}^n$ over U_λ .

A vector bundle of rank n , with its transition maps in GL_n , is such a bundle, as is any torsor under a vector bundle; the trivial example is $X \times \mathbb{A}^n$ itself. In every case p is flat of relative dimension n , so flat pullback $p^*: A_k(X) \rightarrow A_{k+n}(E)$ is defined by definition 1.3.3.

Theorem 1.4.6: Affine-Bundle Invariance

Let $p: E \rightarrow X$ be an affine bundle of relative dimension n over an algebraic scheme X . Then the flat pullback

$$p^*: A_k(X) \xrightarrow{\sim} A_{k+n}(E)$$

is an isomorphism for every k . In particular, for the trivial bundle, $A_k(X) \xrightarrow{\sim} A_{k+n}(X \times \mathbb{A}^n)$.

Proof:

The proof runs the argument directly at relative dimension n ; the fibre dimension enters only through the trivial bundle, where a global tower is available. Note that a global tower of rank-one affine bundles is *not* claimed for a general bundle: the local identification $U \times \mathbb{A}^n = (U \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$ does not glue to a factorization $E \rightarrow X$ over X , since a general affine (or vector) bundle carries no global flag of subbundles. The trivial bundle $X \times \mathbb{A}^n$, being a literal product, *is* such a tower, $X \times \mathbb{A}^n = (X \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$; the construction below invokes the tower only there. Fix n and prove that $p^*: A_k(X) \rightarrow A_{k+n}(E)$ is an isomorphism for all k , by Noetherian induction on the closed subschemes of X . For a closed subscheme $W \subseteq X$ write $E_W = p^{-1}(W)$ and $p_W: E_W \rightarrow W$ for the restricted bundle, again an affine bundle of relative dimension n .

Step 1: reductions. The scheme X is Noetherian, so its closed subschemes satisfy the descending chain condition and it suffices to prove: if p_W^* is an isomorphism for every proper closed subscheme $W \subsetneq X$, then p^* is an isomorphism for X . Chow groups depend only on the reduced structure (example 1.1.6), so take X reduced. If X is reducible, write $X = X_1 \cup X_2$ with X_1 an irreducible component and X_2 the union of the remaining components, both proper closed subschemes; the pullbacks $p_{X_1}^*, p_{X_2}^*, p_{X_1 \cap X_2}^*$ are isomorphisms by hypothesis. The Mayer-Vietoris sequences of corollary 1.4.3 for $X = X_1 \cup X_2$ and for $E = E_{X_1} \cup E_{X_2}$ (whose intersection is $E_{X_1 \cap X_2}$) form a commutative ladder linked by the vertical pullbacks; a diagram chase using the three outer isomorphisms shows p^* is an isomorphism for X . So assume X is a variety (reduced and irreducible), and let $\eta = \text{Spec } k(X)$ be its generic point, with generic fibre $E_\eta \cong \mathbb{A}_{k(X)}^n$.

Step 2: surjectivity of p^ .* It suffices to realize the class of a single $(k+n)$ -dimensional subvariety $V \subseteq E$ as a pullback modulo the image of $A_{k+n}(E_Z)$ for some proper closed $Z \subsetneq X$. Let $W = \overline{p(V)} \subseteq X$. If $W \subsetneq X$ is proper, then $V \subseteq E_W$ is supported over $Z = W$, and $[V] \in \text{im}(A_{k+n}(E_W) \rightarrow A_{k+n}(E))$; by the inductive hypothesis p_W^* is onto, and compatibility of p^* with the closed immersion (the fibre square for p over $W \hookrightarrow X$ gives $p^*i_* = i_*p_W^*$ by proposition 1.3.6) writes $[V]$ as a pullback from X . If instead $W = X$, so V dominates X ,

restrict to the generic fibre. There $V_\eta \subseteq \mathbb{A}_{k(X)}^n$ is a closed subvariety, and its class is controlled by the trivial-bundle case of the theorem over the field $K = k(X)$: the projection $\mathbb{A}_K^n \rightarrow \operatorname{Spec} K$ has surjective flat pullback. This last fact is elementary and needs no base induction, being the trivial bundle over a point: writing the tower $\mathbb{A}_K^n = (\mathbb{A}_K^{n-1}) \times \mathbb{A}^1$ and inducting on n , the base case $A_*(\mathbb{A}_K^1)$ is generated over $A_*(\operatorname{Spec} K)$ by the pullback (every closed point of $\mathbb{A}_K^1$ is the divisor of its minimal polynomial, so $A_0(\mathbb{A}_K^1) = 0$, exactly as in example 1.1.5 with K in place of k), and the tower propagates this up the coordinates. Thus $[V_\eta]$ is rationally equivalent on $\mathbb{A}_{k(X)}^n$ to a pullback $p_\eta^*(a_\eta)$ of a class a_η on $\operatorname{Spec} k(X)$. Spreading out the finitely many rational functions that realize this equivalence, together with a_η , to an open $U \subseteq X$ shows $[V]|_{E_U} = p_U^*(\alpha_U)$ for a class $\alpha_U \in A_k(U)$ and hence $[V] - p^*(\alpha_V)$ (with $\alpha_V \in A_k(X)$ lifting α_U via surjectivity of j^* , theorem 1.4.1) restricts to 0 on E_U , so its support lies over the proper closed $Z = X \setminus U$. In every case

$$[V] = p^*(\alpha_V) + \gamma_V \quad \text{in } A_{k+n}(E), \quad \gamma_V \in \operatorname{im} (A_{k+n}(E_Z) \rightarrow A_{k+n}(E))$$

for some proper closed $Z \subsetneq X$ and $\alpha_V \in A_k(X)$. By the inductive hypothesis p_Z^* is onto $A_*(E_Z)$, so $\gamma_V = i_* p_Z^*(\delta_V) = p^*(i_* \delta_V)$ for a class δ_V on Z , again by proposition 1.3.6. Hence $[V] = p^*(\alpha_V + i_* \delta_V)$ and p^* is surjective.

Surjectivity of p^* is now in hand for every X ; injectivity is the remaining point, and it is not a formal consequence of the localization ladder, since the two rows are only right-exact and provide no left term to chase against. It is proved by exhibiting a left inverse to p^* for the trivial bundle and reducing the general bundle to that case.

Step 3: a left inverse for the trivial bundle. The construction is carried out for a single affine coordinate over an arbitrary base Y ; the trivial bundle $X \times \mathbb{A}^n$ is a tower of n such single coordinates, and the rank- n left inverse is the composite of the n single-coordinate left inverses (Step 3d). Take $E = Y \times \mathbb{A}^1$, with coordinate t on the second factor and projection $p: Y \times \mathbb{A}^1 \rightarrow Y$, and define a homomorphism $\sigma: Z_{k+1}(Y \times \mathbb{A}^1) \rightarrow Z_k(Y)$ on a generator $[V]$, $V \subseteq Y \times \mathbb{A}^1$ a $(k+1)$ -dimensional subvariety, by

$$\sigma[V] = \begin{cases} p_*[\operatorname{div}_V(t|_V)] & t|_V \in k(V)^* \text{ nonconstant} \\ 0 & t|_V \text{ constant} \end{cases}$$

extended additively; here “ $t|_V$ constant” is read to include the degenerate case $t|_V \equiv 0$ (when $V \subseteq Y \times \{0\}$), which likewise contributes 0, since then $t|_V \notin k(V)^*$ and no divisor $\operatorname{div}_V(t|_V)$ is formed. When $t|_V$ is nonconstant, $[\operatorname{div}_V(t|_V)]$ is a k -cycle on V ; the pushforward p_* of this particular cycle is well-defined even though $p|_V: V \rightarrow p(V) \subseteq Y$ need not be proper. The point is that t is a regular function on $Y \times \mathbb{A}^1$, so its restriction $t|_V$ is regular on V and $\operatorname{div}_V(t|_V)$ is effective, with support contained in the zero locus $V \cap (Y \times \{0\}) \subseteq Y \times \{0\}$. On $Y \times \{0\}$ the projection p is the closed immersion identifying $Y \times \{0\}$ with Y , in particular proper, so $p_*[\operatorname{div}_V(t|_V)]$ is a well-defined k -cycle on Y by definition 1.2.1, computed component by component over the locus where $t|_V$ vanishes. Geometrically $\sigma[V]$ is the fibre of V over $t = 0$, projected to Y : the specialization of V to the zero section.

Step 3a: $\sigma \circ p^ = \operatorname{id}$.* For a k -dimensional subvariety $W \subseteq Y$ the pullback is $p^*[W] = [W \times \mathbb{A}^1]$, and t restricts to the coordinate on $W \times \mathbb{A}^1$, which is nonconstant with $[\operatorname{div}_{W \times \mathbb{A}^1}(t)] = [W \times \{0\}]$ (the coordinate t has its single zero along $W \times \{0\}$ and no pole on $W \times \mathbb{A}^1$). Pushing forward, $p_*[W \times \{0\}] = [W]$, so $\sigma(p^*[W]) = [W]$. By additivity $\sigma \circ p^* = \operatorname{id}$ on $Z_k(Y)$, hence on $A_k(Y)$ once σ is known to descend.

Step 3b: σ respects rational equivalence. A generator of $\text{Rat}_{k+1}(Y \times \mathbb{A}^1)$ is $[\text{div}_S(g)]$ for a $(k+2)$ -dimensional subvariety $S \subseteq Y \times \mathbb{A}^1$ and $g \in k(S)^*$, and unwinding the definitions gives $\sigma[\text{div}_S(g)] = p_*(\sum_V \text{ord}_V(g) [\text{div}_V(t|_V)])$, the outer sum over the codimension-one subvarieties $V \subseteq S$ on which $t|_V$ is nonconstant (a V with $t|_V$ constant, including $V \subseteq Y \times \{0\}$ where $t|_V \equiv 0$, drops out, contributing 0). This iterated divisor is symmetric in the two functions g and $t|_S$ modulo a principal divisor on S :

$$\sum_V \text{ord}_V(g) [\text{div}_V(t)] - \sum_V \text{ord}_V(t) [\text{div}_V(g)] \in \text{Rat}_k(S) \quad (1.3)$$

The relation (1.3) is the commutativity of two divisor operations on the $(k+2)$ -dimensional S , equivalently the well-definedness of intersecting a cycle with a Cartier divisor; its proof requires the divisor-theoretic machinery of the next chapter, and is supplied there by theorem 2.3.2 (its surface case is exactly (1.3)), so no tool used here depends on it circularly. Granting (1.3), $\sigma[\text{div}_S(g)]$ differs from $p_*(\sum_V \text{ord}_V(t) [\text{div}_V(g)])$ by the pushforward of a principal divisor, and $\sum_V \text{ord}_V(t) [\text{div}_V(g)]$ is itself a $\mathbb{Z}$ -combination of principal divisors on the subvarieties V , so its pushforward lies in $\text{Rat}_k(Y)$ by theorem 1.2.3. Either way $\sigma[\text{div}_S(g)] \in \text{Rat}_k(Y)$, so σ descends to

$$\sigma: A_{k+1}(Y \times \mathbb{A}^1) \rightarrow A_k(Y), \quad \sigma \circ p^* = \text{id}$$

In particular p^* is injective for the trivial rank-one bundle over any base Y , and being surjective by Step 2, it is an isomorphism there. The left inverse σ commutes with proper pushforward along closed immersions and with flat restriction to opens, since each of p_* and $\text{div}(t)$ does; this compatibility is what carries the trivial-bundle result across the localization sequence in Step 3c.

Step 3d: the trivial rank- n bundle. For $E = X \times \mathbb{A}^n$ with coordinates $t_1, \dots, t_n$, the tower

$$X \times \mathbb{A}^n \xrightarrow{q_n} X \times \mathbb{A}^{n-1} \xrightarrow{q_{n-1}} \dots \xrightarrow{q_1} X$$

factors $p = q_1 \circ \dots \circ q_n$ into single-coordinate trivial rank-one bundles $q_m: (X \times \mathbb{A}^{m-1}) \times \mathbb{A}^1 \rightarrow X \times \mathbb{A}^{m-1}$, so $p^* = q_n^* \circ \dots \circ q_1^*$ by functoriality of flat pullback (proposition 1.3.5). Applying Step 3b with $Y = X \times \mathbb{A}^{m-1}$ gives a left inverse σ_m to each q_m^* ; the composite $\sigma := \sigma_1 \circ \dots \circ \sigma_n: A_{k+n}(X \times \mathbb{A}^n) \rightarrow A_k(X)$ satisfies $\sigma \circ p^* = \text{id}$, so p^* is injective, and with Step 2 it is an isomorphism for the trivial bundle. The compatibility of each σ_m with proper pushforward along closed immersions and with flat restriction to opens passes to the composite, so σ has the same compatibility used in the next step.

Step 3c: injectivity for a general bundle. Let $p: E \rightarrow X$ be an affine bundle of relative dimension n over the variety X and $\alpha \in A_k(X)$ with $p^*\alpha = 0$. Choose a dense open $U \subseteq X$ over which E trivializes, $E_U \cong U \times \mathbb{A}^n$, and set $Z = X \setminus U$, a proper closed subscheme. Flat base change (proposition 1.3.5) gives $p_U^*(j^*\alpha) = j^*(p^*\alpha) = 0$, and p_U^* is injective by Step 3d applied to the trivial bundle over U ; hence $j^*\alpha = 0$. By the localization sequence for $Z \subseteq X$ (theorem 1.4.1), $\alpha = i_*\beta$ for some $\beta \in A_k(Z)$. Pushing β into E through the fibre square over $Z \hookrightarrow X$ (proposition 1.3.6) gives $p^*\alpha = p^*i_*\beta = i_*(p_Z^*\beta) = 0$, so $p_Z^*\beta$ lies in the kernel of $i_*: A_{k+n}(E_Z) \rightarrow A_{k+n}(E)$. Over the total space E the left inverse of Step 3d, restricted to the trivializing locus and extended by $\sigma_Z := (p_Z^*)^{-1}$ over E_Z (the inductive isomorphism), assembles into a retraction $r_E: A_{k+n}(E) \rightarrow A_k(X)$ of p^* ; the compatibility noted at the end of Step 3d makes the square with i_* commute, so that $r_E \circ i_* = i_* \circ \sigma_Z$ on $A_{k+n}(E_Z)$. Evaluating at $p_Z^*\beta$,

$$i_*\beta = i_*(\sigma_Z p_Z^*\beta) = r_E(i_* p_Z^*\beta) = r_E(0) = 0$$

so $\alpha = i_*\beta = 0$. Hence p^* is injective.

Steps 2 and 3 show p^* is an isomorphism for every X and every relative dimension n , completing the Noetherian induction; the tower entered only for the trivial bundle (Step 3d), where it is an actual global factorization, so no global tower was claimed for a general bundle, completing the proof.

The theorem is the algebraic-geometric shadow of a topological triviality: the projection $X \times \mathbb{A}^n \rightarrow X$ is a homotopy equivalence when $\mathbb{A}^n = \mathbb{C}^n$ is given the classical topology, so it induces isomorphisms on singular homology, and Chow groups behave the same way. The degree shift by n is the algebraic bookkeeping for the real dimension shift $2n$ on the topological side, since a k -cycle (complex dimension k) has real dimension $2k$ and A_{k+n} matches $H_{2(k+n)} = H_{2k+2n}$. The proof uses none of this: surjectivity of p^* needs only that every zero-cycle on an affine line over a field is rationally trivial ($A_0(\mathbb{A}_K^1) = 0$), propagated up the base by localization, while injectivity needs a left inverse, the specialization to the zero section. The one step left open, the divisor symmetry (1.3), is exactly the input that makes intersecting with a Cartier divisor well defined on rational equivalence, the payload of the next chapter; the isomorphism here is the first place that well-definedness is needed, and the forward reference is deliberate rather than a gap. It is discharged in theorem 2.3.2. The reason the argument needs Noetherian induction rather than a direct computation is the same reason the localization sequence is not left-exact: one cannot control what rational equivalences a cycle acquires on E without dismantling the base into pieces small enough that the bundle trivializes and the fibrewise vanishing takes over.

Exercise 1.4.1

1. Let $Z \subseteq X$ be closed with complement U . Show directly from theorem 1.4.1 that $A_k(U) \cong A_k(X)/\text{im}(A_k(Z) \rightarrow A_k(X))$, and use this to prove that if $A_k(Z) = 0$ then $j^*: A_k(X) \rightarrow A_k(U)$ is an isomorphism.
2. Let $X = \mathbb{A}^2$ and let $Z = \{0\}$ be the origin, so $U = \mathbb{A}^2 \setminus \{0\}$ is the punctured plane. Using the localization sequence together with the values $A_k(\mathbb{A}^2)$ (namely $A_2(\mathbb{A}^2) = \mathbb{Z}$ and $A_k(\mathbb{A}^2) = 0$ for $k < 2$, which you may take from section 1.5), compute $A_k(U)$ for all k .
3. Let X be a variety of dimension n and $U \subseteq X$ a nonempty open subset. Show that $A_n(X) \rightarrow A_n(U)$ is an isomorphism, and that both are $\mathbb{Z}$, generated by the fundamental class.
4. Compute $A_*(\mathbb{P}^n \times \mathbb{A}^m)$ in terms of $A_*(\mathbb{P}^n)$, assuming the value $A_k(\mathbb{P}^n) = \mathbb{Z}$ for $0 \leq k \leq n$ from section 1.5. State the answer as a graded group.
5. Let $p: E \rightarrow X$ be a vector bundle of rank r over an algebraic scheme X , with zero section $s: X \hookrightarrow E$. Affine-bundle invariance makes $p^*: A_k(X) \rightarrow A_{k+r}(E)$ an isomorphism. Explain why one cannot conclude $A_*(E) \cong A_*(X)$ by inverting p^* with a proper pushforward p_* , identifying the precise obstruction, and describe the degree by which p^* and the closed-immersion pushforward s_* each shift a cycle class.
6. Take X_1 and X_2 to be two distinct lines through the origin in $\mathbb{A}^2$, meeting only at $\{0\}$. Write out the sequence of corollary 1.4.3 in degree 0 and verify its exactness by hand, given that A_0 of a line is 0 and A_0 of a point is $\mathbb{Z}$.

1.5 First Computations

The machinery of the preceding sections is now assembled: cycles, rational equivalence, proper pushforward, flat pullback, and the two fundamental sequences. This section puts it to work on the spaces one meets first. We compute the Chow groups of affine space, of projective space, and of a smooth projective curve, and record the shape the answer takes for a projective bundle. These are the building blocks from which nearly every later computation is assembled, and they already display the two extremes of the theory: affine space, whose cycles almost all sweep away to nothing, and projective space, whose cycles are pinned down by a single integer in each dimension.

The pattern to watch is how the localization sequence and affine-bundle invariance from section 1.4 do the actual work. Affine space is an affine bundle over a point, so its Chow groups collapse; projective space is built from affine space by adding a hyperplane at infinity, so an induction along the localization sequence peels off one $\mathbb{Z}$ per dimension. Every computation below is an instance of one of these two moves.

Affine space is the first computation, and $\mathbb{A}^n$ retracts, in the sense of rational equivalence, all the way down to a point: every subvariety of dimension less than n can be swept off to zero, and only the fundamental class in top dimension survives. This is the higher-dimensional form of the observation $A_0(\mathbb{A}^1) = 0$ from example 1.1.5, and it follows in one line from affine-bundle invariance once one notices that $\mathbb{A}^n$ is an affine bundle over $\text{Spec } \bar{k}$.

Proposition 1.5.1: Chow Groups of Affine Space

For every $n \geq 0$ the Chow groups of affine space are

$$A_k(\mathbb{A}^n) = \begin{cases} \mathbb{Z}[\mathbb{A}^n] & k = n \\ 0 & k \neq n \end{cases}$$

the group $A_n(\mathbb{A}^n)$ being free of rank one on the fundamental class $[\mathbb{A}^n]$.

Proof:

The structure morphism $\pi: \mathbb{A}^n \rightarrow \text{Spec } \bar{k}$ exhibits $\mathbb{A}^n$ as the trivial affine bundle of relative dimension n over a point. Affine-bundle invariance (theorem 1.4.6) states that flat pullback

$$\pi^*: A_k(\text{Spec } \bar{k}) \xrightarrow{\sim} A_{k+n}(\mathbb{A}^n)$$

is an isomorphism for every k . The point $\text{Spec } \bar{k}$ has a single subvariety, itself, of dimension 0, so $A_0(\text{Spec } \bar{k}) = Z_0(\text{Spec } \bar{k}) = \mathbb{Z}[\text{Spec } \bar{k}]$ and $A_k(\text{Spec } \bar{k}) = 0$ for $k \neq 0$. Transporting these through π^* gives $A_{k+n}(\mathbb{A}^n) = 0$ for $k \neq 0$, that is $A_j(\mathbb{A}^n) = 0$ for $j \neq n$, and $A_n(\mathbb{A}^n) = \pi^*(\mathbb{Z}[\text{Spec } \bar{k}])$. Flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space (definition 1.3.3), so $\pi^*[\text{Spec } \bar{k}] = [\mathbb{A}^n]$, and $A_n(\mathbb{A}^n) = \mathbb{Z}[\mathbb{A}^n]$ is free of rank one, completing the proof.

Affine space thus has no cycles in low dimension worth remembering, a fact that reads two ways. From below it says that any k -dimensional subvariety $V \subseteq \mathbb{A}^n$ with $k < n$ bounds: there is a family of cycles, parametrized by the line, sweeping $[V]$ to zero. From above it isolates the one invariant that does survive, the coefficient of the fundamental class, which for a top cycle records the multiplicity

with which $\mathbb{A}^n$ appears in it. The vanishing in low dimension is what makes affine space so poor a stage for enumerative geometry, and simultaneously what makes it the natural chart in which to localize a computation on a larger space. This second role is exactly how projective space is handled next.

The contrast to keep in mind is the affine line. A single point of $\mathbb{A}^1$ is a 0-cycle in dimension $0 < 1$, so proposition 1.5.1 predicts $A_0(\mathbb{A}^1) = 0$, recovering the computation of example 1.1.5: the coordinate function $t - p$ has $[\text{div}(t - p)] = [p]$, presenting the point as a principal divisor and hence as zero in the Chow group. The proposition is the assertion that this collapse happens in every dimension below the top, uniformly and for the same reason.

Projective space is the opposite. Nothing sweeps to zero for free, because a rational function on a subvariety of $\mathbb{P}^n$ has as many zeros as poles, and the surviving invariant in each dimension is an integer. The computation is an induction that peels $\mathbb{P}^n$ apart along a hyperplane: the closed subscheme $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ at infinity, with open complement the affine chart $\mathbb{A}^n$. Affine space contributes nothing below the top by proposition 1.5.1, so the localization sequence forces every class of $\mathbb{P}^n$ in dimension $k < n$ to come from $\mathbb{P}^{n-1}$, and the induction runs.

The motivating principle is that projective space is *cellular*: it is stratified by the chain

$$\mathbb{P}^0 \subset \mathbb{P}^1 \subset \dots \subset \mathbb{P}^{n-1} \subset \mathbb{P}^n$$

of linear subspaces, each obtained from the previous by adding an affine cell $\mathbb{A}^k = \mathbb{P}^k \setminus \mathbb{P}^{k-1}$. The Chow groups turn out to be free on the classes of these cells, one generator in each dimension. This is the algebraic shadow of the cell decomposition that computes the singular homology of $\mathbb{CP}^n$ in [5], where the same chain of linear subspaces yields $H_{2k}(\mathbb{CP}^n; \mathbb{Z}) = \mathbb{Z}$ and $H_{\text{odd}} = 0$.

Theorem 1.5.2: Chow Groups of Projective Space

For every $n \geq 0$ and every $0 \leq k \leq n$, the Chow group $A_k(\mathbb{P}^n)$ is free abelian of rank one, generated by the class $[\mathbb{P}^k]$ of any linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ of dimension k :

$$A_k(\mathbb{P}^n) = \mathbb{Z} [\mathbb{P}^k], \quad 0 \leq k \leq n$$

and $A_k(\mathbb{P}^n) = 0$ for $k > n$. Any two linear subspaces of the same dimension are rationally equivalent, so the generator does not depend on the choice.

Proof:

The argument is an induction on n , run through projection from a point, which realizes the complement of a point in $\mathbb{P}^n$ as a line bundle over $\mathbb{P}^{n-1}$. Write h_k for the class $[\mathbb{P}^k]$ of a linear k -plane; the claim is that $A_k(\mathbb{P}^n) = \mathbb{Z} h_k$ is free of rank one for $0 \leq k \leq n$, and 0 otherwise. Groups $A_k(\mathbb{P}^n)$ with $k > n = \dim \mathbb{P}^n$ vanish because $\mathbb{P}^n$ has no subvariety of that dimension, so only $0 \leq k \leq n$ is at issue.

Step 1: base cases $n = 0$ and $k = 0$. For $n = 0$ the space $\mathbb{P}^0 = \text{Spec } \bar{k}$ is a point, with $A_0(\mathbb{P}^0) = \mathbb{Z} [\mathbb{P}^0]$ and $A_k(\mathbb{P}^0) = 0$ for $k \neq 0$ by proposition 1.5.1, which is the claim. For general n and $k = 0$, every closed point of $\mathbb{P}^n$ is rationally equivalent to every other: two points p, q both lie on a line $\ell \cong \mathbb{P}^1 \subseteq \mathbb{P}^n$, and on ℓ they are rationally equivalent by example 1.1.5, so $[p] \sim [q]$ in $\mathbb{P}^n$. Thus $A_0(\mathbb{P}^n)$ is cyclic, generated by $h_0 = [p]$. Since $\mathbb{P}^n$ is complete, the degree map $\text{deg}: A_0(\mathbb{P}^n) \rightarrow \mathbb{Z}$ of definition 1.2.4 is well defined and sends $h_0 = [p] \mapsto [k(p) : \bar{k}] = 1$, so h_0 has infinite order and $A_0(\mathbb{P}^n) = \mathbb{Z} h_0$ is free of rank one.

Step 2: projection from a point as a line bundle. Fix $n \geq 1$ and a closed point $P \in \mathbb{P}^n$. Choosing coordinates with $P = [0 : \cdots : 0 : 1]$, linear projection away from P is a morphism

$$q: \mathbb{P}^n \setminus \{P\} \rightarrow \mathbb{P}^{n-1}, \quad [x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$$

whose fiber over a point $[a] \in \mathbb{P}^{n-1}$ is the line through P and $[a]$ with P removed, an affine line $\mathbb{A}^1$. This exhibits $\mathbb{P}^n \setminus \{P\}$ as a line bundle over $\mathbb{P}^{n-1}$, the total space of $\mathcal{O}_{\mathbb{P}^{n-1}}(1)$; in particular it is an affine bundle of relative dimension 1. Only the last clause is used below. Affine-bundle invariance (theorem 1.4.6) makes flat pullback

$$q^*: A_{k-1}(\mathbb{P}^{n-1}) \xrightarrow{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \quad (1.4)$$

an isomorphism for every k .

Step 3: removing the point via localization. The point $\{P\}$ is a closed subscheme of $\mathbb{P}^n$ with open complement $\mathbb{P}^n \setminus \{P\}$. The localization sequence (theorem 1.4.1) for this pair reads

$$A_k(\{P\}) \xrightarrow{i_*} A_k(\mathbb{P}^n) \xrightarrow{j^*} A_k(\mathbb{P}^n \setminus \{P\}) \rightarrow 0 \quad (1.5)$$

exact at $A_k(\mathbb{P}^n)$ and $A_k(\mathbb{P}^n \setminus \{P\})$. The point contributes $A_k(\{P\}) = 0$ for every $k \geq 1$ and $A_0(\{P\}) = \mathbb{Z}[P]$. Hence for $k \geq 1$ the left term vanishes and j^* is an isomorphism, which combined with (1.4) gives, for $k \geq 1$,

$$A_k(\mathbb{P}^n) \xrightarrow{j^*} A_k(\mathbb{P}^n \setminus \{P\}) \xleftarrow{q^*} A_{k-1}(\mathbb{P}^{n-1}) \quad (1.6)$$

so $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$ as abelian groups.

Step 4: the induction. Fix k with $1 \leq k \leq n$. Iterating (1.6) k times descends both indices by k :

$$A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1}) \cong \cdots \cong A_0(\mathbb{P}^{n-k})$$

and $A_0(\mathbb{P}^{n-k}) = \mathbb{Z}$ is free of rank one by Step 1. Together with the case $k = 0$ of Step 1, this proves $A_k(\mathbb{P}^n) \cong \mathbb{Z}$ is free of rank one for every $0 \leq k \leq n$. For $k > n$ the group vanishes as noted at the outset.

Step 5: the generator is the class of a linear subspace. It remains to identify the free generator with $h_k = [\mathbb{P}^k]$. Trace a linear $\mathbb{P}^k \subseteq \mathbb{P}^n$ through (1.6). Choose the point $P \in \mathbb{P}^k$; projection of the linear subspace $\mathbb{P}^k$ from the point $P \in \mathbb{P}^k$ is again linear, of one lower dimension, so $q(\mathbb{P}^k \setminus \{P\})$ is a linear $\mathbb{P}^{k-1} \subseteq \mathbb{P}^{n-1}$. Restricting the bundle $q: \mathbb{P}^n \setminus \{P\} \rightarrow \mathbb{P}^{n-1}$ of Step 2 over this $\mathbb{P}^{k-1}$ realizes $\mathbb{P}^k \setminus \{P\}$ as the affine bundle over $\mathbb{P}^{k-1}$; concretely $q^{-1}(\mathbb{P}^{k-1}) = \mathbb{P}^k \setminus \{P\}$ as a reduced irreducible subvariety, so flat pullback gives $q^*[\mathbb{P}^{k-1}] = [q^{-1}(\mathbb{P}^{k-1})] = [\mathbb{P}^k \setminus \{P\}]$ with multiplicity 1. Hence by construction of (1.6) the class $h_k = [\mathbb{P}^k]$ satisfies $j^*h_k = [\mathbb{P}^k \setminus \{P\}] = q^*[\mathbb{P}^{k-1}]$, so h_k corresponds to h_{k-1} under the isomorphism $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$. Descending to $A_0(\mathbb{P}^{n-k})$, the class h_k maps to $h_0 = [\text{pt}]$, which is the free generator by Step 1. Thus h_k is a generator of $A_k(\mathbb{P}^n) = \mathbb{Z}$. The top case $k = n$ is consistent with the fundamental-class computation $A_n(\mathbb{P}^n) = Z_n(\mathbb{P}^n) = \mathbb{Z}[\mathbb{P}^n]$ of example 1.1.6, since $\mathbb{P}^n$ is irreducible and $h_n = [\mathbb{P}^n]$.

Step 6: independence of the linear subspace. Any two linear k -planes $\Lambda, \Lambda' \subseteq \mathbb{P}^n$ have classes differing by an integer multiple of the generator; the previous step assigns each of them the generator h_0 under the descent to $A_0(\mathbb{P}^{n-k})$, so $[\Lambda] = [\Lambda']$ in $A_k(\mathbb{P}^n)$. Concretely a projective linear transformation carrying Λ to Λ' sweeps out a rational equivalence between them, so the generator h_k is independent of the chosen k -plane, completing the induction and the proof.

The theorem says that the Chow groups of $\mathbb{P}^n$ take the simplest possible form: one copy of $\mathbb{Z}$ in each dimension from 0 to n , with a canonical generator, the class of a linear subspace. Writing the total group additively,

$$A_*(\mathbb{P}^n) = \bigoplus_{k=0}^n \mathbb{Z} h_k, \quad h_k = [\mathbb{P}^k]$$

so $A_*(\mathbb{P}^n)$ is free of rank $n+1$. In codimension indexing the generator $h_k = [\mathbb{P}^k]$ has codimension $n-k$, and it is common to write $h = [\mathbb{P}^{n-1}]$ for the hyperplane class and record h_k as the intersection h^{n-k} of $n-k$ hyperplanes; the multiplicative structure that makes this literal is the intersection product developed in a later chapter. For now the additive statement is what matters: on projective space, a k -cycle carries exactly one integer of information, its degree, the coefficient of h_k .

This is the precise sense in which projective space is rigid where affine space is soft. A subvariety $V \subseteq \mathbb{P}^n$ of dimension k and degree d (its degree as a projective variety, the number of points in which it meets a general linear $\mathbb{P}^{n-k}$) has class $[V] = d h_k$, and no amount of rational equivalence can change d . The degree is a complete invariant of the class, and rational equivalence on $\mathbb{P}^n$ is, for irreducible cycles, exactly the relation “same dimension and same degree” (and for arbitrary cycles, “same degree in each dimension”). The comparison with singular homology is exact: under the cycle class map to $H_{2k}(\mathbb{CP}^n)$ studied later, h_k maps to the generator, and the freeness of $A_k(\mathbb{P}^n)$ mirrors the freeness of $H_{2k}(\mathbb{CP}^n; \mathbb{Z})$ ([5]).

The one-dimensional case deserves its own statement, both because curves are where the theory began and because on a curve the Chow group is an object the reader already knows under another name. On a smooth projective curve C the only cycles are 0-cycles and the fundamental class, and the 0-cycles modulo rational equivalence are precisely the divisor classes. This ties the general construction back to the divisor class group $\text{Cl } C$ and the Picard group $\text{Pic } C$ of [4, Divisors: Geometric Representation of Line Bundles].

Proposition 1.5.3: Chow Groups of a Smooth Projective Curve

Let C be a smooth projective curve over $\bar{k}$. Then

$$A_1(C) = \mathbb{Z}[C], \quad A_0(C) \cong \text{Cl } C = \text{Pic } C$$

and $A_k(C) = 0$ for $k \neq 0, 1$. Under the second isomorphism the degree of a zero-cycle corresponds to the degree of a divisor, giving a short exact sequence

$$0 \rightarrow \text{Pic}^0 C \rightarrow A_0(C) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$$

where $\text{Pic}^0 C$ is the group of divisor classes of degree zero.

Proof:

Since $\dim C = 1$, the only nonzero groups are $A_0(C)$ and $A_1(C)$, and $A_k(C) = 0$ for $k \neq 0, 1$. The curve C is irreducible of dimension 1, so by example 1.1.6 its top group is $A_1(C) = Z_1(C) = \mathbb{Z}[C]$, free on the fundamental class.

The zero-cycles. A 0-cycle on C is a finite $\mathbb{Z}$ -combination of closed points, so $Z_0(C)$ is the group $\text{WDiv } C$ of Weil divisors on C . The subgroup $\text{Rat}_0(C)$ of cycles rationally equivalent to zero is generated by the cycles $[\text{div}(r)]$ for $r \in k(W)^*$ ranging over 1-dimensional subvarieties $W \subseteq C$; the only such W is C itself, so $\text{Rat}_0(C)$ is generated by the principal divisors $[\text{div}(r)]$ with

$r \in k(C)^*$. Thus $\text{Rat}_0(C)$ is exactly the group $\text{Prin } C$ of principal divisors in the sense of [4, Principal Divisor], and

$$A_0(C) = Z_0(C)/\text{Rat}_0(C) = \text{WDiv } C / \text{Prin } C = \text{Cl } C$$

is the divisor class group of [4, Divisor Class Group]. Because C is smooth, the divisor-to-line-bundle correspondence of [4, Divisors: Geometric Representation of Line Bundles] identifies $\text{Cl } C$ with $\text{Pic } C$.

The degree sequence. On a smooth projective curve linearly equivalent divisors have equal degree: if $D \sim D'$ then $\deg D = \deg D'$, since the Riemann-Roch theorem ([4, Riemann-Roch for curves]) gives $\chi(\mathcal{L}(D)) = \deg D + 1 - g$, a quantity depending only on the class of D , so $\deg D$ is determined by it. Hence $\deg$ descends to a homomorphism $\deg: A_0(C) \rightarrow \mathbb{Z}$, agreeing under the identification $A_0(C) = \text{Cl } C$ with the degree of a divisor class. It is surjective because a single closed point has degree $[k(p) : \bar{k}] = 1$, and its kernel is the group $\text{Pic}^0 C$ of degree-zero classes by definition. This gives the stated short exact sequence, completing the proof.

The proposition recovers, in the language of Chow groups, the entire divisor theory of a curve. The degree map splits the class group into a discrete part, the copy of $\mathbb{Z}$ recording degree, and a connected part $\text{Pic}^0 C$, the Jacobian, carrying the continuous moduli of divisor classes of a fixed degree. For $C = \mathbb{P}^1$ the Jacobian is trivial and $A_0(\mathbb{P}^1) = \mathbb{Z}$, in agreement with example 1.1.5 and with the case $n = 1$ of theorem 1.5.2; for a curve of genus $g \geq 1$ the group $\text{Pic}^0 C$ is a g -dimensional abelian variety and $A_0(C)$ is strictly larger than $\mathbb{Z}$, so the degree is no longer a complete invariant of a class. The Chow group of a curve is therefore the first place in the theory where an interesting continuous invariant appears, and it is the model on which the higher-dimensional theory is built.

The Chow group $A_0(C)$ also illustrates the failure of the total-degree map to see everything. On $\mathbb{P}^n$ the degree was a complete invariant of a k -cycle class; on a curve of positive genus it is not, because two points $p, q \in C$ need not be rationally equivalent even though $[p]$ and $[q]$ have the same degree. Their difference $[p] - [q]$ is a nonzero class in $\text{Pic}^0 C$ precisely when p and q are not linearly equivalent, which for $g \geq 1$ is the generic situation. This is the geometric content of the statement that a curve of positive genus admits no nonconstant rational function of degree one.

The last computation of the section is stated but not proved, because its proof needs Segre classes, which are developed later. It generalizes the projective-space computation from the trivial bundle to an arbitrary vector bundle: the Chow groups of a projectivized vector bundle are a free module over the Chow groups of the base, on the powers of the tautological class. This is the projective-bundle formula, and it is the engine that will later define Chern classes.

Given a vector bundle E of rank $r + 1$ on a scheme X , write $\mathbb{P}(E) \rightarrow X$ for the associated projective bundle, whose fiber over a point x is the projective space $\mathbb{P}(E_x) \cong \mathbb{P}^r$ of one-dimensional subspaces of the fiber E_x . Let $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in A^1(\mathbb{P}(E))$ be the first Chern class of the Serre-twist line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ (the dual of the tautological subbundle, with the “one-dimensional subspaces” convention above), viewed as an operator on cycles. The projective-bundle formula states that the flat pullback π^* together with capping against powers of ξ makes $A_*(\mathbb{P}(E))$ a free module over $A_*(X)$ on the classes $1, \xi, \dots, \xi^r$:

$$A_*(\mathbb{P}(E)) = \bigoplus_{i=0}^r \xi^i \cap \pi^* A_*(X)$$

Concretely, every class on $\mathbb{P}(E)$ is uniquely $\sum_{i=0}^r \xi^i \cap \pi^* \alpha_i$ with $\alpha_i \in A_*(X)$. Taking $X = \text{Spec } \bar{k}$ and $E = \bar{k}^{n+1}$ recovers $\mathbb{P}(E) = \mathbb{P}^n$ and the free-module statement collapses to $A_*(\mathbb{P}^n) = \bigoplus_{i=0}^n \mathbb{Z} \xi^i$

of theorem 1.5.2, with ξ^i the class of a linear subspace of codimension i . The full statement, and the definition of the Chern class operators it rests on, come in the chapter on cones and Segre classes; the formula is recorded here because it is the natural common generalization of the two computations above, and because it is the shape every later Chow-group computation of a bundle takes.

Before the exercises, the three worked instances of the theory are collected side by side, so the common mechanism is visible in one place.

Example 1.5.4: The Chow Groups of $\mathbb{P}^1$, $\mathbb{P}^2$, and a Plane Curve

1. *The projective line.* For $n = 1$, theorem 1.5.2 gives $A_0(\mathbb{P}^1) = \mathbb{Z}[\text{pt}]$ and $A_1(\mathbb{P}^1) = \mathbb{Z}[\mathbb{P}^1]$, so

$$A_*(\mathbb{P}^1) = \mathbb{Z}[\mathbb{P}^1] \oplus \mathbb{Z}[\text{pt}]$$

The degree-0 part $A_0(\mathbb{P}^1) = \mathbb{Z}$ agrees with example 1.1.5, and with proposition 1.5.3 for the genus-zero curve $\mathbb{P}^1$, whose Jacobian $\text{Pic}^0 \mathbb{P}^1$ is trivial.

2. *The projective plane.* For $n = 2$,

$$A_*(\mathbb{P}^2) = \mathbb{Z}[\mathbb{P}^2] \oplus \mathbb{Z}[\mathbb{P}^1] \oplus \mathbb{Z}[\text{pt}]$$

with $A_2(\mathbb{P}^2) = \mathbb{Z}[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}[\mathbb{P}^1]$ generated by the class of a line, and $A_0(\mathbb{P}^2) = \mathbb{Z}[\text{pt}]$. A plane curve $D \subseteq \mathbb{P}^2$ of degree d has class $[D] = d[\mathbb{P}^1] = d h_1$ in $A_1(\mathbb{P}^2)$, since D is cut out by a form of degree d and hence is rationally equivalent to d lines (the pencil spanned by the form and ℓ^d for a linear form ℓ sweeps one to the other). Two plane curves of degrees c and d therefore have classes meeting, after the intersection product is available, in cd points, which is Bézout's theorem ([1]) read off from $h_1 \cdot h_1 = h_0$.

3. *A plane curve as a curve in its own right.* Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d , so C is a smooth projective curve of genus $g = \binom{d-1}{2}$. Two computations of its Chow groups coexist. Viewed inside $\mathbb{P}^2$, the class $[C] = d h_1 \in A_1(\mathbb{P}^2)$ records only its degree. Viewed intrinsically, proposition 1.5.3 gives $A_0(C) \cong \text{Cl } C = \text{Pic } C$ with its degree sequence $0 \rightarrow \text{Pic}^0 C \rightarrow A_0(C) \rightarrow \mathbb{Z} \rightarrow 0$, and for $d \geq 3$ the genus is positive, so $\text{Pic}^0 C$ is nontrivial and $A_0(C)$ is strictly larger than $\mathbb{Z}$. The pushforward $A_0(C) \rightarrow A_0(\mathbb{P}^2) = \mathbb{Z}$ along the inclusion is the degree map, collapsing all of $\text{Pic}^0 C$ to zero: the intrinsic Chow group of the curve sees the moduli that the ambient projective plane cannot.

Exercise 1.5.1

1. Compute $A_*(\mathbb{A}^2 \setminus \{p\})$, the Chow groups of the affine plane with one closed point removed. (Use the localization sequence for the closed point inside $\mathbb{A}^2$ together with proposition 1.5.1.)
2. Let $H \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ and $U = \mathbb{P}^n \setminus H \cong \mathbb{A}^n$ its complement. Using the localization sequence, show directly that $j^*: A_k(\mathbb{P}^n) \rightarrow A_k(\mathbb{A}^n)$ is an isomorphism for $k = n$ and the zero map for $k < n$, and reconcile this with theorem 1.5.2.
3. Let $Q \subseteq \mathbb{P}^3$ be a smooth quadric surface. Assuming the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ and that $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on the classes of a point, the two rulings, and the fundamental class, determine the rank of each $A_k(Q)$ and describe generators. Compare with $A_*(\mathbb{P}^2)$ and explain why $A_1(Q)$ has rank 2 rather than 1.

4. For $\mathbb{P}^m$ and $\mathbb{P}^n$, use theorem 1.5.2 and the projective-bundle formula (with $\mathbb{P}^m \times \mathbb{P}^n = \mathbb{P}(\mathcal{O}^{\oplus(m+1)})$ over $\mathbb{P}^n$) to show that $A_k(\mathbb{P}^m \times \mathbb{P}^n)$ is free of rank equal to the number of pairs (i, j) with $0 \leq i \leq m$, $0 \leq j \leq n$, and $i + j = k$.
5. Let C be a smooth projective curve of genus 1 over $\bar{k}$. Show that $A_0(C)$ is not finitely generated, and identify the classes $[p] - [q]$ of differences of points inside $\text{Pic}^0 C$. Contrast with $A_0(\mathbb{P}^1)$.
6. Let $X \subseteq \mathbb{A}^3$ be the affine cone $x^2 + y^2 = z^2$ over a conic, a surface singular at the origin. Using $A_k(X) = A_k(X_{\text{red}})$ and the vanishing of $A_k(\mathbb{A}^n)$ in low degree as a guide, determine $A_2(X)$ and explain why the singular point does not contribute a class in $A_0(X)$ that survives rational equivalence.

Divisors, Line Bundles, and the First Chern Class

Chapter 1 built the Chow groups and their two functorialities, proper pushforward and flat pullback, but produced no way to multiply cycles: nothing so far intersects a k -cycle with anything to lower its dimension. The first and most basic such operation is intersection with a divisor. A Cartier divisor D on X , equivalently a line bundle together with a rational section, cuts each subvariety V not contained in its support down to a $(k - 1)$ -cycle $D \cdot [V]$, the divisor of the restricted section. This chapter makes that operation precise, proves it descends to rational equivalence, and establishes its one indispensable property, commutativity: intersecting with D and then with D' agrees with intersecting in the other order.

Commutativity is not a formality. It is the identity whose proof, resting on the vanishing degree of a principal divisor on a complete curve, supplies the well-definedness that the affine-bundle-invariance theorem of section 1.4 took on faith. Packaging intersection with a divisor as the *first Chern class* $c_1(L) \cap -$ of the associated line bundle turns these results into the seed of the entire theory of Chern classes, developed for bundles of higher rank in the chapter that follows.

2.1 Cartier Divisors, Line Bundles, and the Class Map

The Chow group of section 1.1 is graded by dimension, and every construction of sections 1.2 to 1.4 either preserves the grading (proper pushforward, flat pullback of the fundamental class) or shifts it by the fixed relative dimension of a flat map. What is still missing is an operation that *lowers* dimension by one, cutting a k -cycle down to a $(k - 1)$ -cycle by intersecting it with a hypersurface. The classical prototype is Bézout's theorem: a curve of degree d and a curve of degree e in $\mathbb{P}^2$ meet in de points. The intersection is a codimension-one operation, and the object one intersects with is a

divisor. This section sets up the divisor side of that operation. It recalls, from *Algebraic Geometry* [4], the two faces of a divisor (the geometric Weil divisor and the sheaf-theoretic Cartier divisor) and the line bundle that packages the Cartier data, then constructs the first bridge from divisor theory to the Chow group: the class map $\text{Pic}(X) \rightarrow A_{n-1}(X)$ sending a line bundle to the cycle of its rational sections. The intersection operation itself, and the proof that it is well defined on rational equivalence, occupy section 2.2; here the goal is to fix the input and record how it sits inside the Chow group of a fixed variety.

Throughout, X is an algebraic scheme over $\bar{k}$ and V ranges over subvarieties (closed integral subschemes); a point is a closed point, and $n = \dim X$.

Algebraic Geometry [4] developed two theories of divisors, and the reader who has that material in hand can move quickly here. A *Weil divisor* ([4, Weil Divisor]) on a Noetherian integral separated scheme regular in codimension one is a finite formal integer combination $\sum_i n_i [Y_i]$ of prime divisors, where a prime divisor is a codimension-one subvariety. This is precisely a codimension-one algebraic cycle in the sense of section 1.1: when $\dim X = n$, the group $\text{WDiv } X$ of Weil divisors is exactly the cycle group $Z_{n-1}(X)$. A *principal divisor* $\text{div}(f) = \sum_Y \nu_Y(f) [Y]$ ([4, Principal Divisor]) records the zeros and poles of a rational function $f \in K(X)^*$, weighted by the order ν_Y measured in the discrete valuation ring $\mathcal{O}_{X, \eta_Y}$; this is the same order-along-a-subvariety $\text{ord}_Y(f)$ that defines rational equivalence in definition 1.1.1. The divisor class group $\text{Cl } X = \text{WDiv } X / \text{PDiv}(X)$ of [4, Divisor Class Group] is therefore literally the top Chow group $A_{n-1}(X)$ in the case where the only cycles rationally equivalent to zero come from functions on X itself, that is, on a variety.

A *Cartier divisor* ([4, Cartier Divisor]) encodes the same data locally and sheaf-theoretically: it is a global section of $\mathcal{K}_X^* / \mathcal{O}_X^*$, represented by an open cover $\{U_i\}$ with rational functions $f_i \in \Gamma(U_i, \mathcal{K}_X^*)$ whose ratios f_i / f_j are units on overlaps. A Cartier divisor is *principal* when it comes from a single global rational function; two Cartier divisors are linearly equivalent when they differ by a principal one. The gluing data $\{(U_i, f_i)\}$ builds a line bundle $\mathcal{O}_X(D)$, the sub- $\mathcal{O}_X$ -module of $\mathcal{K}_X$ generated locally by f_i^{-1} , and this assignment is the heart of the theory: $D \mapsto \mathcal{O}_X(D)$ is a homomorphism $\text{CDiv } X \rightarrow \text{Pic } X$ that carries addition of divisors to tensor product of line bundles, sends $-D$ to the dual, and vanishes exactly on principal divisors. On an integral scheme it induces an isomorphism from the Cartier divisor class group $\text{CaCl } X$ of [4, Cartier Divisor Class Group] onto the whole Picard group (all of this is developed in [4, Divisors: Geometric Representation of Line Bundles]),

$$\text{CaCl } X \xrightarrow{\sim} \text{Pic } X$$

so a line bundle is the same thing as a Cartier divisor up to linear equivalence. This is the sense in which the section title lists “Cartier divisors” and “line bundles” as two names for one object.

The reader should keep both faces in view. The Cartier divisor is what one intersects with, because its local equations f_i give the equations to cut against; the line bundle $\mathcal{O}_X(D)$ is the coordinate-free invariant, the thing that survives when the choice of local equations is forgotten. The class map built below is precisely the passage from the line bundle back to a well-defined cycle class.

The two theories agree under a regularity hypothesis, and pinning down that hypothesis is what makes the class map land in the Chow group cleanly. The point is one of local principality: a Weil divisor is Cartier exactly when each of its prime divisors is locally cut out by a single equation. Regularity in codimension one is enough to send a Cartier divisor to a Weil divisor; local factoriality is what makes the map an isomorphism.

Proposition 2.1.1: Weil-Cartier Comparison

Let X be a variety of dimension n over $\bar{k}$.

1. If X is regular in codimension one (for instance X normal), there is a homomorphism

$$\mathrm{CDiv} X \longrightarrow \mathrm{WDiv} X = Z_{n-1}(X), \quad D \longmapsto [D]$$

carrying principal Cartier divisors to principal Weil divisors. In local data $D = \{(U_i, f_i)\}$ the associated Weil cycle is $[D] = \sum_Y \nu_Y(f_i)[Y]$, the coefficient of a prime divisor Y being $\nu_Y(f_i)$ for any index i with $\eta_Y \in U_i$.

2. If X is locally factorial (every local ring $\mathcal{O}_{X,x}$ is a unique factorization domain; smooth varieties qualify), this map is an isomorphism $\mathrm{CDiv} X \xrightarrow{\sim} \mathrm{WDiv} X$ carrying principal divisors to principal divisors, hence an isomorphism of class groups

$$\mathrm{Pic} X \cong \mathrm{CaCl} X \cong \mathrm{Cl} X = A_{n-1}(X)$$

Proof:

Both parts are proved in full in [4, Divisors: Geometric Representation of Line Bundles] and only recalled here. The homomorphism of part (1) is [4, Cartier Divisors to Weil Divisors], and its upgrade to an isomorphism under local factoriality (part (2)) is [4, Weil and Cartier Divisors Under Locally Factorial Hypotheses]; the resulting chain of class-group isomorphisms $\mathrm{Pic} X \cong \mathrm{CaCl} X \cong \mathrm{Cl} X$ is [4, Divisor Class Group and Picard Group]. Only the final equality $\mathrm{Cl} X = A_{n-1}(X)$ is new to this Part, and it is the identification made in the paragraph below: on a variety the codimension-one cycles modulo rational equivalence are exactly the Weil divisors modulo principal divisors, both quotients of $Z_{n-1}(X)$ by $\mathrm{PDiv}(X)$, as we sought to show.

The equality $\mathrm{Cl} X = A_{n-1}(X)$ deserves a moment's reflection, because it is easy to mistake for a definition. Rational equivalence on $Z_{n-1}(X)$ is generated by cycles $[\mathrm{div}(r)]$ where r is a rational function on a subvariety $W \subseteq X$ of dimension n . On a variety X , the only n -dimensional subvariety is X itself, so those generators are exactly the principal Weil divisors $\mathrm{div}(r)$ with $r \in K(X)^*$. Hence $A_{n-1}(X) = Z_{n-1}(X)/\mathrm{Rat}_{n-1}(X)$ and $\mathrm{Cl} X = \mathrm{WDiv} X/\mathrm{PDiv}(X)$ are the same quotient of the same group. Linear equivalence of divisors *is* rational equivalence of codimension-one cycles. This is the seam that will let the class map below produce Chow classes, not just divisor classes.

Example 2.1.2: The Quadric Cone: Cartier Strictly Inside Weil

The comparison map need not be surjective when factoriality fails. On the affine quadric cone $X = \mathrm{Spec} \bar{k}[x, y, z]/(xy - z^2)$, a normal surface singular at the vertex, [4, Cartier Divisors] computes that the ruling $Y = V(y, z)$ is a Weil divisor which is not Cartier at the vertex, while its double $2[Y] = \mathrm{div}(y)$ is, so

$$\mathrm{CaCl} X = 0 \subsetneq \mathrm{Cl} X = \mathbb{Z}/2\mathbb{Z} = A_1(X)$$

the final equality being the identification $\mathrm{Cl} X = A_{n-1}(X)$ of proposition 2.1.1. Normality supplies part (1), but the vertex obstructs local principality, so part (2) fails; on a smooth variety no such gap exists and Weil, Cartier, Picard, and A_{n-1} all coincide.

On a smooth variety the isomorphism $\mathrm{Pic} X \cong A_{n-1}(X)$ of proposition 2.1.1 already provides a bridge from line bundles to Chow classes. To intersect and to compute Chern classes on *arbitrary*

algebraic schemes, one needs this bridge without any regularity hypothesis, and one needs it stated at the level of line bundles rather than Cartier divisors, so that the input is a coordinate-free invariant. The construction is direct: a line bundle L has rational sections, a rational section has a divisor of zeros and poles, and that divisor is a codimension-one cycle whose class in $A_{n-1}(X)$ turns out not to depend on the section chosen.

Fix a variety X of dimension n with function field $K(X)$, and a line bundle L on X . A *rational section* of L is a nonzero section of $L \otimes_{\mathcal{O}_X} K(X)$ over X , equivalently a section of L over some nonempty open subset. Since X is a variety, $L \otimes K(X)$ is a one-dimensional $K(X)$ -vector space, so rational sections exist and any two differ by multiplication by an element of $K(X)^*$. On the open set where s is defined and nonvanishing, s trivializes L ; along each prime divisor Y one may compare s with a local trivialization of L near η_Y and read off an integer $\text{ord}_Y(s)$, the order of vanishing (or of the pole) of s along Y . Concretely, if t is a local generator of L near η_Y and $s = g \cdot t$ with $g \in K(X)^*$, set $\text{ord}_Y(s) := \nu_Y(g)$; a different generator $t' = ut$ with $u \in \mathcal{O}_{X,\eta_Y}^*$ changes g to $u^{-1}g$ and leaves $\nu_Y(g)$ unchanged, so $\text{ord}_Y(s)$ is well defined. The *divisor of s* is the Weil divisor

$$\text{div}(s) = \sum_Y \text{ord}_Y(s) [Y] \in Z_{n-1}(X)$$

finite because s is a regular nonvanishing section over some nonempty open $U \subseteq X$, where it trivializes L and so $\text{ord}_Y(s) = 0$ for every prime divisor Y meeting U ; the only possible contributors are the finitely many prime divisors in the proper closed complement $X \setminus U$. This is the same finiteness that makes the principal divisor $\text{div}(f)$ of [4, Principal Divisor] a well-defined cycle. If $L = \mathcal{O}_X(D)$ for a Cartier divisor D and s is the canonical rational section (the constant $1 \in \mathcal{K}_X$, which against the local generator f_i^{-1} of $\mathcal{O}_X(D)|_{U_i}$ is written $1 = f_i \cdot f_i^{-1}$, giving $\text{ord}_Y(s) = \nu_Y(f_i)$), then $\text{div}(s)$ is exactly the Weil cycle $[D]$ of proposition 2.1.1.

Proposition 2.1.3: The Class Map

Let X be a variety of dimension n . For a line bundle L on X , choose any nonzero rational section s of L and set

$$c(L) := [\text{div}(s)] \in A_{n-1}(X)$$

Then $c(L)$ is independent of the choice of s , and the resulting assignment

$$c: \text{Pic } X \longrightarrow A_{n-1}(X), \quad L \longmapsto c(L)$$

is a group homomorphism. When X is locally factorial (in particular smooth), c is the isomorphism $\text{Pic } X \cong A_{n-1}(X)$ of proposition 2.1.1; on a general variety c need be neither injective nor surjective.

Proof:

Step 1: independence of the section. Let s, s' be two nonzero rational sections of L . Both lie in the one-dimensional $K(X)$ -vector space $L \otimes K(X)$, so $s' = gs$ for a unique $g \in K(X)^*$. Along each prime divisor Y , choose a local generator t of L near η_Y and write $s = at$, $s' = a't$ with $a, a' \in K(X)^*$; then $a' = ga$ and

$$\text{ord}_Y(s') = \nu_Y(a') = \nu_Y(g) + \nu_Y(a) = \nu_Y(g) + \text{ord}_Y(s)$$

because ν_Y is a valuation. Summing over Y gives $\text{div}(s') = \text{div}(g) + \text{div}(s)$ in $Z_{n-1}(X)$, where $\text{div}(g) = \sum_Y \nu_Y(g)[Y]$ is the principal Weil divisor of $g \in K(X)^*$. Since $\text{div}(g)$ is a principal

divisor on the variety X , it is rationally equivalent to zero in $A_{n-1}(X)$ by the identification $\text{Rat}_{n-1}(X) = \text{PDiv}(X)$ recorded after proposition 2.1.1. Hence $[\text{div}(s')] = [\text{div}(s)]$ in $A_{n-1}(X)$, and $c(L)$ is well defined.

Step 2: homomorphism property. Let L, L' be line bundles with rational sections s, s' . Then $s \otimes s'$ is a nonzero rational section of $L \otimes L'$, and along each prime divisor Y , choosing local generators t of L and t' of L' near η_Y and writing $s = at$, $s' = a't'$, the product $s \otimes s' = (aa')(t \otimes t')$ is expressed against the local generator $t \otimes t'$ of $L \otimes L'$. Thus

$$\text{ord}_Y(s \otimes s') = \nu_Y(aa') = \nu_Y(a) + \nu_Y(a') = \text{ord}_Y(s) + \text{ord}_Y(s')$$

so $\text{div}(s \otimes s') = \text{div}(s) + \text{div}(s')$, and passing to classes gives $c(L \otimes L') = c(L) + c(L')$. The trivial bundle $\mathcal{O}_X$ has the rational section 1 with $\text{div}(1) = 0$, so $c(\mathcal{O}_X) = 0$, and $c(L^{-1}) = -c(L)$ follows from $L \otimes L^{-1} = \mathcal{O}_X$. Therefore c is a homomorphism.

Step 3: comparison with the smooth case. When X is locally factorial, every line bundle is $\mathcal{O}_X(D)$ for a Cartier divisor D , and the canonical rational section of $\mathcal{O}_X(D)$ has $\text{div}(s) = [D]$, the Weil cycle of D . Under $\text{Pic } X \cong \text{CaCl } X$ ([4, Divisors: Geometric Representation of Line Bundles]) and $\text{CaCl } X \cong \text{Cl } X = A_{n-1}(X)$ (proposition 2.1.1), the assignment $L = \mathcal{O}_X(D) \mapsto [D]$ is exactly that isomorphism, so c is an isomorphism. On a variety where these hypotheses fail, c can lose injectivity or surjectivity, as example 2.1.2 shows for the class-group comparison. This completes the proof.

The class map is the first of the two ways line bundles act on the Chow group. It records *where* a line bundle is, as a codimension-one class $c(L) \in A_{n-1}(X)$, but it does not yet say how L intersects cycles of other dimensions. That is the job of the operator $c_1(L) \cap -$ built in section 2.4, whose value on the fundamental class $[X]$ recovers $c(L)$: writing ahead, $c_1(L) \cap [X] = c(L)$. The class map is thus the top-dimensional shadow of the full first Chern class, and everything in this chapter can be read as promoting c from a map on line bundles to an operator on all of $A_*(X)$.

The value of stating c at the level of line bundles, rather than resting on the isomorphism $\text{Pic} \cong A_{n-1}$, is that c makes sense with no regularity hypothesis on X : a rational section and its order along a prime divisor require only that X be a variety, and the group $A_{n-1}(X)$ is defined for every algebraic scheme. The intersection theory of section 2.2 is built for arbitrary X , and it needs c in this general form.

Example 2.1.4: The Class Map on Projective Space

Projective space is where every ingredient can be computed by hand. Since $\mathbb{P}^n$ has all local rings unique factorization domains, it is locally factorial, so $\text{Pic } \mathbb{P}^n \cong \text{Cl } \mathbb{P}^n \cong \mathbb{Z}$, generated by $\mathcal{O}_{\mathbb{P}^n}(1)$, and by theorem 1.5.2 the top Chow group is $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}[\mathbb{P}^{n-1}]$, generated by the class of a hyperplane. The class map identifies the two.

A rational section of $\mathcal{O}_{\mathbb{P}^n}(1)$ is a nonzero linear form. Take $s = x_0$, the section whose zero scheme is the hyperplane $H = V_+(x_0)$, whose Cartier data $\{(U_i, x_0/x_i)\}$ was recorded in [4, Cartier Divisors]. On the standard chart $U_i = D_+(x_i) \cong \mathbb{A}^n$ the sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$ is trivialized by x_i , and $s = x_0 = (x_0/x_i)x_i$, so s corresponds to the rational function $x_0/x_i \in K(\mathbb{P}^n)$ against the generator x_i . Along the prime divisor H , the generic point lies in charts U_i with $i \neq 0$, where x_0/x_i is a local equation cutting out H with order one, so $\text{ord}_H(s) = 1$; along every other prime divisor x_0/x_i is a unit near the generic point, contributing zero. Hence

$$\text{div}(x_0) = [H] = [\mathbb{P}^{n-1}], \quad c(\mathcal{O}_{\mathbb{P}^n}(1)) = [\mathbb{P}^{n-1}] \in A_{n-1}(\mathbb{P}^n)$$

Choosing instead $s = x_1$ gives the hyperplane $V_+(x_1)$, a different divisor but the same class: $x_1 = (x_1/x_0)x_0$ differs from x_0 by the rational function x_1/x_0 , and $\text{div}(x_1) - \text{div}(x_0) = \text{div}(x_1/x_0) \sim 0$, confirming the independence of proposition 2.1.3 in this case. More generally, $\mathcal{O}_{\mathbb{P}^n}(d)$ has rational sections given by degree- d forms; the section x_0^d has divisor $d[H]$, so $c(\mathcal{O}_{\mathbb{P}^n}(d)) = d[\mathbb{P}^{n-1}]$, and $c: \mathbb{Z} \rightarrow \mathbb{Z}$ is the identity. The generator $\mathcal{O}(1)$ of Pic maps to the generator $[\text{hyperplane}]$ of A_{n-1} , and the class map is the announced isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$.

There is one further refinement to record before intersecting, and it addresses a bookkeeping problem rather than a new geometric idea. When a Cartier divisor D is intersected with a subvariety V , one wants the resulting cycle to live on $V \cap |D|$, the part of V where D cuts, not just somewhere on V . Localizing the answer to the support is what makes the intersection product refined enough to carry information (for instance, an excess-intersection or a self-intersection formula) and what makes it functorial under the localization sequence of theorem 1.4.1. The device that carries this localized data is the *pseudo-divisor*: a line bundle, a chosen closed support, and a trivialization of the bundle away from that support.

Definition 2.1.5: Pseudo-Divisor

Let X be an algebraic scheme. A *pseudo-divisor* on X is a triple

$$\mathfrak{D} = (L, Z, s)$$

consisting of a line bundle L on X , a closed subset $Z \subseteq X$ (the *support*, written $|\mathfrak{D}|$), and a trivialization $s: \mathcal{O}_{X \setminus Z} \xrightarrow{\sim} L|_{X \setminus Z}$ of L over the complement of Z . The line bundle L is denoted $\mathcal{O}_X(\mathfrak{D})$.

A Cartier divisor D on X determines a pseudo-divisor $(\mathcal{O}_X(D), |D|, s_D)$ whose support is the support $|D|$ of D and whose trivialization s_D is the canonical rational section of $\mathcal{O}_X(D)$, which is a nonvanishing regular section away from $|D|$. Two Cartier divisors D, D' with the same support and the same associated line bundle give the same pseudo-divisor when their canonical sections agree off the support.

The three pieces of data are exactly what a divisor supplies for cutting. The line bundle L is the coordinate-free invariant whose class map value is $c(L)$; the support Z pins down where the cutting happens, so the product lands in $A_*(Z)$ rather than $A_*(X)$; and the trivialization s off Z is what lets one measure orders of vanishing, because on the locus where L is trivialized by s there is nothing to cut, and all the divisorial content is concentrated on Z . A Cartier divisor is the special case where $Z = |D|$ is as small as possible, but allowing Z to be larger is what gives the flexibility to restrict, pull back, and add pseudo-divisors while keeping supports under control. The next section takes a subvariety V not contained in the support and defines $\mathfrak{D} \cdot [V]$ as a class in $A_{k-1}(V \cap Z)$; when $V \subseteq Z$, the restricted line bundle $L|_V$ supplies the answer through its own class map. Here the pseudo-divisor is introduced only to fix the input; its use is the content of section 2.2.

Example 2.1.6: The Pseudo-Divisor of a Hyperplane

Let $X = \mathbb{P}^n$ and $H = V_+(x_0)$ the hyperplane at $x_0 = 0$. The associated pseudo-divisor is

$$\mathfrak{H} = (\mathcal{O}_{\mathbb{P}^n}(1), H, s)$$

with support the hyperplane H and trivialization s given by the section x_0 over the complement $\mathbb{P}^n \setminus H = D_+(x_0) \cong \mathbb{A}^n$, where x_0 is nonvanishing and so trivializes $\mathcal{O}(1)$. The line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$ carries the class $c(\mathcal{O}(1)) = [H] \in A_{n-1}(\mathbb{P}^n)$ computed in example 2.1.4; the support restricts any

intersection against $\mathfrak{H}$ to H ; and the trivialization records that off H the hyperplane cuts nothing. Intersecting $\mathfrak{H}$ with a subvariety $V \not\subseteq H$ will, in section 2.2, produce the cycle $[V \cap H]$ with its multiplicities, supported on $V \cap H$, which is the algebraic incarnation of Bézout's degree count.

The class map and the pseudo-divisor together fix the two ways a line bundle enters intersection theory: as a class $c(L) \in A_{n-1}(X)$, which is its total content, and as a localized cutting datum (L, Z, s) , which is how it acts. With the input settled, the operation $\mathfrak{D} \cdot -$ can be defined and shown well defined on rational equivalence, which is the subject of section 2.2.

Exercise 2.1.1

1. Let $X = \mathbb{P}^n$ and let $Q = V_+(q)$ be a smooth quadric hypersurface, cut out by a nondegenerate quadratic form q . Compute $c(\mathcal{O}_{\mathbb{P}^n}(Q))$, the class map value of the line bundle of Q , in $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}[\mathbb{P}^{n-1}]$, and explain the answer in terms of the degree of Q .
2. Let X be a variety and $r \in K(X)^*$ a nonzero rational function. Show that the line bundle $\mathcal{O}_X(\text{div}(r))$ associated to the principal Cartier divisor $\text{div}(r)$ is trivial, and deduce that $c(\mathcal{O}_X(\text{div}(r))) = 0$ directly from proposition 2.1.3. Which step of the class-map construction is this computing?
3. On the affine quadric cone $X = \text{Spec } \bar{k}[x, y, z]/(xy - z^2)$ of example 2.1.2, the class map $c: \text{Pic } X \rightarrow A_1(X)$ is a homomorphism $\text{CaCl } X \rightarrow \text{Cl } X$. Using the values $\text{CaCl } X = 0$ and $\text{Cl } X = \mathbb{Z}/2\mathbb{Z}$ from that example, describe c explicitly and state whether it is injective, surjective, or neither. What does this say about the necessity of the factoriality hypothesis in proposition 2.1.1(2)?
4. Pseudo-divisors add: given $\mathfrak{D} = (L, Z, s)$ and $\mathfrak{D}' = (L', Z', s')$, define $\mathfrak{D} + \mathfrak{D}' := (L \otimes L', Z \cup Z', s \otimes s')$. Verify that this is a well-defined pseudo-divisor, that $\mathcal{O}_X(\mathfrak{D} + \mathfrak{D}') = \mathcal{O}_X(\mathfrak{D}) \otimes \mathcal{O}_X(\mathfrak{D}')$, and that on the line-bundle level this addition is compatible with the homomorphism property of the class map in proposition 2.1.3.
5. Let L be a line bundle on a variety X , s a nonzero rational section, and Y a prime divisor. Show that the order $\text{ord}_Y(s) := \nu_Y(a)$, defined by writing $s = at$ against a local generator t of L near η_Y with $a \in K(X)^*$, does not depend on the choice of generator t . Then show that if s is a regular nonvanishing section of L near η_Y , then $\text{ord}_Y(s) = 0$.

2.2 Intersecting with a Divisor

The class map of section 2.1 sends a line bundle to a cycle class in codimension one, but it says nothing about how that cycle meets the other cycles already living on X . This section supplies the missing operation. A Cartier divisor D acts on cycles by cutting each subvariety down along the zeros and poles of a rational section, an operation that lowers dimension by one and records where the cut takes place. The construction is the first instance of a general principle of the subject, that a line bundle acts on the Chow group as an operator, and it is the operation whose associativity and commutativity the rest of the chapter is built to establish.

The geometry is the one already visible in codimension one. A Cartier divisor is locally the divisor of a single rational function f_i on an open U_i , and the transition ratios f_i/f_j are units, so the zeros and poles of the local equations patch to a well-defined codimension-one cycle on X . To intersect D with

a k -dimensional subvariety V , restrict this local data to V and read off the resulting $(k-1)$ -cycle on V . Two features force themselves on the construction. First, the answer is supported on the intersection $V \cap |D|$, since away from the support of D the local equations are units and contribute nothing. Second, the recipe stumbles when V lies inside $|D|$, where the local equations $f_i|_V$ may vanish identically and $\operatorname{div}(f_i|_V)$ is undefined; the trivialization off $|D|$ is exactly what is lost. The definition below treats the two cases separately and unifies them through the restricted line bundle.

Definition 2.2.1: Intersection with a Cartier Divisor

Let D be a Cartier divisor on X with support $|D|$ and associated line bundle $\mathcal{O}_X(D)$, and let $V \subseteq X$ be a k -dimensional subvariety. The *intersection class* $D \cdot [V] \in A_{k-1}(V \cap |D|)$ is defined as follows.

1. If $V \not\subseteq |D|$, then the local equations of D restrict to a nonzero rational function s_V on V : choosing a representative $\{(U_i, f_i)\}$ of D , the ratios $f_i|_V$ agree up to units on overlaps and glue to a rational section s_V of $\mathcal{O}_X(D)|_V$ that is nonzero on the dense open $V \setminus |D|$. Its divisor is a $(k-1)$ -cycle supported on $V \cap |D|$, and one sets

$$D \cdot [V] = [\operatorname{div}(s_V)] \in A_{k-1}(V \cap |D|)$$

2. If $V \subseteq |D|$, then no such nonzero section exists, and one instead uses the restricted line bundle. The class map of proposition 2.1.3 applied to $\mathcal{O}_X(D)|_V$ produces a class in $A_{k-1}(V)$; since $V \subseteq |D|$ gives $V \cap |D| = V$, one sets

$$D \cdot [V] = c(\mathcal{O}_X(D)|_V) \in A_{k-1}(V) = A_{k-1}(V \cap |D|)$$

Extending $\mathbb{Z}$ -linearly over the subvarieties appearing in a cycle $\alpha = \sum_i n_i [V_i]$ gives the *intersection cycle* $D \cdot \alpha = \sum_i n_i (D \cdot [V_i])$, a class in $A_{k-1}(|D| \cap \operatorname{Supp} \alpha)$, where $\operatorname{Supp} \alpha = \bigcup_i V_i$.

Two conventions in the definition deserve unpacking, since both recur throughout the theory. The first is that the intersection class is recorded on the intersection $V \cap |D|$, not on all of X . This is a refinement one does not want to discard prematurely: the intersection of a curve with a divisor is a set of points on the curve, and remembering that the points lie in $V \cap |D|$ is exactly what makes localized intersection numbers and excess-intersection formulas possible later. When only the class in $A_{k-1}(X)$ is wanted, push forward along the inclusion $V \cap |D| \hookrightarrow X$, a proper map. The second convention is the treatment of $V \subseteq |D|$. Here the naive recipe fails because the local equations restrict to zero, so there is no rational section to take the divisor of; but $\mathcal{O}_X(D)|_V$ is still a line bundle on V , and its class under the class map $c: \operatorname{Pic}(V) \rightarrow A_{k-1}(V)$ of proposition 2.1.3 is a $(k-1)$ -cycle class. Both clauses are the same construction, taking the class of the restricted bundle $\mathcal{O}_X(D)|_V$. When $V \not\subseteq |D|$ the bundle comes with a preferred rational section s_V and the class is computed as $[\operatorname{div}(s_V)]$; when $V \subseteq |D|$ that section degenerates and only the bundle survives. The packaging of the whole operation as the first Chern class $c_1(\mathcal{O}_X(D)) \cap -$ is the subject of section 2.4.

That the operation lowers dimension by exactly one is built into the definition: the divisor of a rational section on a k -dimensional variety is a codimension-one cycle, hence $(k-1)$ -dimensional. Geometrically, $D \cdot [V]$ is the divisor cut out on V by the equation defining D , the higher-dimensional form of substituting a curve into the equation of a hypersurface and reading off the intersection points with multiplicity. The next example makes this concrete in the case a reader already knows from the classical degree of a plane curve.

Example 2.2.2: A Hyperplane Meeting a Plane Curve

Let $X = \mathbb{P}^2$ with homogeneous coordinates $[x_0 : x_1 : x_2]$, let $H = \{x_0 = 0\}$ be the line at infinity, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(H) = \mathcal{O}_{\mathbb{P}^2}(1)$, and let $V \subseteq \mathbb{P}^2$ be a smooth conic, say the closure of $\{x_1^2 = x_0x_2\}$, a subvariety of dimension 1 not contained in H . The unique intersection point turns out to be $p = [0 : 0 : 1]$, which lies in the chart $U_2 = \{x_2 \neq 0\}$; work there, with affine coordinates $u = x_0/x_2$ and $v = x_1/x_2$. On U_2 the conic is $V = \{u = v^2\}$ (from $x_1^2 = x_0x_2$), the divisor $H = \{x_0 = 0\}$ is $\{u = 0\}$, and against the local generator x_2 of $\mathcal{O}_{\mathbb{P}^2}(1)$ the section x_0 representing H is the function $x_0/x_2 = u$. Its restriction to V is $s_V = u|_V = v^2$, which vanishes exactly where V meets H : $u = 0$ forces $v = 0$, so p is the only intersection point, and there V is tangent to H . In the local coordinate v on V near p , the function $s_V = v^2$ has a double zero (the conic meets its tangent line to order two), so

$$H \cdot [V] = [\operatorname{div}(s_V)] = 2[p] \in A_0(V \cap H)$$

Pushing forward to $\mathbb{P}^2$ and taking degrees gives $\deg(H \cdot [V]) = 2$, the degree of the conic. The intersection class remembers more than the number 2: it records that both units of intersection are concentrated at the single point p , information the bare degree discards.

The example illustrates the two things the intersection class is meant to capture, the total count and its distribution, and shows how the classical intersection number of two plane curves reappears as the degree of $D \cdot [V]$. For a curve V of degree d and a line H in general position the same computation gives d distinct simple points, and $\deg(H \cdot [V]) = d$ in every case; this is the statement, once the well-definedness below is in hand, that intersecting with a hyperplane computes the degree. Before turning to that, the second worked example records the complementary phenomenon, what happens when V lies inside the support of D , so that the second clause of definition 2.2.1 is in force.

Example 2.2.3: A Divisor Restricted to a Component of Its Support

Let $X = \mathbb{P}^2$ again and let $D = L$ be a line, say $L = \{x_0 = 0\}$, so $|D| = L$ and $\mathcal{O}_{\mathbb{P}^2}(L) = \mathcal{O}_{\mathbb{P}^2}(1)$. Take $V = L$ itself, a 1-dimensional subvariety contained in $|D|$. The first clause does not apply, since x_0 restricts to zero on L ; the second clause instructs one to compute the class of the restricted line bundle $\mathcal{O}_{\mathbb{P}^2}(1)|_L$. Now $L \cong \mathbb{P}^1$, and $\mathcal{O}_{\mathbb{P}^2}(1)|_L = \mathcal{O}_{\mathbb{P}^1}(1)$ by the adjunction of the hyperplane bundle to a linear subspace. The class map $\operatorname{Pic}(\mathbb{P}^1) = \mathbb{Z} \rightarrow A_0(\mathbb{P}^1) = \mathbb{Z}$ of proposition 2.1.3 carries $\mathcal{O}_{\mathbb{P}^1}(1)$ to the class of a single point $[q]$, so

$$L \cdot [L] = c(\mathcal{O}_{\mathbb{P}^2}(1)|_L) = [q] \in A_0(L)$$

the class of one point on L . This is the self-intersection $L \cdot L = 1$ of a line in the plane, the positive-degree shadow of the fact that two distinct lines meet in one point: moving L to a nearby line L' and intersecting $L \cdot [L'] = [L \cap L']$ recovers the same class, now visibly one transverse point. The second clause of the definition is what lets self-intersection make sense at all, since a divisor cannot literally be substituted into a variety it defines.

The two examples exhibit the operation on both sides of the containment condition and already hint at the compatibility the operation must have with rational equivalence. In example 2.2.3 the value $L \cdot [L]$ was computed by restricting the bundle, but it was also claimed to agree with $L \cdot [L']$ for a nearby line L' , obtained by the first clause. Since $L \sim L'$ as divisors, these two computations must give the same class if the operation is to depend only on the rational-equivalence class of the cycle, and only then does the self-intersection number deserve its name. Establishing that agreement in

general is the content of the section, and it is the property that will let the intersection class be defined on the Chow group rather than on cycles alone.

The stake is higher than notational convenience. Intersection with a divisor is meant to be an operator on $A_*(X)$, so that $D \cdot -$ carries $A_k(X)$ to $A_{k-1}(X)$ and can be iterated; without invariance under rational equivalence the operation lives only on cycles, where iteration is not controlled and no ring structure can form. The following proposition is the foundation of everything downstream, and it is one half of the identity (1.3) that the affine-bundle-invariance theorem of section 1.4 deferred to this chapter, the half asserting that $D \cdot -$ is well defined; the commutativity half is section 2.3.

Proposition 2.2.4: Rational Invariance of Divisor Intersection

Let D be a Cartier divisor on X . If $\alpha \in Z_k(X)$ is rationally equivalent to zero and no component of α lies in $|D|$, then $D \cdot \alpha \sim 0$ in $A_{k-1}(|D|)$. Consequently, for any class $a \in A_k(X)$ the intersection $D \cdot a \in A_{k-1}(|D|)$ is well defined by choosing a representative cycle none of whose components lies in $|D|$, and $D \cdot -$ descends to a homomorphism

$$D \cdot - : A_k(X) \rightarrow A_{k-1}(|D|)$$

independent of all choices.

The proof is the technical heart of the divisor calculus, and it is worth naming the obstacle before entering it. A cycle rationally equivalent to zero is a sum of divisors $[\text{div}(r)]$ of rational functions r on $(k+1)$ -dimensional subvarieties W . Intersecting with D replaces each such W -divisor by its trace under the section s_W of $\mathcal{O}_X(D)|_W$, and the claim is that this trace is again a boundary. On the surface W there are now two rational functions in play, the given r and the local equation of D , and the argument turns on the symmetry between them: the order of r measured along the zeros of D matches the order of the equation of D measured along the zeros of r , up to a principal divisor. That symmetry is a statement about a two-dimensional surface, and it is proved by pushing everything down to a complete curve, where a principal divisor has degree zero. Three inputs carry the argument: this degree-zero fact, proposition 1.2.2; the reciprocity lemma on the auxiliary curve that turns it into the two-function symmetry; and the localization sequence theorem 1.4.1 reducing the boundary strata to lower dimension. The degree-zero fact is why the argument could not be given in section 1.4 before the divisor machinery was in place.

Proof:

By linearity it suffices to treat a single generator of $\text{Rat}_k(X)$, so let $W \subseteq X$ be a $(k+1)$ -dimensional subvariety and $r \in k(W)^*$, and set $\alpha = [\text{div}(r)]$. The hypothesis that no component of α lies in $|D|$ is used only to make each $D \cdot [V]$ fall under the first clause of definition 2.2.1; the reduction below produces $D \cdot \alpha$ as the divisor of a rational function, and the containment cases are absorbed at the end.

Step 1: Reduction to the surface W . Replacing X by W and D by its restriction $D|_W$ changes neither side: the components V of $[\text{div}(r)]$ are codimension-one subvarieties of W , the section s_V used to form $D \cdot [V]$ is the restriction to V of the section s_W of $\mathcal{O}_X(D)|_W$, and the inclusion $W \hookrightarrow X$ carries $A_{k-1}(|D|_W) \rightarrow A_{k-1}(|D|)$ compatibly. It therefore suffices to prove the statement with $X = W$, so that D is a Cartier divisor on the $(k+1)$ -dimensional variety W , $r \in k(W)^*$, and $D \cdot [\text{div}(r)]$ is to be shown rationally equivalent to zero in $A_{k-1}(W)$. Restrict further to the dense open $W^\circ \subseteq W$ where D is principal, given by a single rational function $g \in k(W)^*$ with $\mathcal{O}_W(D)|_{W^\circ} = \mathcal{O}_{W^\circ} \cdot g^{-1}$; the codimension-one subvarieties of W meeting W°

are exactly those on which the computation is nontrivial, and those contained in the complement $W \setminus W^\circ$ contribute to both a principal divisor and the divisor of D and are handled by the localization sequence in Step 4. On W° the section s_V of clause (1) is $g|_V$, so

$$D \cdot [\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r) [\operatorname{div}_V(g|_V)]$$

the outer sum over the codimension-one subvarieties $V \subseteq W$ with r having a zero or pole along V , and the inner divisor formed on each such V .

Step 2: The two-function symmetry. The right-hand side is not visibly symmetric in r and g , but its symmetrization is the object that vanishes. Consider the two iterated sums

$$\Phi(r, g) = \sum_V \operatorname{ord}_V(r) [\operatorname{div}_V(g|_V)], \quad \Phi(g, r) = \sum_V \operatorname{ord}_V(g) [\operatorname{div}_V(r|_V)]$$

both $(k-1)$ -cycles on W , where in each the outer sum runs over codimension-one subvarieties $V \subseteq W$ and the inner divisor is formed on V of the restricted function. The claim of this step is the tame symmetry

$$\Phi(r, g) - \Phi(g, r) \sim 0 \quad \text{in } A_{k-1}(W)$$

Granting it, the proof is nearly complete: $\Phi(g, r) = \sum_V \operatorname{ord}_V(g) [\operatorname{div}_V(r|_V)]$ is a $\mathbb{Z}$ -linear combination of divisors of the rational functions $r|_V$ on the k -dimensional subvarieties V , hence lies in $\operatorname{Rat}_{k-1}(W)$ by definition 1.1.3; combined with the symmetry, $\Phi(r, g) = D \cdot [\operatorname{div}(r)]$ is rationally equivalent to $\Phi(g, r) \sim 0$, which is the assertion.

Step 3a: Localizing the symmetry at a codimension-two locus. The symmetry is a statement whose content is concentrated in codimension two on W , so fix a codimension-two subvariety $Z \subseteq W$, a $(k-1)$ -dimensional variety, and compare the coefficients of $[Z]$ on the two sides. A codimension-one $V \subseteq W$ contributes to the coefficient of $[Z]$ in $\Phi(r, g)$ only when $Z \subseteq V$, that is when Z is a prime divisor of V ; write the coefficient of $[Z]$ in $\Phi(r, g)$ as

$$c_Z(r, g) = \sum_{Z \subseteq V} \operatorname{ord}_V(r) \operatorname{ord}_{Z,V}(g|_V)$$

the sum over codimension-one V with $Z \subseteq V \subseteq W$, and $\operatorname{ord}_{Z,V}$ the order along Z computed on V . The task is to show $c_Z(r, g) = c_Z(g, r)$ modulo the coefficient of $[Z]$ in a principal divisor, and this is a computation on the two-dimensional local geometry of W along Z . Passing to the local ring $\mathcal{O}_{W,Z}$, a two-dimensional local domain with fraction field $L = k(W)$ and residue field $K = k(Z)$, the codimension-one V containing Z correspond to the height-one primes $\mathfrak{p} \subseteq \mathcal{O}_{W,Z}$; each carries a discrete valuation $\operatorname{ord}_{\mathfrak{p}} = \operatorname{ord}_V$ on L , and the residue field $k(\mathfrak{p})$ of $\mathfrak{p}$ is the function field $k(V)$ of the divisor V . In these terms

$$c_Z(r, g) = \sum_{\mathfrak{p}} \operatorname{ord}_{\mathfrak{p}}(r) \operatorname{ord}_Z(g \bmod \mathfrak{p})$$

where $g \bmod \mathfrak{p}$ is the image of g in $k(\mathfrak{p}) = k(V)^*$ and ord_Z on $k(V)$ is the order along Z regarded as a prime divisor of V . The two factors play asymmetric roles: $\operatorname{ord}_{\mathfrak{p}}(r)$ is the transverse order of r along the entire divisor V , a single integer, while $\operatorname{ord}_Z(g \bmod \mathfrak{p})$ is the order of the residue $g|_V$ along the codimension-one subvariety $Z \subseteq V$. Swapping r and g interchanges these two roles rather than permuting equal factors, so $c_Z(r, g) = c_Z(g, r)$ is a genuine reciprocity between a transverse divisorial order and a residual order, not the commutativity of a product.

Step 3b: The exceptional curve and the tame symbol. The two integers in each term of $c_Z(r, g)$, the transverse order $\text{ord}_{\mathfrak{p}}(r)$ and the residual order $\text{ord}_Z(g \bmod \mathfrak{p})$, are the two ends of a single valuation datum, and completeness enters by resolving the two-dimensional geometry of $\mathcal{O}_{W,Z}$ into a curve. Choose a proper birational morphism $\pi: S \rightarrow \text{Spec } \mathcal{O}_{W,Z}$ with S a regular two-dimensional scheme (a resolution of the localized surface, by normalization and blow-up), and let $E = \pi^{-1}(\mathfrak{m})$ be its exceptional fibre over the closed point, a complete curve over K whose integral components are $E_1, \dots, E_m$, each a complete curve over K with function field $k(E_i)$. The coefficient $c_Z(r, g)$ is invariant under π : since π is proper and birational, proposition 1.2.2 identifies the coefficient of $[Z]$ downstairs with the coefficient of the exceptional fibre upstairs, and among the prime divisors of S only the exceptional E_i map into Z , so the strict transforms of the V drop out and the coefficient is carried by $E_1, \dots, E_m$ alone. Along each exceptional component E_i the function r has a transverse order $a_i = \text{ord}_{E_i}(r)$ and g a transverse order $b_i = \text{ord}_{E_i}(g)$, both single integers, and once these orders are cleared r and g restrict to rational functions on E_i ; their zeros and poles on E_i are exactly the points where the strict transforms $\tilde{V}$ meet E_i . Attach to E_i the *tame symbol* of r and g , the residue element

$$\tau_i(r, g) = (-1)^{a_i b_i} \left(\frac{r^{b_i}}{g^{a_i}} \right) \Big|_{E_i} \in k(E_i)^*$$

which is a well-defined nonzero rational function on E_i because r^{b_i}/g^{a_i} has order zero along E_i . Reading off its order at a closed point $P \in E_i$ gives $\text{ord}_P(\tau_i(r, g)) = b_i \text{ord}_P(r|_{E_i}) - a_i \text{ord}_P(g|_{E_i})$. The contribution of (i, P) to $c_Z(r, g)$ is the transverse order of r against the residual order of g , namely $a_i \text{ord}_P(g|_{E_i})$, and its contribution to $c_Z(g, r)$ is $b_i \text{ord}_P(r|_{E_i})$; their difference is thus $-\text{ord}_P(\tau_i(r, g))$. Summing over S and using the π -invariance above, the two coefficients organize as

$$c_Z(r, g) - c_Z(g, r) = - \sum_{i=1}^m \sum_{P \in E_i} [k(P) : K] \text{ord}_P(\tau_i(r, g))$$

the double sum over the components E_i of the exceptional curve and their closed points P . The pairing is asymmetric: the summand pairs the transverse order of one function along E_i against the residual order of the other at P , and swapping r and g replaces $\tau_i(r, g)$ by $\tau_i(r, g)^{-1}$ up to sign, so the difference is the sum of the point orders of the tame symbols, not a rearrangement of a symmetric product $\text{ord}_P(r) \text{ord}_P(g)$.

Step 3c: Assembling by degree zero. Each tame symbol $\tau_i(r, g)$ is a rational function on the complete curve E_i over K , so its divisor $\text{div}_{E_i}(\tau_i(r, g))$ is a principal divisor on a complete curve. By proposition 1.2.2(1) applied to the proper structure map $E_i \rightarrow \text{Spec } K$, its degree over K vanishes: $\deg_K \text{div}_{E_i}(\tau_i(r, g)) = \sum_{P \in E_i} [k(P) : K] \text{ord}_P(\tau_i(r, g)) = 0$ for every i . Summing over the finitely many components,

$$c_Z(r, g) - c_Z(g, r) = - \sum_{i=1}^m \deg_K \text{div}_{E_i}(\tau_i(r, g)) = 0$$

so $c_Z(r, g) = c_Z(g, r)$ for every codimension-two Z . This is the reciprocity that a naive term-by-term rearrangement would miss: it pairs the transverse order of one function against the residual order of the other, and its vanishing rests on completeness of each exceptional component E_i through the degree-zero property alone, with no external tame-symbol machinery imported. Consequently $\Phi(r, g) = \Phi(g, r)$ coefficient by coefficient on the locus where D is principal, any residual difference being a principal divisor supported off W° . This is the symmetry claimed in Step 2, as needed.

Step 4: The boundary components. It remains to account for the codimension-one subvarieties $V \subseteq W \setminus W^\circ$, where D was not represented by the single function g . Let $U = W^\circ$ and $Y = W \setminus W^\circ$ be its closed complement, of codimension at least one in W . The localization sequence (theorem 1.4.1) for $Y \subseteq W$,

$$A_{k-1}(Y) \rightarrow A_{k-1}(W) \rightarrow A_{k-1}(U) \rightarrow 0$$

is right exact, and Steps 1 through 3 established $D \cdot [\text{div}(r)] \sim \Phi(g, r) \sim 0$ after restriction to U , that is, the class of $D \cdot [\text{div}(r)]$ dies in $A_{k-1}(U)$. By exactness its class in $A_{k-1}(W)$ comes from $A_{k-1}(Y)$; but Y has dimension at most k , and the restriction $D|_Y$ is again a Cartier divisor, so the boundary class is computed by the same recipe on the lower-dimensional Y . To pin the inductive object down, decompose $Y = \bigcup_j Y_j$ into irreducible components, each of dimension at most k , with inclusions $\iota_j: Y_j \hookrightarrow W$. The class in $A_{k-1}(Y)$ produced by the boundary of the localization sequence is represented by $\sum_j (\iota_j)_* (D|_{Y_j} \cdot [\text{div}(r_j)])$, where $r_j := r|_{Y_j} \in k(Y_j)^*$ is the restriction of r to the generic point of Y_j , a nonzero rational function since the zero and pole components of r were split off in Steps 1 through 3, and $[\text{div}(r_j)]$ is the principal cycle it cuts on the variety Y_j of dimension at most k . Each summand is thus $D|_{Y_j}$ intersected with a principal cycle on Y_j , the exact object to which the proposition applies one dimension lower. A Noetherian induction on $\dim W$ closes the argument: the base case $\dim W = 0$ is vacuous, and for the inductive step each summand $D|_{Y_j} \cdot [\text{div}(r_j)]$ is rationally trivial on Y_j by the inductive hypothesis applied to the Cartier divisor $D|_{Y_j}$ and the principal cycle $[\text{div}(r_j)]$ on the lower-dimensional Y_j ; pushing forward along ι_j preserves rational equivalence, so the whole boundary class in $A_{k-1}(Y)$ vanishes. Hence $D \cdot [\text{div}(r)] \sim 0$ in $A_{k-1}(W)$, and pushing forward along $W \hookrightarrow X$ gives $D \cdot [\text{div}(r)] \sim 0$ in $A_{k-1}(|D|)$.

Step 5: Well-definedness of $D \cdot a$. Fix a class $a \in A_k(X)$. Every rational-equivalence class has a representative cycle none of whose finitely many components lies in $|D|$, since a component inside $|D|$ can be replaced by a rationally equivalent cycle whose components meet $|D|$ properly; on such a representative every $D \cdot [V]$ falls under the first clause of definition 2.2.1. Let $\alpha \sim \alpha'$ be two such representatives, so their difference lies in $\text{Rat}_k(X)$, a finite $\mathbb{Z}$ -combination of generators $[\text{div}(r)]$ with $r \in k(W)^*$ on $(k+1)$ -dimensional W . Steps 1 through 4 give $D \cdot [\text{div}(r)] \sim 0$ in $A_{k-1}(|D|)$ for each generator, so summing gives $D \cdot (\alpha - \alpha') \sim 0$, and $D \cdot a := [D \cdot \alpha]$ is independent of the chosen representative. Additivity in a is immediate from additivity of $D \cdot -$ on cycles, so $D \cdot -: A_k(X) \rightarrow A_{k-1}(|D|)$ is a homomorphism, completing the proof.

The proposition is the license to speak of $D \cdot a$ for a class a , and with it the intersection with a divisor becomes an operator on the Chow group. Read backward, the argument explains why the well-definedness was out of reach in section 1.4: the entire content is the tame symmetry of Step 2, and that symmetry rests on the degree-zero property of a principal divisor on a complete curve, a fact about the divisor calculus that section did not yet have. The specialization map used there to invert affine-bundle pullback was built from exactly the operation $\text{div}(t) \cdot -$ for the coordinate t , and its descent to Chow groups is this proposition applied to $D = \{t = 0\}$; the forward reference in that proof pointed here.

Two features of the argument recur and are worth isolating. The reduction of Step 1, replacing X by the $(k+1)$ -dimensional W on which r lives, is the standard move that turns a statement about intersecting a principal cycle into a statement about a single surface with two functions on it, and it will reappear when commutativity is proved in section 2.3. The symmetry of Step 3, that pairing the zeros of r against g equals pairing the zeros of g against r , is the algebraic form of Weil reciprocity, here proved from the ground up rather than cited: the tame symbol is constructed by hand along

each component of the exceptional curve, and its reciprocity is reduced to the vanishing degree of a principal divisor on that complete curve, so no external tame-symbol machinery is imported. The same reciprocity, promoted one dimension higher, is what makes $D \cdot D' = D' \cdot D$, and the two proofs share their engine.

The operator now available, $D \cdot - : A_k(X) \rightarrow A_{k-1}(|D|)$, invites the computation that motivated the whole construction. Intersecting repeatedly with a hyperplane on projective space should descend through the dimensions and terminate in a point, recovering the degree; the next example carries this out and previews the Chern-class computation of section 2.4.

Example 2.2.5: Iterated Hyperplane Sections on Projective Space

Let $X = \mathbb{P}^n$ and $H = \{x_0 = 0\}$ a hyperplane, so $\mathcal{O}_{\mathbb{P}^n}(H) = \mathcal{O}_{\mathbb{P}^n}(1)$ and $|H| \cong \mathbb{P}^{n-1}$. By theorem 1.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}h_k$ with h_k the class of a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$. To compute $H \cdot h_k$, represent h_k by a linear $\mathbb{P}^k$ meeting H transversally, that is by an $L \cong \mathbb{P}^k$ not contained in H ; then $L \cap H$ is a linear $\mathbb{P}^{k-1}$, and the restricted section x_0/x_j on L has its zero divisor equal to $L \cap H$ with multiplicity one. Hence

$$H \cdot h_k = [L \cap H] = h_{k-1} \in A_{k-1}(\mathbb{P}^{n-1})$$

pushed into $\mathbb{P}^n$, the class of a linear $\mathbb{P}^{k-1}$. Intersecting n times starting from the fundamental class $h_n = [\mathbb{P}^n]$,

$$\underbrace{H \cdot H \cdots H}_k \cdot [\mathbb{P}^n] = h_{n-k}$$

and in particular $H^n \cdot [\mathbb{P}^n] = h_0 = [\text{pt}]$, a single reduced point of degree one. The iterated intersection walks down the tower of linear subspaces one step at a time, and its terminal value is the class of a point, whose degree 1 is the degree of $\mathbb{P}^n$ under the embedding $\mathcal{O}(1)$. The choice of a transversal representative L at each stage is legitimated precisely by proposition 2.2.4: the answer h_{k-1} does not depend on which $\mathbb{P}^k$ in the class h_k is used, so long as it meets H properly.

The computation is the prototype of every enumerative count the theory produces. Reinterpreted through the first Chern class in section 2.4, it reads $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = h_{n-k}$, and its terminal instance $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\text{pt}]$ is the statement that the top self-intersection of the hyperplane class on $\mathbb{P}^n$ is one, the algebraic form of the assertion that n general hyperplanes meet in a single point. That the intermediate steps are independent of the representatives chosen is the payoff of this section, and it is what turns a naive count of intersection points into an invariant.

Exercise 2.2.1

1. On $X = \mathbb{P}^2$ let $D = \{x_0x_1 = 0\}$ be the union of two lines, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(D) = \mathcal{O}_{\mathbb{P}^2}(2)$, and let V be the conic $\{x_2^2 = x_0x_1\}$. Compute $D \cdot [V] \in A_0(V \cap |D|)$ explicitly as a sum of points with multiplicities, and verify that its degree equals 4. Explain how the multiplicities distribute among the intersection points.
2. Let $V \subseteq \mathbb{P}^2$ be a smooth curve of degree d and H a line meeting V transversally in d distinct points. Using definition 2.2.1, show that $H \cdot [V]$ is the sum of those d points each with multiplicity one, and deduce $\deg(H \cdot [V]) = d$. Then explain why the value $\deg(H \cdot [V]) = d$ holds for every line H , transversal or not, without recomputing, citing proposition 2.2.4.
3. Let $D = \text{div}(f)$ be a principal Cartier divisor on a variety X , so $\mathcal{O}_X(D) \cong \mathcal{O}_X$ is trivial. Show that $D \cdot a = 0$ in $A_{k-1}(X)$ for every class $a \in A_k(X)$. (Push $D \cdot a$ forward from $|D|$ into X ;

the point is that a global trivialization of $\mathcal{O}_X(D)$ makes the restricted section extend to all of V .)

4. On $\mathbb{P}^3$ let H be a hyperplane and let V be a smooth cubic surface, regarded through its inclusion as a subvariety, with the induced very ample bundle $\mathcal{O}_V(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_V$. Compute $H \cdot [V]$ as a class in $A_1(V \cap H)$, identify $V \cap H$ geometrically, and compute the degree of the further intersection $H \cdot (H \cdot [V])$.
5. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$ and let $D = f_1 = \{a\} \times \mathbb{P}^1$ be a ruling of the first kind, with $\mathcal{O}_Q(D) = \text{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(1)$. Compute $D \cdot [f_1]$ and $D \cdot [f_2]$, where $f_2 = \mathbb{P}^1 \times \{b\}$ is a ruling of the second kind, and interpret the results as the self-intersection f_1^2 and the cross-intersection $f_1 \cdot f_2$. Which case uses the second clause of definition 2.2.1?
6. Give an explicit example on $X = \mathbb{P}^1$ of a Cartier divisor D and two rationally equivalent zero-cycles $\alpha \sim \alpha'$ for which the naive formula would assign different answers if one ignored the requirement that components avoid $|D|$, and explain how definition 2.2.1 and proposition 2.2.4 together resolve the apparent discrepancy. (Here $k = 0$, so $D \cdot \alpha$ lands in $A_{-1} = 0$; use this to frame what the proposition is and is not asserting in the lowest dimension.)

2.3 Commutativity and the Projection Formula

Intersection with a divisor, made precise in section 2.2, is an operation $D \cdot -$ that lowers dimension by one and, by proposition 2.2.4, descends to rational equivalence. Nothing so far says these operations may be applied in either order. Given two Cartier divisors D and D' on X and a k -cycle α , one can form $D \cdot (D' \cdot \alpha)$ and $D' \cdot (D \cdot \alpha)$, each a class in $A_{k-2}(X)$; that they agree is the single algebraic fact on which the entire intersection calculus rests. Its proof does not follow from the definitions by formal manipulation, because the two orders of intersection produce different cycles before passing to rational equivalence. The content is that the difference is always a boundary, and the reason it is a boundary is the vanishing of the degree of a principal divisor on a complete curve, established back in proposition 1.2.2.

This is also the point at which a debt from chapter 1 comes due. The proof of affine-bundle invariance (theorem 1.4.6) deferred one identity, the divisor symmetry (1.3), on the grounds that the tools were not yet in place; that identity is exactly the commutativity proved here. This section discharges it in full and then draws the two functorial consequences, compatibility of $D \cdot -$ with proper pushforward and with flat pullback, that make the operation part of the same covariant/contravariant package as pushforward and pullback themselves.

The obstruction to commutativity lives on a surface, and the mechanism that removes it is a symmetry between two rational functions. Before the theorem, the mechanism deserves to be isolated as a lemma. Let S be a complete variety of dimension two and let $f, g \in k(S)^*$ be nonzero rational functions. Each has a divisor $[\text{div}(f)] = \sum_W \text{ord}_W(f) [W]$, the sum over the codimension-one subvarieties (the curves) $W \subseteq S$, and on each such curve W the other function g restricts, wherever it is neither zero nor a pole identically along W , to a rational function $g|_W$ with its own divisor of points. The two-function symmetry lemma compares the two zero-cycles obtained by cutting first with f and then with g , against the same two cuts in the opposite order.

The statement needs the order of one function measured against the divisor of another. For a curve $W \subseteq S$ with $\text{ord}_W(g) = 0$, so that g is a unit at the generic point of W and restricts to a nonzero

rational function $g|_W \in k(W)^*$, write $[\operatorname{div}_W(g)] = \sum_{P \in W} \operatorname{ord}_P(g|_W) [P]$ for the divisor of that restriction, a zero-cycle on W and hence on S . The lemma is a reciprocity between the iterated cuts $\sum_W \operatorname{ord}_W(f) [\operatorname{div}_W(g)]$ and $\sum_W \operatorname{ord}_W(g) [\operatorname{div}_W(f)]$.

Lemma 2.3.1: Two-Function Reciprocity on a Surface

Let S be a complete variety of dimension two and let $f, g \in k(S)^*$. Then

$$\sum_W \operatorname{ord}_W(f) [\operatorname{div}_W(g)] \sim \sum_W \operatorname{ord}_W(g) [\operatorname{div}_W(f)] \quad \text{in } A_0(S)$$

where each sum runs over the curves $W \subseteq S$ on which the relevant restriction is defined, and the divisors $[\operatorname{div}_W(g)]$, $[\operatorname{div}_W(f)]$ are the zero-cycles of the restricted functions. In fact both sides have the same degree, and their difference is a sum of principal divisors of rational functions on the curves $W \subseteq S$, hence rationally trivial.

The lemma is the two-dimensional heart of commutativity, and its proof is the one place where the degree-zero property of principal divisors on complete curves does the decisive work. The strategy is to reduce the equality of two zero-cycles to an equality of degrees along each curve, and then to recognize each such degree as the value at a point of the *tame symbol*, a bimultiplicative pairing on pairs of rational functions whose reciprocity across a complete curve is precisely the statement that a principal divisor has degree zero.

Proof:

Step 1: the tame symbol at a curve. Fix a curve $W \subseteq S$ and let $A = \mathcal{O}_{S,W}$ be the local ring of S at the generic point of W . Because S is a variety and W has codimension one, A is a one-dimensional local domain with fraction field $k(S)$ and residue field $k(W)$; its normalization is a discrete valuation ring, and after replacing S by its normalization along W (which changes neither side of the asserted identity, both being computed by orders and residues that are birational invariants) one may assume A is itself a discrete valuation ring with valuation $v_W = \operatorname{ord}_W$ and uniformizer π . Define the *tame symbol* of f, g at W to be the residue class

$$\partial_W(f, g) = (-1)^{v_W(f) v_W(g)} \frac{f^{v_W(g)}}{g^{v_W(f)}} \Big|_W \in k(W)^*$$

the restriction to W of the indicated unit, which lies in A and is a unit there because its valuation

$$v_W \left((-1)^{v_W(f) v_W(g)} f^{v_W(g)} g^{-v_W(f)} \right) = v_W(g) v_W(f) - v_W(f) v_W(g) = 0$$

vanishes. The construction is bimultiplicative in f and g and antisymmetric, $\partial_W(g, f) = \partial_W(f, g)^{-1}$, both directly from the formula. The point of the sign is that this antisymmetry holds on the nose, not just up to squares.

Step 2: the tame symbol computes the difference of iterated divisors, pointwise. Fix a closed point $P \in S$ and a curve W through P . When $\operatorname{ord}_W(g) = 0$ the restriction $g|_W$ is a nonzero function on W and the coefficient of $[P]$ in $\operatorname{ord}_W(f) [\operatorname{div}_W(g)]$ is $\operatorname{ord}_W(f) \operatorname{ord}_P(g|_W)$; symmetrically for the other cut. Comparing the two iterated cuts at P amounts to comparing, over all curves $W \ni P$,

$$\sum_{W \ni P} \operatorname{ord}_W(f) \operatorname{ord}_P(g|_W) \quad \text{against} \quad \sum_{W \ni P} \operatorname{ord}_W(g) \operatorname{ord}_P(f|_W)$$

The tame symbol reconciles them, but the reconciliation is delicate at the curves where both functions have nonzero order, and getting it right is the whole content of the step. For a single curve W with $\text{ord}_W(f) = a$ and $\text{ord}_W(g) = b$, write $f = \pi^a u$ and $g = \pi^b w$ with u, w units at W . The tame symbol of *Step 1* then simplifies to

$$\partial_W(f, g) = (-1)^{ab} \bar{u}^b \bar{w}^{-a} \in k(W)^*$$

the powers of π cancelling, so its order at P is always

$$\text{ord}_P(\partial_W(f, g)) = b \text{ord}_P(\bar{u}) - a \text{ord}_P(\bar{w}) \quad (2.1)$$

regardless of whether a or b vanishes. The iterated cuts, on the other hand, see only the restricted functions, and $f|_W = \bar{u}$ is defined precisely when $a = 0$ while $g|_W = \bar{w}$ is defined precisely when $b = 0$. Compare the two curve-by-curve.

If $a = 0$ and $b \neq 0$: then $\text{ord}_W(g) [\text{div}_W(f)]$ contributes $b \text{ord}_P(\bar{u})$ at P while $\text{ord}_W(f) [\text{div}_W(g)] = 0$, so the iterated-cut difference is $b \text{ord}_P(\bar{u})$, which is exactly (2.1) since $a = 0$ kills the second term. If $a \neq 0$ and $b = 0$: symmetrically the difference is $-a \text{ord}_P(\bar{w})$, again (2.1). If $a = b = 0$: both restrictions are units, both sides vanish. In all three of these *non-mixed* cases,

$$\left(\text{ord}_W(g) \text{ord}_P(f|_W) - \text{ord}_W(f) \text{ord}_P(g|_W) \right) = \text{ord}_P(\partial_W(f, g))$$

the left side being the contribution of W to the iterated-cut difference at P , where a term whose restricted function is undefined is read as zero because its coefficient $\text{ord}_W(g)$ or $\text{ord}_W(f)$ then vanishes. The remaining case is the *mixed* one, $a \neq 0$ and $b \neq 0$: here *neither* $f|_W$ nor $g|_W$ is defined, so W contributes nothing to either iterated cut, yet $\partial_W(f, g) = (-1)^{ab} \bar{u}^b \bar{w}^{-a}$ is a genuine nonzero rational function on W whose divisor is generally nonzero. Thus on a mixed curve the iterated-cut difference and $\text{ord}_P(\partial_W)$ do *not* agree; the tame symbol overcounts by exactly the mixed contribution. Summing the non-mixed agreement over the curves through P ,

$$\left(\sum_W \text{ord}_W(g) [\text{div}_W(f)] - \sum_W \text{ord}_W(f) [\text{div}_W(g)] \right)_P = \sum_{\substack{W \ni P \\ \text{non-mixed}}} \text{ord}_P(\partial_W(f, g)) \quad (2.2)$$

the sum on the right running only over curves $W \ni P$ on which at least one of f, g has order zero, and the left side denoting the coefficient of $[P]$ in the difference of the two iterated cuts. The mixed curves are deliberately excluded from the right side; they carry no iterated-cut contribution, and Step 3 will show their tame symbols are individually harmless.

Step 3: reciprocity along each curve kills the total. Fix a curve $W \subseteq S$. The tame symbol $\partial_W(f, g)$ is a nonzero rational function on the complete curve W (it is complete because S is complete and W is closed in S). Its divisor $[\text{div}(\partial_W(f, g))] = \sum_P \text{ord}_P(\partial_W(f, g)) [P]$ is a principal divisor on W , so by proposition 1.2.2(1) applied to the structure map $W \rightarrow \text{Spec } \bar{k}$, which is proper surjective with target of strictly smaller dimension, its pushforward vanishes: $\deg[\text{div}(\partial_W(f, g))] = 0$. This holds for *every* curve W , mixed or not. Summing (2.2) over all closed points P and reorganizing the double sum by curves,

$$\deg \left(\sum_W \text{ord}_W(g) [\text{div}_W(f)] - \sum_W \text{ord}_W(f) [\text{div}_W(g)] \right) = \sum_{\text{non-mixed } W} \deg[\text{div}(\partial_W(f, g))] = 0$$

each summand vanishing individually. Thus the difference of the two iterated zero-cycles has degree zero on S .

Step 4: from degree zero to rational equivalence. A zero-cycle of degree zero on the complete variety S need not be rationally equivalent to zero in general, but the difference here is more than a degree-zero cycle: (2.2), summed over all closed points P and reorganized by curves, exhibits it as a sum of principal divisors,

$$\sum_W \text{ord}_W(g) [\text{div}_W(f)] - \sum_W \text{ord}_W(f) [\text{div}_W(g)] = \sum_{\text{non-mixed } W} [\text{div}(\partial_W(f, g))]$$

the right side running over the curves on which one of f, g has order zero, each term the divisor of the rational function $\partial_W(f, g)$ on the complete curve $W \subseteq S$. Each $[\text{div}(\partial_W(f, g))]$ is by definition rationally equivalent to zero on W , hence on S under the proper pushforward along $W \hookrightarrow S$, which preserves rational equivalence by theorem 1.2.3. Therefore the difference is a finite sum of cycles each ~ 0 , so it is itself ~ 0 in $A_0(S)$, which is the asserted reciprocity, as we sought to show.

The lemma is the algebraic incarnation of a classical reciprocity, but the mechanism it uses is elementary: on each curve W the tame symbol $\partial_W(f, g)$ is a single rational function, its divisor on the complete curve W is principal and so rationally trivial, and the difference of the two iterated cuts is a finite sum of such divisors. No appeal to Weil reciprocity or to the point-level tame symbol $\prod_P \partial_P(f, g) = 1$ as an external input is needed, and indeed none is available in this book; the entire force of the lemma comes from the single fact that a rational function on a complete curve has as many zeros as poles, counted with multiplicity and residue degree, applied one curve at a time. That fact is proposition 1.2.2(1), proved in chapter 1 by the model computation on $\mathbb{P}^1$. The reader should notice that the surface S never needed to be smooth; normalizing along each curve reduced every local computation to a discrete valuation ring, which is all the tame symbol requires.

With the reciprocity lemma in hand, commutativity is a matter of reducing the general assertion to the surface case the lemma governs. The reduction is entirely formal, following the pattern used throughout chapter 1: the operation $D \cdot -$ is a homomorphism, so it suffices to check the identity on the class of a single subvariety, and the intersection with D and D' is supported where the two divisors meet the subvariety, so one may replace the ambient scheme by the subvariety itself.

Theorem 2.3.2: Commutativity of Divisor Intersection

Let D and D' be Cartier divisors on an algebraic scheme X , and let $\alpha \in A_k(X)$. Then

$$D \cdot (D' \cdot \alpha) = D' \cdot (D \cdot \alpha) \quad \text{in } A_{k-2}(X)$$

More precisely, both sides are represented by well-defined classes in $A_{k-2}(|D| \cap |D'| \cap \text{supp} \alpha)$, and they coincide there. This identity is the divisor symmetry (1.3) deferred in the proof of theorem 1.4.6.

Proof:

Step 1: reduction to a variety. Both $D \cdot -$ and $D' \cdot -$ are homomorphisms $A_k \rightarrow A_{k-1}$ by definition 2.2.1 and proposition 2.2.4, so both iterated operations are homomorphisms $A_k(X) \rightarrow A_{k-2}(X)$. It therefore suffices to verify the identity on a generator $\alpha = [V]$, the class of a k -dimensional subvariety $V \subseteq X$. Restricting the line bundles $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ and their rational sections to V , the entire computation of $D \cdot (D' \cdot [V])$ and $D' \cdot (D \cdot [V])$ takes place

on V using the restricted divisors $D|_V$ and $D'|_V$. Replacing X by V , one is reduced to proving

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X])$$

for X itself a k -dimensional variety and D, D' Cartier divisors on it, with the common value a class in $A_{k-2}(|D| \cap |D'|)$.

Step 2: choosing sections and unwinding the two orders. Let s and s' be the rational sections of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ cutting out D and D' , so that on the variety X ,

$$D \cdot [X] = [\operatorname{div}(s)] = \sum_W \operatorname{ord}_W(s) [W], \quad D' \cdot [X] = [\operatorname{div}(s')] = \sum_{W'} \operatorname{ord}_{W'}(s') [W']$$

the sums over the codimension-one subvarieties $W, W' \subseteq X$. To compute the double intersection one then intersects each $[W']$ with D and each $[W]$ with D' . On a codimension-one subvariety W not contained in $|D'|$, the section s' restricts to a rational section $s'|_W$ of $\mathcal{O}_X(D')|_W$, whose divisor is $[\operatorname{div}_W(s')]$, and by definition 2.2.1,

$$D' \cdot (D \cdot [X]) = \sum_W \operatorname{ord}_W(s) [\operatorname{div}_W(s')], \quad D \cdot (D' \cdot [X]) = \sum_{W'} \operatorname{ord}_{W'}(s') [\operatorname{div}_{W'}(s)]$$

as cycles supported on $|D| \cap |D'|$. The two expressions are the two iterated cuts of the reciprocity lemma, with the roles of the two divisors interchanged.

Step 3: the difference as a sum of tame symbols. The difference

$$\Delta = \sum_{W'} \operatorname{ord}_{W'}(s') [\operatorname{div}_{W'}(s)] - \sum_W \operatorname{ord}_W(s) [\operatorname{div}_W(s')]$$

is a $(k-2)$ -cycle supported on $|D| \cap |D'|$, and the claim is $\Delta \sim 0$ in $A_{k-2}(|D| \cap |D'|)$. The strategy is to write Δ as a $\mathbb{Z}$ -combination of divisors of rational functions on $(k-1)$ -dimensional subvarieties of X , using the tame symbol of lemma 2.3.1. For a $(k-1)$ -dimensional subvariety $W \subseteq X$ on which one of the two sections has nonzero order, choose a common local trivialization of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ near the generic point of W (a line bundle is locally trivial, and one open set trivializes both), turning s, s' into rational functions in the local ring $\mathcal{O}_{X,W}$, a two-dimensional local domain. The tame symbol

$$\partial_W(s, s') = (-1)^{\operatorname{ord}_W(s) \operatorname{ord}_W(s')} \frac{s^{\operatorname{ord}_W(s')}}{s'^{\operatorname{ord}_W(s)}} \Big|_W \in k(W)^*$$

is then a nonzero rational function on the $(k-1)$ -dimensional variety W , defined exactly as in the lemma. Localizing X along W removes the trivialization ambiguity: two trivializations differ by a unit, which does not change $\partial_W(s, s')$ up to a rational function that is regular and nonvanishing at the generic point of W , so its divisor on W is well defined.

Step 4: applying reciprocity, one codimension-two point at a time. Fix a $(k-2)$ -dimensional subvariety $Z \subseteq |D| \cap |D'|$ and compute the coefficient of $[Z]$ on both sides of the asserted identity $\Delta = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(s, s'))]$, the sum running over $(k-1)$ -dimensional W on which one of s, s' has order zero. That coefficient is determined at the generic point ζ of Z , where $\mathcal{O}_{X,\zeta}$ has dimension two; the two-dimensional local situation there is governed by lemma 2.3.1, whose (2.2) reads, at ζ ,

$$\left(\sum_W \operatorname{ord}_W(s') [\operatorname{div}_W(s)] - \sum_W \operatorname{ord}_W(s) [\operatorname{div}_W(s')] \right)_{\zeta} = \sum_{\substack{W \ni \zeta \\ \text{non-mixed}}} \operatorname{ord}_{\zeta}(\partial_W(s, s'))$$

the right sum over the $(k-1)$ -dimensional subvarieties W through ζ on which one of s, s' has order zero, exactly as in the lemma; the mixed W , on which both sections have nonzero order, are excluded because they contribute to neither iterated cut. This is the coefficient of $[Z]$ in Δ set equal to the coefficient of $[Z]$ in $\sum_{\text{non-mixed } W} [\text{div}(\partial_W(s, s'))]$. Since the two sides agree at every such Z , they agree as cycles:

$$\Delta = \sum_{\text{non-mixed } W} [\text{div}(\partial_W(s, s'))]$$

Each term $[\text{div}(\partial_W(s, s'))]$ is the divisor of a rational function on the $(k-1)$ -dimensional subvariety W , hence rationally equivalent to zero on W , and its proper pushforward to $|D| \cap |D'|$ along $W \hookrightarrow |D| \cap |D'|$ preserves that equivalence by theorem 1.2.3. As a finite sum of cycles each ~ 0 , therefore $\Delta \sim 0$, so

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X]) \quad \text{in } A_{k-2}(|D| \cap |D'|)$$

Pushing forward to X along the inclusion, the identity holds in $A_{k-2}(X)$ for $\alpha = [X]$, and by Step 1 for every $\alpha \in A_k(X)$. The surface case $X = S$, in which the difference of the two orders is exhibited as a sum of principal cycles, is precisely the deferred divisor symmetry (1.3), with the two rational functions $s|_S$ and $s'|_S$ here playing the roles of g and the coordinate t there, completing the proof.

Commutativity is the property that makes the phrase “intersection number” unambiguous. Without it, the number attached to a chain of divisors would depend on the order in which they are imposed, and no well-defined product on cycles could exist. With it, one may write $D_1 \cdot D_2 \cdots D_r \cdot \alpha$ for an iterated intersection and speak of it without specifying an order, and the class of a complete intersection $D_1 \cap \cdots \cap D_r$ in A_* is symmetric in the defining divisors. The proof also settles the debt from chapter 1: the identity (1.3) that theorem 1.4.6 took on faith is the surface case here, so affine-bundle invariance is now unconditional. The tame symbol was the only tool required beyond the definitions, and its reciprocity was nothing more than the degree-zero property of principal divisors, propagated from curves to surfaces.

A first reading of the theorem should note what it does *not* require. The divisors need not be effective, the sections need not be regular, and X need not be smooth or even normal; the entire argument localizes to one-dimensional local rings, where a divisor is controlled by a single valuation. This is why intersection with a Cartier divisor is available on every algebraic scheme, in contrast to the intersection product of arbitrary cycles, which will demand the deformation-to-the-normal-cone machinery of a later chapter and works cleanly only on smooth varieties.

The second half of the section records how $D \cdot -$ interacts with the two functorialities of the Chow group. Both statements are what one would guess from the topological analogy, in which cap product with a cohomology class is natural for maps: pushforward absorbs the pullback of a divisor, and flat pullback commutes with intersection outright. The proofs run through the same reduction to a single subvariety and the same input, the pushforward of a principal divisor, that powered commutativity.

For a proper morphism $f: X \rightarrow Y$ and a Cartier divisor D on Y , the pullback f^*D is defined whenever no component of X maps into $|D|$, by pulling back the line bundle $\mathcal{O}_Y(D)$ and its rational section; it is the Cartier divisor $\text{div}(f^\#s)$ of the pulled-back section. The projection formula compares intersecting upstairs and pushing down against pushing down and intersecting downstairs.

Proposition 2.3.3: Projection Formula for Divisors

Let D be a Cartier divisor on Y .

1. *Proper pushforward.* If $f: X \rightarrow Y$ is proper and $\alpha \in A_k(X)$ is a cycle no component of whose support maps into $|D|$, so that $f^*D \cdot \alpha$ is defined, then

$$f_*(f^*D \cdot \alpha) = D \cdot f_*\alpha \quad \text{in } A_{k-1}(Y)$$

2. *Flat pullback.* If $g: X' \rightarrow Y$ is flat of relative dimension n and $\beta \in A_k(Y)$, then

$$g^*(D \cdot \beta) = g^*D \cdot g^*\beta \quad \text{in } A_{k+n-1}(X')$$

where g^*D is the pullback Cartier divisor $\text{div}(g^\#s)$.

Proof:

Both statements are linear in α and β , so it suffices to verify each on the class of a subvariety, and both reduce, by the argument that opened the commutativity proof, to a computation with the divisor of a rational function that is settled by proposition 1.2.2 and proposition 1.3.6.

1. *Proper pushforward.* Take $\alpha = [V]$ for a k -dimensional subvariety $V \subseteq X$, and set $W = f(V) \subseteq Y$, so $f_*[V] = \deg(V/W)[W]$ if $\dim W = k$ and $f_*[V] = 0$ if $\dim W < k$, by definition 1.2.1. Restricting to V and W , one may assume $X = V$, $Y = W$, and $f: V \rightarrow W$ proper surjective with V a variety. Let s be the rational section of $\mathcal{O}_Y(D)$ cutting out D ; its pullback $f^\#s$ is a rational section of $f^*\mathcal{O}_Y(D)$ cutting out f^*D , and $f^*D \cdot [V] = [\text{div}(f^\#s)]$ while $D \cdot [W] = [\text{div}(s)]$, both by definition 2.2.1. There are two cases.

If $\dim W < k$, then $f_*[V] = 0$, so the right side $D \cdot f_*[V]$ vanishes. On the left, $[\text{div}(f^\#s)]$ is a $(k-1)$ -cycle on V , and f pushes it into Y ; but $f(V) = W$ has dimension $< k$, so $f_*[\text{div}(f^\#s)]$ is a cycle of dimension $k-1$ supported on the $(< k)$ -dimensional W , and proposition 1.2.2(1) applied to $f: V \rightarrow W$ (proper surjective, target of strictly smaller dimension) gives $f_*[\text{div}(f^\#s)] = 0$. Both sides vanish.

If $\dim W = k$, then $f: V \rightarrow W$ is proper surjective of equal dimension, generically finite of degree $d = \deg(V/W) = [k(V) : k(W)]$. Now $f^\#s$ is a rational section whose local expression, in a trivialization of $\mathcal{O}_Y(D)$ near a curve of W , is a rational function $r \in k(V)^*$ pulled back from $k(W)$, and proposition 1.2.2(2) gives $f_*[\text{div}(f^\#s)] = [\text{div}(N(f^\#s))]$ with N the norm along $k(V)/k(W)$. Because $f^\#s$ is pulled back from s , its norm is $N(f^\#s) = s^d$ up to the local trivialization, so $[\text{div}(N(f^\#s))] = d[\text{div}(s)] = d(D \cdot [W])$. Hence

$$f_*(f^*D \cdot [V]) = f_*[\text{div}(f^\#s)] = d(D \cdot [W]) = D \cdot (d[W]) = D \cdot f_*[V]$$

as required.

2. *Flat pullback.* Take $\beta = [V]$ for a k -dimensional subvariety $V \subseteq Y$. Both sides are supported over $V \cap |D|$, so restricting to V one may assume $Y = V$ is a variety and D a Cartier divisor on it with cutting section s , so $D \cdot [V] = [\text{div}(s)]$. Then $g^*(D \cdot [V]) = g^*[\text{div}(s)]$, and by the flat-pullback divisor formula established in proposition 1.3.7, flat pullback commutes with

taking the divisor of a rational function: $g^*[\operatorname{div}(s)] = [\operatorname{div}(g^\#s)]$ componentwise, with the multiplicities of the components of $g^{-1}(V)$ accounted by flatness. On the other side, g^*D is the Cartier divisor $\operatorname{div}(g^\#s)$ on X' by definition, and $g^*[V] = [g^{-1}(V)] = \sum_j m_j [X'_j]$ over the components X'_j of the preimage with their flat multiplicities m_j by proposition 1.3.5. Intersecting,

$$g^*D \cdot g^*[V] = g^*D \cdot \sum_j m_j [X'_j] = \sum_j m_j [\operatorname{div}((g^\#s)|_{X'_j})] = [\operatorname{div}(g^\#s)] = g^*(D \cdot [V])$$

the middle equality being definition 2.2.1 applied on each component and the last the same componentwise divisor formula. This is the claim; the compatibility of these classes with the base change is proposition 1.3.6, which guarantees the two routes land in the same class of $A_{k+n-1}(X')$. This establishes both parts, completing the proof.

The projection formula is the mechanism by which a computation on a resolved or covering space transfers to the base. In its pushforward form, intersecting the pullback of D upstairs and pushing the answer down is the same as pushing the cycle down and intersecting with D there, which is exactly what one does when computing an intersection number on Y by passing to a modification $X \rightarrow Y$ where the geometry is transparent. The scalar $d = \deg(V/W)$ in the equal-dimensional case is the reason degrees multiply under finite covers, and it is the algebraic source of formulas like $\deg(f^*D) = (\deg f)(\deg D)$ for a finite map of curves. In its flat form, the formula says intersection with a divisor is a local construction, insensitive to flat base change: pulling back the divisor and pulling back the cycle and then intersecting reproduces the pullback of the intersection. Together the two halves make $D \cdot -$ compatible with the covariant/contravariant package (f_*, g^*) of chapter 1, the bimodule structure that all of intersection theory is built to respect.

The abstractions above are best seen at work on the projective plane, where every class is a known multiple of a linear space and the intersection numbers can be read off by hand. The following example verifies commutativity concretely and recovers the intersection number of two lines.

Example 2.3.4: Two Lines on the Projective Plane

Let $X = \mathbb{P}^2$ over $\bar{k}$, with homogeneous coordinates $[x_0 : x_1 : x_2]$. By theorem 1.5.2, $A_2(\mathbb{P}^2) = \mathbb{Z}[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}[L]$ generated by the class of a line L , and $A_0(\mathbb{P}^2) = \mathbb{Z}[P]$ generated by the class of a point. Take the two Cartier divisors $D = \{x_0 = 0\}$ and $D' = \{x_1 = 0\}$, each a line, with $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(D') \cong \mathcal{O}(1)$ and cutting sections $s = x_0$, $s' = x_1$ (rational sections of $\mathcal{O}(1)$ in the trivialization on the chart $U_2 = \{x_2 \neq 0\}$, where the affine coordinates are $u = x_0/x_2$, $v = x_1/x_2$).

First order. Intersecting $[\mathbb{P}^2]$ with D gives $D \cdot [\mathbb{P}^2] = [\operatorname{div}(s)] = [D] = [L]$, the line $\{x_0 = 0\}$. Now intersect with D' . Since $D = \{x_0 = 0\}$ is not contained in $|D'| = \{x_1 = 0\}$, the section $s'|_D = x_1|_D$ is a rational section of $\mathcal{O}(1)|_D$; on the line $D \cong \mathbb{P}^1$ with coordinate $[x_1 : x_2]$ it vanishes to order one at the single point $x_1 = 0$, namely $[0 : 0 : 1]$. Thus

$$D' \cdot (D \cdot [\mathbb{P}^2]) = D' \cdot [L] = [\operatorname{div}_D(s')] = [[0 : 0 : 1]]$$

the class of the point where the two lines meet.

Reverse order. Intersecting $[\mathbb{P}^2]$ with D' first gives $D' \cdot [\mathbb{P}^2] = [\operatorname{div}(s')] = [D'] = [L]$, the line $\{x_1 = 0\}$. Intersecting with D , the section $s|_{D'} = x_0|_{D'}$ on $D' \cong \mathbb{P}^1$ with coordinate $[x_0 : x_2]$ vanishes to order one at $x_0 = 0$, again $[0 : 0 : 1]$, so

$$D \cdot (D' \cdot [\mathbb{P}^2]) = D \cdot [L] = [\operatorname{div}_{D'}(s)] = [[0 : 0 : 1]]$$

The two orders produce the same zero-cycle, the reduced point $[0 : 0 : 1]$, on the nose, which is theorem 2.3.2 made visible: here no rational equivalence was even needed, because the two lines are transverse and the tame symbol $\partial_W(x_0, x_1)$ has no zeros or poles. Taking degrees on the complete $\mathbb{P}^2$,

$$\deg(D \cdot D' \cdot [\mathbb{P}^2]) = 1$$

the intersection number of two distinct lines: they meet in exactly one point. This is the case $c = d = 1$ of Bézout's theorem ($[1]$), recovered from the divisor calculus.

Where reciprocity would enter. Replacing D' by D itself, one must intersect $[L]$ with the same line, which requires moving D within its linear system: since $\mathcal{O}(1)$ has many sections, choose instead $s'' = x_1$, whose divisor $\{x_1 = 0\}$ is linearly equivalent to $\{x_0 = 0\}$, and the previous computation gives $D \cdot D \cdot [\mathbb{P}^2] = [[0 : 0 : 1]]$ of degree 1, so the self-intersection $L^2 = 1$ on $\mathbb{P}^2$. The passage from $s = x_0$ to the linearly equivalent $s'' = x_1$ changes the cutting section by the rational function x_1/x_0 , and that the answer is unchanged is precisely proposition 2.2.4, the well-definedness whose commutativity companion is proved in this section.

The example is the enumerative payoff in miniature. The number 1 is the degree of a zero-cycle, the two lines meet where their defining sections simultaneously vanish, and the independence of the answer from the order of intersection and from the choice of representative in each linear system is exactly what theorem 2.3.2 and proposition 2.2.4 guarantee. Iterating, r general hyperplanes in $\mathbb{P}^n$ intersect in $c_1(\mathcal{O}(1))^r \cap [\mathbb{P}^n] = [\mathbb{P}^{n-r}]$, and taking $r = n$ recovers a single reduced point of degree one, the statement that n general hyperplanes meet in one point. This is the computation the next section packages as the first Chern class.

Exercise 2.3.1

1. On $\mathbb{P}^2$, let $D = \{x_0x_1 = 0\}$ be the union of two coordinate lines (a divisor in the class $2L$) and let $D' = \{x_2 = 0\}$ be a third line. Compute $D \cdot (D' \cdot [\mathbb{P}^2])$ and $D' \cdot (D \cdot [\mathbb{P}^2])$ as zero-cycles, verify they agree, and compute the degree of the product. Confirm the answer is $2 = \deg D \cdot \deg D'$, consistent with Bézout.
2. Let $S = \mathbb{P}^2$ with affine coordinates $u = x_0/x_2$, $v = x_1/x_2$ on the chart $x_2 \neq 0$. For the rational functions $f = u$ and $g = v$, compute the tame symbol $\partial_W(f, g)$ at each coordinate curve $W \in \{u = 0\}, \{v = 0\}$ and at the line at infinity, and verify directly that $\sum_W \text{ord}_W(f) [\text{div}_W(g)] = \sum_W \text{ord}_W(g) [\text{div}_W(f)]$ as zero-cycles.
3. Let $f: \mathbb{P}^1 \rightarrow \mathbb{P}^1$ be the degree-two map $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (characteristic $\neq 2$) and let $D = [\infty]$ be the divisor of the point at infinity on the target. Using proposition 2.3.3(1), compute both $f_*(f^*D \cdot [\mathbb{P}^1])$ and $D \cdot f_*[\mathbb{P}^1]$, identify the scalar $d = \deg f$, and confirm they agree.
4. Let $g: \mathbb{A}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$ be the projection, flat of relative dimension one, and let $D = [0]$ be the origin on $\mathbb{P}^1$. Compute $g^*(D \cdot [\mathbb{P}^1])$ and $g^*D \cdot g^*[\mathbb{P}^1]$ and verify proposition 2.3.3(2) directly.
5. Deduce from theorem 2.3.2 that for any Cartier divisor D on a complete variety X of dimension n , the iterated intersection $D^n \cdot [X] \in A_0(X)$ is symmetric under any reordering of the n copies of D (trivially, but state precisely why the number $\deg(D^n \cdot [X])$ is well defined), and use theorem 1.5.2 to compute $H^n \cdot [\mathbb{P}^n]$ for H a hyperplane.
6. Explain why the proof of theorem 2.3.2 does not require X to be normal, identifying the exact step where normalization is used and why it changes neither side of the identity. (One

sentence on each.)

2.4 The First Chern Class of a Line Bundle

The three previous sections built a single operation and studied it thoroughly: a Cartier divisor D on X acts on cycles by $\alpha \mapsto D \cdot \alpha$, this action descends to rational equivalence (proposition 2.2.4), and two such actions commute (theorem 2.3.2). The divisor D , however, is not the most invariant object in sight. What acts on $A_*(X)$ is not D itself but its linear-equivalence class, since D and $D + \text{div}(r)$ produce the same operator, and a linear-equivalence class of Cartier divisors is exactly a line bundle. The right name for the operator is therefore the *first Chern class* of the associated line bundle $\mathcal{O}_X(D)$, written $c_1(\mathcal{O}_X(D)) \cap -$.

Renaming $D \cdot -$ as $c_1(L) \cap -$ is more than cosmetic. It records that the operator depends only on $L = \mathcal{O}_X(D)$, so it is defined for a line bundle whether or not a divisor is at hand; it makes the additive structure transparent, because tensoring line bundles adds their Chern classes; and it is the first instance of a construction that in the next chapter will attach to a vector bundle E of rank r a whole family of operators $c_1(E), \dots, c_r(E)$, the higher Chern classes. Everything the first Chern class does is already contained in the divisor calculus of the preceding sections; this section repackages that calculus into its final, functorial form and reads off the consequences.

The starting point is that the operator $D \cdot -$ sees only the line bundle $\mathcal{O}_X(D)$. Two Cartier divisors with isomorphic associated bundles are linearly equivalent, that is, they differ by the divisor of a rational function, and a principal divisor intersects every subvariety in a principal divisor, hence acts as zero on $A_*(X)$; proposition 2.4.2 below records this and is the independence needed to base the definition on L rather than on D .

Definition 2.4.1: The First Chern Class of a Line Bundle

Let L be a line bundle on an algebraic scheme X , and suppose $L \cong \mathcal{O}_X(D)$ for some Cartier divisor D on X . The *first Chern class* of L is the operator on Chow groups

$$c_1(L) \cap - : A_k(X) \longrightarrow A_{k-1}(X), \quad c_1(L) \cap \alpha := D \cdot \alpha$$

defined for every k , where $D \cdot \alpha$ is the intersection with the Cartier divisor D of definition 2.2.1. The class $c_1(L)$ is the operator itself; the notation $c_1(L) \cap \alpha$ reads “ $c_1(L)$ capped with α ”.

The definition names D but must not depend on it: a line bundle admits many divisors of sections, and the Chern class has to be the same for all of them. The following proposition is the price of the definition, and its proof is short because the work was done in section 2.2.

Proposition 2.4.2: Independence of the Divisor

Let L be a line bundle on X and let D, D' be Cartier divisors with $\mathcal{O}_X(D) \cong L \cong \mathcal{O}_X(D')$. Then $D \cdot \alpha = D' \cdot \alpha$ in $A_{k-1}(X)$ for every $\alpha \in A_k(X)$. Consequently $c_1(L) \cap -$ of definition 2.4.1 is well defined, independent of the choice of D .

Proof:

An isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ of line bundles is exactly a linear equivalence $D \sim D'$ of Cartier divisors: the tautological rational section of $\mathcal{O}_X(D)$ and that of $\mathcal{O}_X(D')$, compared through the isomorphism, differ by a global invertible section r of the sheaf of total quotient rings $\mathcal{K}_X^*$, so $D - D' = \text{div}(r)$ is principal (on a variety $\mathcal{K}_X^* = k(X)^*$ and r is a rational function; on a general algebraic scheme r is a section of the total-quotient sheaf, which is all the definition of a principal Cartier divisor requires). It therefore suffices to show that a principal Cartier divisor $\text{div}(r)$ acts as zero on rational equivalence classes.

Fix $\alpha \in A_k(X)$ and represent it by a cycle $\sum_i n_i [V_i]$. Since $\text{div}(r) \cdot -$ is defined component by component and extended $\mathbb{Z}$ -linearly (definition 2.2.1), it suffices to show $\text{div}(r) \cdot [V] = 0$ in $A_{k-1}(X)$ for a single k -dimensional subvariety V . The associated bundle of the principal divisor $\text{div}(r)$ is $\mathcal{O}_X$, and the rational section of $\mathcal{O}_X$ that represents $\text{div}(r)$ is r itself; unwinding definition 2.2.1 for this section gives

$$\text{div}(r) \cdot [V] = [\text{div}(r|_V)] \in A_{k-1}(V)$$

where $r|_V \in k(V)^*$ is the restriction of r to V (a nonzero rational function when $V \not\subseteq |\text{div}(r)|$, and otherwise computed through the restricted trivial bundle $\mathcal{O}_X|_V \cong \mathcal{O}_V$, whose section is again a rational function on V). In either case $\text{div}(r|_V)$ is the divisor of a rational function on the variety V , hence a *principal* Weil divisor on V , which is rationally equivalent to zero by the very definition of rational equivalence (definition 1.1.3). Its class in $A_{k-1}(V)$, and so its image in $A_{k-1}(X)$, is therefore 0: intersecting a subvariety with a principal divisor produces a principal divisor, which dies in the Chow group. This is the same computation that drives the well-definedness of proposition 2.2.4, read now with the roles reversed, the divisor rather than the cycle being principal. Summing over the components of α gives $\text{div}(r) \cdot \alpha = 0$, and so

$$D \cdot \alpha - D' \cdot \alpha = (D - D') \cdot \alpha = \text{div}(r) \cdot \alpha = 0$$

giving $D \cdot \alpha = D' \cdot \alpha$, as we sought to show.

The proposition is the reason the notation $c_1(L)$ is legitimate at all. It shows the divisor served only to build the operator: once the operator exists, it is recorded by the line bundle alone. From here on a divisor is used to compute $c_1(L)$, never to define it, and the choice of divisor is a matter of convenience. On a variety, every line bundle admits a Cartier divisor, so the definition applies to all line bundles; on a general scheme one takes L to be of the form $\mathcal{O}_X(D)$, which covers every line bundle whose class lies in the image of the Cartier divisor class group. The definition and the properties below are stated for a general algebraic scheme, but every worked example and exercise that follows runs on a variety, where a rational section and its function field $k(V)$ are always at hand; the general-scheme statements specialize to these illustrations by taking X integral.

A first sanity check grounds the definition before its formal properties: the Chern class of the trivial bundle is the zero operator, and the two constructions available for a hyperplane bundle agree.

Example 2.4.3: The Trivial Bundle and the Hyperplane Bundle

1. *The trivial bundle.* For $L = \mathcal{O}_X$ one may take $D = 0$, the empty divisor. Then $c_1(\mathcal{O}_X) \cap \alpha = 0 \cdot \alpha = 0$ for every α , so $c_1(\mathcal{O}_X)$ is the zero operator. This matches the reading of c_1 as a measure of the twisting of L : a trivial bundle has no twisting to record.

2. *The hyperplane bundle on $\mathbb{P}^n$.* Let $L = \mathcal{O}_{\mathbb{P}^n}(1)$ and let $H \subseteq \mathbb{P}^n$ be a hyperplane, cut out by a linear form; then $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$. The class map of proposition 2.1.3 sends $\mathcal{O}_{\mathbb{P}^n}(1) \mapsto [H] \in A_{n-1}(\mathbb{P}^n)$, the generator of theorem 1.5.2 in codimension one. Capping with $[\mathbb{P}^n]$ recovers this class,

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^n] = [H]$$

since $\mathbb{P}^n \not\subseteq H$ and the restricted section of $\mathcal{O}_{\mathbb{P}^n}(1)$ to $\mathbb{P}^n$ has divisor exactly H . The first Chern class of $\mathcal{O}(1)$, capped against the fundamental class, is the hyperplane class.

With the operator anchored, its formal properties follow. Each is a translation of a divisor identity already proved, and the point of collecting them is to see the first Chern class as a well-behaved endomorphism of the graded group $A_*(X)$: additive in α , commuting with its own kind, additive in L , compatible with pushforward and pullback, and eventually nilpotent. The statement below is the working summary of the theory of c_1 , and every clause is used repeatedly in the chapters that follow.

Proposition 2.4.4: Properties of the First Chern Class

Let X be an algebraic scheme, L, L' line bundles on X , and $\alpha \in A_k(X)$.

1. *Homomorphism.* $c_1(L) \cap - : A_k(X) \rightarrow A_{k-1}(X)$ is a group homomorphism: $c_1(L) \cap (\alpha + \beta) = c_1(L) \cap \alpha + c_1(L) \cap \beta$.
2. *Commutativity.* For line bundles L, L' ,

$$c_1(L) \cap (c_1(L') \cap \alpha) = c_1(L') \cap (c_1(L) \cap \alpha)$$

in $A_{k-2}(X)$.

3. *Additivity.* $c_1(L \otimes L') = c_1(L) + c_1(L')$ as operators; that is, $c_1(L \otimes L') \cap \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$. In particular $c_1(L^\vee) = -c_1(L)$.
4. *Projection formula.* For a proper morphism $f: X' \rightarrow X$, a line bundle L on X , and $\alpha' \in A_k(X')$,

$$f_*(c_1(f^*L) \cap \alpha') = c_1(L) \cap f_*\alpha'$$

and for a flat morphism $g: X'' \rightarrow X$ of relative dimension m and $\beta \in A_k(X)$,

$$c_1(g^*L) \cap g^*\beta = g^*(c_1(L) \cap \beta)$$

in $A_{k+m-1}(X'')$.

5. *Nilpotence.* If X is a variety of dimension n , then for any line bundle L ,

$$c_1(L)^{n+1} \cap [X] = 0$$

where $c_1(L)^{n+1} \cap [X]$ means $c_1(L)$ capped $n+1$ times with the fundamental class $[X]$.

Proof:

Choose Cartier divisors D, D' with $\mathcal{O}_X(D) \cong L$ and $\mathcal{O}_X(D') \cong L'$; by proposition 2.4.2 the choices do not affect any of the claims. Each part translates a divisor identity from section 2.2 or section 2.3.

1. *Homomorphism.* The intersection $D \cdot -$ is defined on cycles component by component and extended $\mathbb{Z}$ -linearly (definition 2.2.1), so it is additive on $Z_k(X)$, and proposition 2.2.4 makes it descend to a well-defined additive map on $A_k(X)$. Hence $c_1(L) \cap - = D \cdot -$ is a homomorphism $A_k(X) \rightarrow A_{k-1}(X)$.
2. *Commutativity.* By definition $c_1(L) \cap (c_1(L') \cap \alpha) = D \cdot (D' \cdot \alpha)$ and $c_1(L') \cap (c_1(L) \cap \alpha) = D' \cdot (D \cdot \alpha)$. These are equal in $A_{k-2}(X)$ by theorem 2.3.2, the commutativity of intersection with two Cartier divisors.
3. *Additivity.* The associated-bundle construction of [4, Divisors: Geometric Representation of Line Bundles] converts the sum of Cartier divisors into the tensor product of line bundles, $\mathcal{O}_X(D+D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$, so $D+D'$ is a Cartier divisor with $\mathcal{O}_X(D+D') \cong L \otimes L'$. Using this divisor to compute the Chern class of $L \otimes L'$,

$$c_1(L \otimes L') \cap \alpha = (D + D') \cdot \alpha = D \cdot \alpha + D' \cdot \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$$

where the middle equality is the additivity of $- \cdot \alpha$ in the divisor, and it holds already at the level of cycles, on *every* subvariety V , with no repositioning of α required. Since $D \cdot -$ is defined component by component and extended $\mathbb{Z}$ -linearly (definition 2.2.1), it suffices to check $(D+D') \cdot [V] = D \cdot [V] + D' \cdot [V]$ for a single k -dimensional subvariety V . Restricting the tensor-product isomorphism above to V gives $\mathcal{O}_X(D+D')|_V \cong \mathcal{O}_X(D)|_V \otimes_{\mathcal{O}_V} \mathcal{O}_X(D')|_V$, so a rational section s_V of $\mathcal{O}_X(D+D')|_V$ representing $(D+D')|_V$ is the product $s_V = s_V^D \cdot s_V^{D'}$ of representing rational sections of $\mathcal{O}_X(D)|_V$ and $\mathcal{O}_X(D')|_V$. By definition 2.2.1 the cap is the class of the divisor of the restricted section computed inside V , and the order function along a codimension-one subvariety W of V is additive on a product of rational sections, $\text{ord}_W(s_V^D \cdot s_V^{D'}) = \text{ord}_W(s_V^D) + \text{ord}_W(s_V^{D'})$, since ord_W is a valuation. Summing over such W gives $[\text{div}(s_V)] = [\text{div}(s_V^D)] + [\text{div}(s_V^{D'})]$ in $A_{k-1}(V)$, that is $(D+D') \cdot [V] = D \cdot [V] + D' \cdot [V]$; pushing forward to X and summing over the components of α gives the middle equality. This is a cycle-level identity holding for each V , so no moving lemma repositioning α off $|D|$ and $|D'|$ is needed, and none exists in general. Taking $L' = L^\vee$, so that $\mathcal{O}_X \cong L \otimes L^\vee$ and $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, gives $c_1(L) + c_1(L^\vee) = 0$, hence $c_1(L^\vee) = -c_1(L)$.

4. *Projection formula.* The pullback along f of the Cartier divisor D , in the pseudo-divisor formalism of definition 2.1.5, is a pseudo-divisor f^*D with $f^*\mathcal{O}_X(D) \cong \mathcal{O}_{X'}(f^*D)$; so $c_1(f^*L) \cap \alpha' = f^*D \cdot \alpha'$. The proper case is the divisor projection formula $f_*(f^*D \cdot \alpha') = D \cdot f_*\alpha'$ of proposition 2.3.3, which rewritten with Chern-class notation is exactly $f_*(c_1(f^*L) \cap \alpha') = c_1(L) \cap f_*\alpha'$. The flat case is the compatibility $g^*(D \cdot \beta) = g^*D \cdot g^*\beta$ of the same proposition, and since $c_1(g^*L) \cap g^*\beta = g^*D \cdot g^*\beta$ this reads $c_1(g^*L) \cap g^*\beta = g^*(c_1(L) \cap \beta)$.
5. *Nilpotence.* Each cap with $c_1(L)$ lowers dimension by one, since $c_1(L) \cap -: A_j(X) \rightarrow A_{j-1}(X)$ by definition 2.4.1, so repeatedly capping $[X] \in A_n(X)$ gives

$$c_1(L)^{n+1} \cap [X] \in A_{n-(n+1)}(X) = A_{-1}(X)$$

and $A_j(X) = 0$ for $j < 0$, there being no subvarieties of negative dimension. Hence the class is zero.

| Each clause has been reduced to a result of the preceding sections, completing the proof.

Parts (1) through (4) say that c_1 is a homomorphism of graded groups compatible with all the structure already on A_* , and that $L \mapsto c_1(L)$ turns tensor product into addition, so $c_1 : \text{Pic}(X) \rightarrow \text{End}(A_*(X))$ is a homomorphism from the Picard group into the operators lowering degree by one. Part (2), the commutativity, is the fact that makes these operators generate a commutative subring of $\text{End}(A_*(X))$; it is the discharged form of the divisor symmetry that theorem 1.4.6 in section 1.4 took on faith, now proved through theorem 2.3.2. Part (5), the nilpotence, is a dimension count: it says a single line bundle can only cut down the fundamental class n times before the result must vanish, and it is what makes the top self-intersection number $c_1(L)^n \cap [X]$, a zero-cycle, the natural degree-type invariant of L on a complete variety.

The nilpotence has a converse worth flagging: while $c_1(L)^{n+1} \cap [X]$ always vanishes, the intermediate power $c_1(L)^n \cap [X] \in A_0(X)$ need not, and its degree $\int_X c_1(L)^n$ is the numerical invariant of a line bundle on an n -dimensional complete variety. For $n = 1$ this is the degree of a line bundle on a curve; for a surface it is the self-intersection L^2 . These invariants are the raw material of the intersection numbers computed throughout this book.

The first substantial computation is the self-intersection of $\mathcal{O}(1)$ on projective space, which reproduces the entire linear-subspace basis of theorem 1.5.2 from a single line bundle. It shows the first Chern class already generates the Chow ring of $\mathbb{P}^n$, and it is the model computation the higher Chern classes generalize.

Example 2.4.5: Powers of the Hyperplane Class on $\mathbb{P}^n$

Let $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$ act on $A_*(\mathbb{P}^n)$. By theorem 1.5.2 the Chow group $A_*(\mathbb{P}^n) = \bigoplus_{j=0}^n \mathbb{Z} [\mathbb{P}^j]$ is free on the classes of linear subspaces, with $[\mathbb{P}^j]$ the class of a j -plane. The claim is

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \quad 0 \leq k \leq n$$

so the k -th self-cap of the hyperplane bundle is the class of a linear subspace of codimension k .

The proof is induction on k . For $k = 0$ the left side is $[\mathbb{P}^n]$, the fundamental class, and $[\mathbb{P}^{n-0}] = [\mathbb{P}^n]$. Assume $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for some $k < n$, so the class is represented by the subvariety $\mathbb{P}^{n-k} \subseteq \mathbb{P}^n$, a linear subspace of dimension $n - k$. Cap once more with H . There is a hyperplane H_0 with $\mathbb{P}^{n-k} \not\subseteq H_0$, because some linear form does not vanish identically on the nonempty $\mathbb{P}^{n-k}$ (equivalently, $\mathbb{P}^{n-k}$ is not contained in every hyperplane); its zero locus is such an H_0 , and $H_0 \cap \mathbb{P}^{n-k}$ is then a hyperplane inside $\mathbb{P}^{n-k}$, of dimension $n - k - 1$. Thus $\mathbb{P}^{n-k} \not\subseteq |H_0|$, and by definition 2.2.1 the cap $H \cdot [\mathbb{P}^{n-k}]$ is the divisor of the restricted linear form on $\mathbb{P}^{n-k}$, namely the hyperplane section $H_0 \cap \mathbb{P}^{n-k}$, a linear subspace $\mathbb{P}^{n-k-1}$ of dimension $n - k - 1$:

$$c_1(\mathcal{O}(1))^{k+1} \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^{n-k}] = [H_0 \cap \mathbb{P}^{n-k}] = [\mathbb{P}^{n-k-1}]$$

completing the induction. In particular $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathbb{P}^0] = [\text{pt}]$ is the class of a single point, and its degree is

$$\int_{\mathbb{P}^n} c_1(\mathcal{O}_{\mathbb{P}^n}(1))^n = 1$$

the statement that n general hyperplanes in $\mathbb{P}^n$ meet in a single reduced point. Capping once more, $c_1(\mathcal{O}(1))^{n+1} \cap [\mathbb{P}^n] = H \cdot [\text{pt}] = 0$, in agreement with the nilpotence of proposition 2.4.4(5). Every generator $[\mathbb{P}^j]$ of $A_*(\mathbb{P}^n)$ is thus a power of $c_1(\mathcal{O}(1))$ applied to $[\mathbb{P}^n]$, so the first Chern class of the hyperplane bundle generates the Chow group of projective space as a module over the operators.

The computation is the first appearance of the ring structure that intersection theory is heading toward: on $\mathbb{P}^n$ the operators H^k multiply the way linear subspaces intersect, $[\mathbb{P}^a] \cdot [\mathbb{P}^b] = [\mathbb{P}^{a+b-n}]$ when $a + b \geq n$ and zero otherwise, and $\int_{\mathbb{P}^n} H^n = 1$ is the normalization. The Chow group $A_*(\mathbb{P}^n)$ is the truncated polynomial ring $\mathbb{Z}[H]/(H^{n+1})$ with $H = c_1(\mathcal{O}(1))$, and the nilpotence relation $H^{n+1} = 0$ is exactly proposition 2.4.4(5). This ring will be built in general later in this Part, once the intersection product is available; here it is visible for projective space through the single line bundle $\mathcal{O}(1)$.

The second computation identifies c_1 on a curve with the classical degree, tying the abstract operator to the invariant every reader already knows. On a smooth projective curve the first Chern class is the degree map, and capping with $c_1(L)$ is subtraction of that degree from the point count.

Example 2.4.6: The First Chern Class on a Curve is the Degree

Let C be a smooth projective curve over $\bar{k}$ and L a line bundle on C . Since $\dim C = 1$, the only nonzero cap is on $A_1(C) = \mathbb{Z}[C]$, and it lands in $A_0(C)$. Write $L \cong \mathcal{O}_C(D)$ for a Cartier divisor $D = \sum_P n_P [P]$, which on a smooth curve is the same as a Weil divisor, a finite $\mathbb{Z}$ -combination of closed points. Then

$$c_1(L) \cap [C] = D \cdot [C] = \sum_P n_P [P] \in A_0(C)$$

by definition 2.2.1, the section of $\mathcal{O}_C(D)$ having divisor D on the curve C itself. Pushing this zero-cycle to a point through $\pi: C \rightarrow \text{Spec } \bar{k}$ and taking degrees (definition 1.2.4),

$$\int_C c_1(L) \cap [C] = \sum_P n_P [\bar{k}(P) : \bar{k}] = \sum_P n_P = \deg D = \deg L$$

since every closed point of a variety over the algebraically closed field $\bar{k}$ has residue field $\bar{k}$. The number $\int_C c_1(L)$ is the degree of the line bundle. Under the identification $A_0(C) \cong \text{Cl } C = \text{Pic } C$ of proposition 1.5.3, capping with $c_1(L)$ against $[C]$ is the class map $L \mapsto [D]$, and its composite with the degree $A_0(C) \xrightarrow{\deg} \mathbb{Z}$ recovers the degree of the divisor class, fitting into the sequence $0 \rightarrow \text{Pic}^0 C \rightarrow A_0(C) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$ of proposition 1.5.3.

The two computations frame the whole theory. On projective space $c_1(\mathcal{O}(1))$ generates the Chow ring and its top power counts a single point; on a curve $c_1(L)$ is the degree, the invariant that governs Riemann-Roch through $\chi(L) = \deg L + 1 - g$ ([4, Riemann-Roch for curves]). Both are shadows of one construction: the first Chern class turns a line bundle into an operator on cycles, and its powers are the numerical invariants that intersection theory computes. The chapter that follows generalizes the construction from line bundles to vector bundles of arbitrary rank: a rank- r bundle E carries Chern classes $c_1(E), \dots, c_r(E)$, built so that c_1 of a line bundle is the operator of this section, and so that the total Chern class is multiplicative in short exact sequences. The machine that produces them, Segre classes of the projectivized bundle $\mathbb{P}(E)$, is the subject of the next chapter, and the divisor calculus of this chapter is the rank-one base case on which it is built.

Exercise 2.4.1

1. Compute $\int_{\mathbb{P}^1} c_1(\mathcal{O}_{\mathbb{P}^1}(d))$ for every integer d , both by the curve-degree formula of example 2.4.6 and directly from a divisor of a section of $\mathcal{O}_{\mathbb{P}^1}(d)$. Confirm that $c_1: \text{Pic}(\mathbb{P}^1) \rightarrow A_0(\mathbb{P}^1) = \mathbb{Z}$ is an isomorphism.

2. Let S be a smooth projective surface and $L = \mathcal{O}_S(C)$ for a curve $C \subseteq S$. Express $c_1(L)^2 \cap [S] \in A_0(S)$ and its degree $\int_S c_1(L)^2$ in terms of the divisor C , and identify $\int_S c_1(L)^2$ with the self-intersection number $C \cdot C$ of the surface theory of [4, Intersection Theory on Surfaces]. Why does proposition 2.4.4(5) say $c_1(L)^3 \cap [S] = 0$?
3. Using proposition 2.4.4(3), show that $c_1 : \text{Pic}(X) \rightarrow \text{End}(A_*(X))$ is a group homomorphism from the Picard group to the additive group of degree-lowering operators. Deduce that $c_1(L^{\otimes m}) = m c_1(L)$ for every integer m , and that c_1 factors through the numerical class of L in the sense that linearly equivalent line bundles have equal first Chern class.
4. On $\mathbb{P}^1 \times \mathbb{P}^1$, let p_1, p_2 be the two projections and $L = p_1^* \mathcal{O}(a) \otimes p_2^* \mathcal{O}(b)$ the line bundle of bidegree (a, b) . Using the projection formula proposition 2.4.4(4) and additivity, compute $\int_{\mathbb{P}^1 \times \mathbb{P}^1} c_1(L)^2$ in terms of a and b . (You may use $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ free on the classes of a point, the two rulings, and the fundamental class, with each ruling of self-intersection zero and the two rulings meeting in one point.)
5. Let $f : X \rightarrow Y$ be a proper surjective morphism of varieties with $\dim X = \dim Y = n$ and generic degree δ , and let L be a line bundle on Y . Show that $\int_X c_1(f^* L)^n = \delta \int_Y c_1(L)^n$. (Cap the projection formula n times and use $f_*[X] = \delta[Y]$.) Then explain why $c_1(f^* L)^{n+1} \cap [X] = 0$ holds regardless of δ .

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